A Modern Handbook of Physics

B. M. Yavorsky, A.A. Detlaf

Contents

Preface v

I Mechanics 1

1 Kinematics 3
- 1.1 Mechanical Motion. The Subject Matter of Mechanics 3
- 1.2 Frames of Reference. Path, Path Length 5
- 1.3 Velocity 11
- 1.4 Acceleration 15
- 1.5 Translational and Rotary Motion of a Rigid Body 19

2 Newton's Laws 29
- 2.1 Newton's First Law. Inertial Frames of Reference 29
- 2.2 Force 31
- 2.3 Mass. Momentum 35
- 2.4 Newton's Second Law 38

2.5 Newton's Third Law. Motion of the Centre of Mass 41
2.6 Motion of a Body of Variable Mass 43
2.7 Law of Conservation of Momentum 46
2.8 Galilean Transformations 49

3 Work And Mechanical Energy 57
3.1 Energy, Work and Power 57
3.2 Kinetic Energy 65
3.3 Potential Energy 68
3.4 Law of Conservation of Mechanical Energy 74
3.5 Perfectly Elastic and Inelastic Collisions 80

4 Dynamics of Rotary Motion 87
4.1 Moment of Force and Angular Momentum 87
4.2 Moment of Inertia 94
4.3 The Fundamental Law in the Dynamics of Rotary Motion 97
4.4 Law of Conservation of Angular Momentum 103

5 Fundamentals of the Special Theory of Relativity 109
5.1 Postulates of the Special Theory of Relativity 109
5.2 Simultaneity of Events. Synchronization of Clocks 113
5.3 Lorentz's Transformations 116
5.4 Relativity of Lengths and Time Intervals. 119
5.5 Transformation of Velocities and Accelerations 128
5.6 Basic Law of Relativistic Dynamics 133
5.7 Mass-Energy Relation 136

6 Gravitation **143**
6.1 Law of Universal 143
6.2 Gravitational Field 146
6.3 Kepler's Laws. Space Velocities 154

7 Motion in non-inertial Frames of Reference **159**
7.1 Kinematics of Relative Motion 159
7.2 Inertial Forces 162
7.3 Frame of Reference Fixed to the Earth 166
7.4 Principle of Equivalence 172

II Fundamentals of Molecular Physics and Thermodynamics **177**

8 Ideal Gases **179**
8.1 Subject Matter of Molecular Physics. Thermal Motion 179
8.2 Statistical and Thermodynamic Methods 182
8.3 Thermodynamic Variables 184
8.4 Equation of State of an Ideal Gas 189

9 First law of Thermodynamics **193**
9.1 Total and Internal Energy 193
9.2 Heat and Work 196
9.3 First Law of Thermodynamics 202
9.4 Graphical Representation 204
9.5 Heat Capacity of Matter 207

10 Kinetic Theory of Gases **219**
10.1 Certain Information on Classical Statistical Physics 219
10.2 Basic Equation 221

10.3 Maxwell's Distribution Law 224
10.4 Boltzmann Distribution 230
10.5 Mean Free Path of Molecules 233
10.6 Equipartition of Energy 235

Preface

The basic sciences and physics, in particular, are of prime importance today in the training of engineers for the various branches of the national economy. This has led to fundamental changes in recent years in the teaching of physics in engineer ing institutes, and to the students of other educational institutions in which they do not major in physics. The scope and scientific level of physics courses have been substantially supplemented and cover the main trends in the development of modern physics. Consequently, the physics textbooks for engineering students have inevitably become three-volume editions of almost fifteen hundred pages. The need has arisen, in this connection, for a concise handbook on this subject. The aim of the authors was to fulfill this need.

In scope and depth this handbook includes all of the definitions, formulas and information covered in the most comprehensive and up-to-date physics courses of engineering institutes and the physics departments of universities and colleges. Physical laws are concisely formulated, all the necessary explanations are given and, in many cases, derivations as well. Though it plays a vi-

tal role in the teaching of physics, experimental material has been omitted. This is due only to a lack of space. All the units and symbols comply with the requirements of the SI. Units of physical quantities and systems of units are listed and dealt with in a short appendix.

This handbook is designed primarily for engineering students, as well as college and university students studying, but not majoring in physics. It can be used to advantage by engineers and graduate students, as well as by instructors and lecturers of intermediate schools and colleges.

Mathematical knowledge required in using the handbook is within the scope of the ordinary mathematics courses of engineering institutes. The detailed index and the numerous cross references, indicating the chapter, section and subsection, are of aid in finding any required information. The authors are sincerely grateful to the translator, N. Weinstein, and the science editor, T. Baranovskaya, for their comprehensive and expert work in the publication of the English edition of this handbook, as well as for their valuable suggestions, which should facilitate its use by the reader.

B. Yavorsky
A. Detlaf

Parts One, Four, Five and the appendices were written by Prof. A. A. Detlaf; Parts Two, Three, Six, Seven and Eight, by Prof. B. M. Yavorsky.

Part I

Mechanics

1 Kinematics

1.1 Mechanical Motion. The Subject Matter of Mechanics

1.1.1

The simplest form of motion in nature is *mechanical motion*. It consists of changes in the relative positions of bodies or their parts in space in the course of time. The branch of physics dealing with the laws of mechanical motion is called *mechanics*. In a more narrow sense of the word, this is often under stood to mean *classical mechanics*, which deals with the motion of macroscopic bodies travelling at velocities a great many times less than that of light in free space. Classical mechanics is based on Newton's laws and is therefore frequently referred to as *Newtonian mechanics*. The laws of motion of bodies at velocities close to that of light in free space are taken up in *relativistic mechanics* (5.1.1), and the laws of motion of micro particles (for example, the electrons in atoms, molecules, crystals, etc.) are the subject matter of *quantum mechanics* (**??**).

1.1.2

Classical mechanics comprises three basic branches: statics, kinematics and dynamics. Statics deals with the laws for the composition of forces and the conditions for the equilibrium of bodies. *Kinematics* provides a mathematical description of all possible kinds of mechanical motion, regardless of what causes each particular kind of motion. *Dynamics* investigates the effect of the interaction between bodies on their mechanical motion.

1.1.3

The mechanical properties of bodies are determined by their chemical nature, internal structure and state. These are beyond the scope of mechanics and are studied in other branches of physics. Hence, in accordance with the conditions of each specific problem on hand, various simplified models are employed to describe real bodies in motion. These include the particle, the perfectly rigid body, the perfectly elastic body, the perfectly inelastic body and others.

A *material point*, or *particle*, is a body whose shape and size are of no consequence under the conditions of the problem being considered. For example, in a first approximation, the motion of a ship from one point to another can be dealt with as the motion of a particle. If, however, it becomes necessary to take into account such details of this motion as the pitching and rolling of the ship by swells or heavy seas, the ship should be dealt with as an extended body of definite shape. In the literature the word particle is more frequently used than material point.

Any extended body or system of such bodies, making up the *mechanical system* being investigated, can be regarded as a *system of particles*. For this purpose, ail the bodies of the system can be conceived of as being broken down into such a large number of parts that the size of each part is negligibly small compared to the sizes of the bodies themselves.

1.1.4

A *perfectly rigid body*, or simply *rigid body*, is one whose deformation can be neglected under the conditions of the given problem. The distance between any two points of a rigid body remains constant no matter what action the body is subjected to. A perfectly rigid body can be regarded as a system of particles rigidly connected to one another.

A *perfectly elastic body* is one whose deformation obeys Hooke's law (**??**). After the external forces applied to such a body have been removed, its initial size and shape are completely restored.

A *perfectly inelastic body* is one that completely retains its deformed state, caused by external forces applied to the body, when these forces have been removed.

1.2 Frames of Reference. Path, Path Length and Displacement Vector of a Particle

1.2.1

The position of a body in space can be specified only in reference to other bodies. It makes sense, for instance, to speak of the location of the planets with respect to the sun, that of an air liner or steamship with respect to the earth, but we cannot indicate their position in space "in general" without reference to some specific body. A perfectly rigid body in which a coordinate system is rigidly fixed, furnished with a "clock" and employed to determine the position of bodies and particles being investigated in space at various instants of time, is called a frame of reference. Sometimes the frame of reference is the coordinate system itself if it is provided with a "clock" and the solid in which the system is rigidly fixed is called the body of reference. In each specific problem the frame of reference is selected so as to simplify the problem to the maximum extent. Inertial frames of reference (2.1.2) are the most commonly used in physics.

1.2.2

Most extensively employed is the Cartesian rectangular system of coordinates (Figure 1.1) whose orthonormal (normal orthogonal) basis is formed by the three vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$, which are unit vectors with respect to their magnitudes, mutually orthogonal and pass through the origin O of coordinates. The position of an arbitrary point M is specified by the radius vector $\mathbf{r}$, which connects the origin O of coordinates to point M. Vector $\mathbf{r}$ can be resolved along the axes $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ of the basis as follows:

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

where $x\mathbf{i}$, $y\mathbf{j}$ and $z\mathbf{k}$ are the components of vector $\mathbf{r}$ along the coordinate axes. The coefficients of resolution, x, y and z, are the Cartesian coordinates of point M and, due to the orthogonality of the vectors of the basis, the projections of radius vector $\mathbf{r}$ on the corresponding coordinate axes.

The motion of a particle is completely specified if three continuous and single-valued functions of the time t,

$$x = x(t),\ y = y(t) \text{ and } z = z(t),$$

are given to describe the changes in the coordinates of the particle with time.

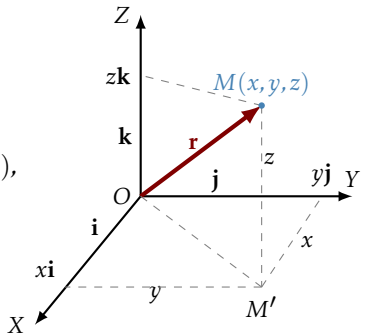


Figure 1.1: Cartesian system of coordinates.

These equations are said to be the *kinematic equations of motion of a particle*. They are equivalent to the single vector equation $\mathbf{r} = \mathbf{r}(t)$, for a moving particle.

1.2.3

The line described by a moving particle in space is called its *path*. The kinematic equations of motion of a particle specify the equation of its path in parametric form (the parameter being the time t). Depending upon the shape of the path, distinction is made between *rectilinear* and

curvilinear motion. The motion of a particle is said to be *plane* if its path lies completely in a single plane.

The mechanical motion of a body is relative, i.e. the nature of the motion and, in particular, the kind of path travelled by the points of the body depend upon the chosen frame of reference.

1.2.4

In the general case, the path of a moving particle is a spatial, rather than plane, curve. Introduced for such curves is the concept of an *osculating plane*. The osculating plane at the arbitrary point M of a curve is in the limiting position of a plane passing through any three points of the curve when these points approach point M without limit.

An *osculating circle* at point M of a curve is the limit of a circle passing through three points of the curve being considered when these points approach point M without limit. The osculating circle lies in the osculating plane. The centre of the osculating circle and its radius are called the *centre of curvature* and *radius of curvature*, at point M, of the curve being considered. The straight line connecting point M to the centre of curvature is called the *principal normal* to the curve at point M. The tangent to the curve at point M is perpendicular to the principal normal and also lies in the osculating plane.

1.2.5

The *path length* is the sum of the lengths of all the portions of the path over which the particle passes m the given interval of time. The instant of time $t = t_0$, from which we begin to consider the motion of a particle, is called the initial instant of time, and the position of the particle at this instant (point A in Figure 1.2) is called its initial position. Owing to the arbitrary nature of the time reference point it is usually assumed that $t_0 = 0$.

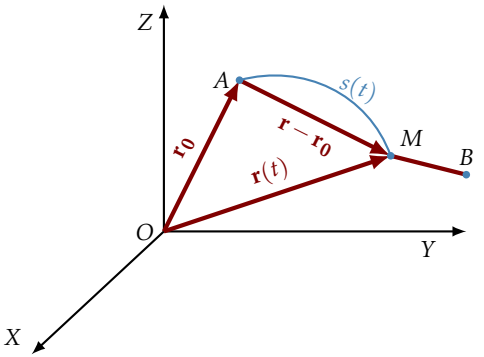


Figure 1.2: A particle moving on a curve.

The path length s, travelled by the particle from its initial position, is a scalar function of time: $s = s(t)$. As is obvious from its definition, the path length of a particle cannot be a negative quantity. If the particle travels along an arc of the path AB (Figure 1.2), always in the same direction and is at point M at the instant of time t,then $s(t) = \widehat{AM}$. If the particle travels along the path in a more complicated way, for example if it trav-

els from A to B, reaching B at the instant of time $t_1 < t$, and then reverses, returning to point M at instant t, then $s(t) = \widehat{AB} + \widehat{BM}$.

1.2.6

The displacement vector of a moving particle during the time interval from $t = t_1$ to $t = t_2$ is the vector from the, position of the particle at the instant of time t_1 to its position at instant t_2. It is equal to the increment in the radius vector of the particle during the time interval being considered:

$$\mathbf{r_2} - \mathbf{r_1} = \mathbf{r}(t_2) - \mathbf{r}t_1$$

The displacement vector is always directed along the chord subtending the corresponding portion of the path.

Shown in Figure 1.2 is the displacement vector of the particle for the time interval from t_0 to t. It is equal to $\mathbf{r} - \mathbf{r_0} = \mathbf{r}(t) - \mathbf{r}(t_0)$.

The displacement vector of a moving particle for the time interval from t to $t + \Delta t$ equals

$$\Delta r = r(t + \Delta t) - r(t) = \Delta x \times \mathbf{i} + \Delta y \times \mathbf{j} + \Delta z \times \mathbf{k}$$

where Δx, Δy and Δz are the increments (changes) in the coordinates of the particle during the time interval being considered.

1.2.7

A particle travelling freely in space can accomplish only three *independent motions*, i.e. motions that cannot be

represented as combinations of the other two. As a matter of fact, the motion of a particle along each of the axes of a Cartesian rectangular coordinate system cannot be obtained by its motion along the other two axes. The number of independent motions that a mechanical system can accomplish is called the *number of degrees of freedom* of the system. Hence, a free particle has three degrees of freedom.

1.3 Velocity

1.3.1

The concept of velocity is introduced in mechanics to characterize the rate of motion of bodies. It is the time rate of change in the position of moving bodies. The *average velocity* of a moving particle in the time interval from t to $t+\Delta t$ is the vector $\mathbf{v_{av}}$, equal to the ratio of the increment $\Delta\mathbf{r}$ in the radius vector of the particle during this time interval to the length of the time interval Δt:

$$\mathbf{v_{av}} = \frac{\Delta \mathbf{r}}{\Delta t}$$

Vector $\mathbf{v_{av}}$ has the same direction as $\Delta\mathbf{r}$, i.e. along the chord subtending the corresponding portion of the particle's path.

1.3.2

The *velocity* (or *instantaneous velocity*) of a moving particle is the vector quantity $\mathbf{v}$, equal to the first time deriva-

tive of the radius vector $\mathbf{r}$ of the particle being considered:

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$$

The velocity of the particle at the instant of time t is the limiting value that the average velocity $\mathbf{v_{av}}$ approaches for an infinitely small value of the time interval Δt. Thus

$$\mathbf{v} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{r}}{\Delta t} = \lim_{\Delta t \to 0} \mathbf{v_{av}}$$

The velocity vector $\mathbf{v}$ of the particle has the direction along a tangent to the path toward the motion in the same way as the vector $d\mathbf{r} = \mathbf{v}dt$ of infinitely small displacement of the particle during an infinitely short time interval dt.

The path length ds travelled by the particle during the length of time dt is equal to the magnitude of the displacement vector: $ds = |d\mathbf{r}|$. Therefore the magnitude of the velocity vector of a moving particle is equal to the first time derivative of the path length:

$$v = |\mathbf{v}| = \frac{\mathrm{d}s}{\mathrm{d}t}$$

1.3.3

The resolution of vector $\mathbf{v}$ along the basis of a Cartesian rectangular coordinate system is of the following form:

$$\mathbf{v} = v_x \times \mathbf{i} + v_y \times \mathbf{j} + v_z \times \mathbf{k}$$

The projections of the velocity of a moving particle on the coordinate axes are equal to the first time derivative

of the corresponding coordinates of the particle:

$$v_x = \frac{\mathrm{d}x}{\mathrm{d}t},\ v_y = \frac{\mathrm{d}y}{\mathrm{d}t},\ v_z = \frac{\mathrm{d}z}{\mathrm{d}t}$$

and the magnitude of the velocity vector, sometimes called the *speed*, is

$$v = |\mathbf{v}| = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2}$$

1.3.4

In rectilinear motion of a particle, the direction of its velocity vector remains constant. The motion of a particle is said to be *uniform* if the magnitude of its velocity does not change with time: $v = ds/dt = \text{const}$. The path length s travelled by a uniformly moving particle is a linear function of time: $s = vt$ (provided that $t_0 = 0$, see (1.2.5)).

If the magnitude of the particle's velocity, i.e. its speed, increases with time ($dv/dt > 0$), the motion is said to be *accelerated*; if the speed decreases with time ($dv/dt < 0$), the motion is said to be *decelerated*.

1.3.5

The *average velocity of nonuniform motion* of a particle over a given portion of its path is the scalar v_{av}, called the *travelling speed*, which is equal to the ratio of the length Δs of this portion to the length Δt of the time interval required to travel over this portion:

$$v_{\text{av}} = \frac{\Delta s}{\Delta t}$$

It is equal to the speed of uniform motion in which the time required to travel the same path length As is the same as in the nonuniform motion being considered.

In curvilinear motion of a particle $|\Delta\mathbf{r}| < \Delta s$. Therefore, in the general case, the average velocity (travelling speed) v_{av} of the particle is not equal to the magnitude of its average velocity $\mathbf{v_{av}}$ over the same portion of its path (1.3.1): $v_{\text{av}} \geqslant |\mathbf{v}|$ where the equality sign corresponds to a rectilinear portion of the path.

1.3.6

In the case of plane motion of a particle M (1.2.3), it often proves convenient to use the polar coordinates r and φ, where r is the distance from pole O to particle M and φ is the vectorial angle measured from polar axis OA (Figure 1.3). The velocity $\mathbf{v}$ of particle M can be resolved

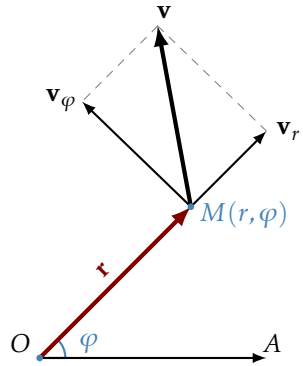


Figure 1.3: Polar coordinates of a moving particle.

into two mutually perpendicular components: the *radial velocity* $\mathbf{v}_r$ and the *transverse velocity* $\mathbf{v}_\varphi$:

$$\mathbf{v} = \mathbf{v}_r + \mathbf{v}_\varphi$$

in which

$$\mathbf{v}_r = \frac{1}{r}\frac{\mathrm{d}r}{\mathrm{d}t}\mathbf{r}, \text{ and } \mathbf{v}_\varphi = \frac{\mathrm{d}\varphi}{\mathrm{d}t}[\mathbf{kr}]$$

where $\mathbf{r}$ is the polar radius vector of particle M, and $\mathbf{k}$ is the unit vector directed perpendicular to the plane of motion of the particle so that the rotation of radius vector $\mathbf{r}$ upon an increase in angle φ is seen, from the head of unit vector $\mathbf{k}$, to be counterclockwise.

The magnitude v of the velocity vector of particle M, in plane motion, is

$$v = \sqrt{\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + \left(r\frac{\mathrm{d}\varphi}{\mathrm{d}t}\right)^2}$$

During the infinitely short length of time dt, polar radius vector $\mathbf{r}$ of the particle in plane motion sweeps over a sector of a circle, the area of the sector being $dS = r^2\,d\varphi/2$. Hence, the quantity

$$\sigma = \frac{\mathrm{d}S}{\mathrm{d}t} = \frac{1}{2}r^2\frac{\mathrm{d}\varphi}{\mathrm{d}t} = \frac{1}{2}rv_\varphi$$

is called the *areal* velocity.

1.4 Acceleration

1.4.1

The concept of acceleration is introduced in mechanics to characterize the rate of change of the velocity vector of a mov ing particle. The *average acceleration* in the time interval from t to $t + \Delta t$ is the vector $\mathbf{a}_{\text{av}}$, which is equal to the ratio of the increment $\Delta\mathbf{v}$ of the velocity vector of the particle, during the lime interval, to the length of the time interval Δt:

$$\mathbf{v}_{\text{av}} = \frac{\Delta\mathbf{v}}{\Delta t}$$

1.4.2

Acceleration (or *instantaneous acceleration*) of a moving particle is the vector quantity $\mathbf{a}$, equal to the first time derivative of the velocity $\mathbf{v}$ of the particle being considered or, other wise, the second time derivative of the particle's radius vector $\mathbf{r}$:

$$\mathbf{a} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}$$

The acceleration of a moving particle at the instant of time t is the limiting value that the average acceleration $\mathbf{a}_{\text{av}}$ approaches for an infinitely small time interval Δt. Thus

$$\mathbf{a} = \lim_{\Delta t \to 0} \frac{\Delta\mathbf{v}}{\Delta t} = \lim_{\Delta t \to 0} \mathbf{a}_{\text{av}}$$

1.4.3

The resolution of vector $\mathbf{a}$ along the basis of a rectangular Cartesian coordinate system is of the form:

$$\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

The projections of the acceleration vector of a moving particle on the coordinate axes are equal to the first time derivatives of the corresponding velocity projections or, otherwise, the second time derivatives of the corresponding coordinates of the particle:

$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2},\ a_y = \frac{dv_y}{dt} = \frac{d^2y}{dt^2},\ a_z = \frac{dv_z}{dt} = \frac{d^2z}{dt^2}$$

The magnitude of the acceleration vector is

$$a = |a| = \sqrt{\left(\frac{dv_x}{dt}\right)^2 + \left(\frac{dv_y}{dt}\right)^2 + \left(\frac{dv_z}{dt}\right)^2}$$
$$= \sqrt{\left(\frac{d^2x}{dt^2}\right)^2 + \left(\frac{d^2y}{dt^2}\right)^2 + \left(\frac{d^2z}{dt^2}\right)^2}$$

1.4.4

The acceleration vector of a moving particle lies in the osculating plane (1.2.4) passing through point M of the path being considered and is direct ed toward the concavity of the path BC (Figure 1.4). In this plane the acceleration vector $\mathbf{a}$ can be resolved into two mutually perpendicular components $\mathbf{a}_\tau$ and $\mathbf{a}_n$:

$$\mathbf{a} = \mathbf{a}_\tau + \mathbf{a}_n$$

1.4.5

Component $\mathbf{a}_\tau$ is called the *tangential acceleration* of the moving particle. It is directed along the tangent to the

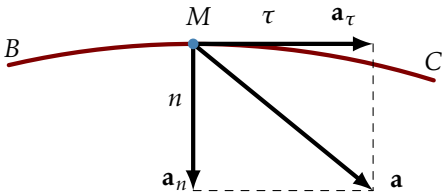


Figure 1.4: Tangential and normal components of the acceleration of a moving particle.

path and equals

$$\mathbf{a}_\tau = \frac{\mathrm{d}v}{\mathrm{d}t}\tau \text{ and } a_t = \frac{\mathrm{d}v}{\mathrm{d}t}$$

where $\tau = \mathbf{v}/v$ is the unit vector of the tangent, from point M of the path in the direction of velocity $\mathbf{v}$ of the particle, and a_τ is the projection of the tangential acceleration on the direction of vector $\mathbf{v}$. The tangential acceleration characterizes the rate of change of the magnitude of the particle's velocity vector. Vectors $\mathbf{a}_\tau$ and $\mathbf{v}$ coincide in direction, i.e. $a_\tau > 0$, in accelerated motion (1.3.4); vectors $\mathbf{a}_\tau$ and $\mathbf{v}$ are opposite in direction $a_\tau < 0$, in decelerated motion of the particle; and $a_\tau = 0$ in its uniform motion. If $a_\tau = \text{const} \neq 0$, the motion is said to be uniformly variable. In uniformly variable motion the magnitude of the particle's velocity is linearly time-dependent:

$$v = v_0 + a_\tau t$$

where $v_0 = v(0)$ is the magnitude of the initial velocity, i.e. the velocity at the initial instant of time $t = 0$. If $a_\tau = \text{const} > 0$, the motion of the particle is said to be

uniformly accelerated, whereas if $a_\tau = \text{const} < 0$, it is said to be *uniformly decelerated*.

1.4.6

Component $\mathbf{a}_n$ of acceleration $\mathbf{a}$ of the moving particle is called its *normal acceleration*. It is directed along the principal normal to the path at point M being considered, toward the centre of curvature (1.2.4) of the path. Therefore, an is often called the *centripetal acceleration* of the particle. The normal acceleration equals

$$\mathbf{a}_n = \frac{v^2}{R}\mathbf{n}$$

where $\mathbf{n}$ is the unit vector of the principal normal, and R is the radius of curvature of the path. The normal acceleration characterizes the rate of change in direction of the particle's velocity vector.

If the particle travels in a straight line the normal acceleration $\mathbf{a}_n = 0$ and the acceleration is equal to the tangential acceleration of the particle: $\mathbf{a} = \mathbf{a}_\tau$.

1.5 Translational and Rotary Motion of a Rigid Body[1]

1.5.1

Translation is the kind of motion of a rigid body in which any straight line rigidly fixed in the body (for instance,

[1] Only perfectly rigid bodies are dealt with in this section; for the sake of brevity, they have been called simply "rigid bodies".

line AB in Figure 1.5), remains parallel to its initial position (A_0B_0). With respect to the earth, a lift cabin (elevator car), lathe tool, and many other items have translational motion. In the translational motion of a rigid rigid

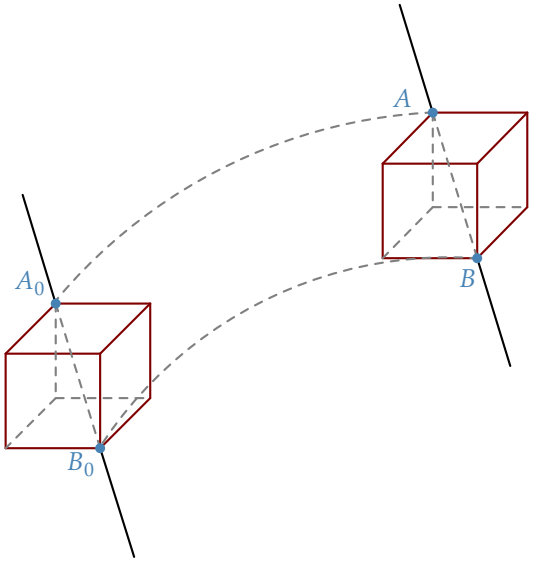


Figure 1.5: Translational motion of a rigid body.

body all of its points are displaced in an identical way.

During the infinitely short time dt the radius vectors of these points are changed by the same amount dr. Hence the velocity of all points of the body is the same in each instant of time and is equal to d$\mathbf{r}$/dt and consequently, their accelerating is also the same. For this reason, a kinematic discussion of the translational mo-

tion of a rigid body reduces to the investigation of the motion of any one of its points, or particles. Dynamics usually deals with the motion of the centre of mass of the body (2.3.3). A rigid body travel ling freely in space has three translational degrees of freedom (1.2.7), corresponding to its translational motions along the three coordinate axes.

1.5.2

The motion of a rigid body in which two of its points, A and B, remain fixed is called *rotation (or rotary motion) of the body about a fixed axis*. The fixed straight line AB is the axis of rotation. In the rotation of a rigid body about a fixed axis all the points of the body describe circles whose centres lie on the axis of rotation and whose planes are perpendicular to this axis. The rotors of turbines, motors and generators, for example, have rotary motion with respect to the earth if these machines are rigidly mounted.

A rigid body rotating about a fixed axis has only one degree of freedom (1.2.7). Its position in space can be completely defined by specifying the angle of rotation (position angle) φ from some definite (initial) position.

1.5.3

The angular velocity serves to characterize the rate and direction of rotation of a body about an axis. The *angular velocity* is the vector $\boldsymbol{\omega}$, which is equal in magnitude to the first derivative of the angle of rotation φ with respect to time t and is directed along the fixed axis of rotation

in such a way that from its head the rotation of the body is counterclockwise (Fig. 1.6)[2]:

$$\boldsymbol{\omega} = \frac{\mathrm{d}\boldsymbol{\varphi}}{\mathrm{d}t} \text{ and } \omega = \frac{\mathrm{d}\varphi}{\mathrm{d}t}$$

Here $\mathrm{d}\boldsymbol{\varphi}$ is the vector of an element of rotation of the body during the time dt. This vector is directed along the axis of rotation, also according to the right-hand screw rule (Figure 1.6). Axial vectors $\mathrm{d}\boldsymbol{\varphi}$ and $\boldsymbol{\omega}$ have no definite points of application: they can be laid off from any point on the axis of rotation. In Figure 1.6 they are laid off from a certain point O of the fixed axis of rotation. This point is also taken as the origin of the reference frame.

Rotation of a body is said to be *uniform* if the magnitude of its angular velocity remains constant with time $\omega = \text{const}$. In this case the angle of rotation of the body is linearly dependent on time: $\varphi = \omega t$.

1.5.4

An arbitrary point M of a rigid body rotating about fixed axis OZ with the angular velocity $\boldsymbol{\omega}$ describes a circle of radius ρ with its centre at point O' (Figure 1.7). As distinct from the angular velocity of the body, the velocity

[2] The direction of vector $\boldsymbol{\omega}$ can also be determined by the *right-hand screw rule*. It coincides with the direction of transla tional motion of a right-hand screw rotating together with the body. Vectors like $\boldsymbol{\omega}$, whose direction is related to the direction of rotation and is reversed in transforming from a right-handed to a left-handed system of coordinates, are called *pseudovectors*, or *axial vectors* (in distinction to ordinary, *polar vectors*, which are not reversed in the above-mentioned transformation of coor dinates). For example, the vector product of two polar vectors is a pseudovector, and that of a pseudovector and a polar vector is a polar vector.

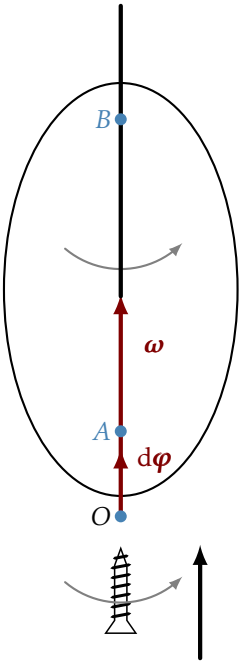


Figure 1.6: Angular velocity and right-hand rule.

v of point M is often called the *linear velocity*. It is perpendicular both to the axis of rotation (i.e. to vector $\boldsymbol{\omega}$) and to radius vector $\boldsymbol{\rho}$, from the centre O' of the circle to point M, and is equal to their vector product:

$$\mathbf{v} = [\boldsymbol{\omega}\boldsymbol{\rho}] = [\boldsymbol{\omega}\mathbf{r}] \text{ and } v = \omega\rho$$

Here $\mathbf{r} = \mathbf{OO'} + \boldsymbol{\rho}$ and is the radius vector of point M, from point O of the axis of rotation. Point O is taken as

the origin of coordinates.

1.5.5

The *period of revolution* is the time interval T required for a body, rotating uniformly at the angular velocity ω, to complete one revolution about its axis of rotation (i.e. to rotate through the angle $\varphi = 2\pi$):

$$T = \frac{2\pi}{\omega}$$

The *rotational speed* $n = 1/T = \omega/2\pi$ is the number of revolutions made by a body in unit time in uniform rotation at the angular velocity ω.

1.5.6

The rotation of a rigid body in which one of its points remains fixed is said to be *rotation about a fixed point*. This point is usually taken as the origin of the fixed reference frame. In such rotation, all the points of the body travel along the surfaces of concentric spheres whose centre is at the fixed point.

At each instant of time, this motion of the rigid body can be regarded as rotation about a certain axis that passes through the fixed point and is called the instantaneous axis of rotation. In the general case, the position of the *instantaneous axis of rotation* varies with respect to both a fixed reference frame and one rigidly fixed in the rotating body.

The velocity $\mathbf{v}$ of arbitrary point M of such a body is

$$\mathbf{v} = [\boldsymbol{\omega}\mathbf{r}] \text{ and } v = \omega\rho.$$

Here $\boldsymbol{\omega} = \mathrm{d}\boldsymbol{\varphi}/\mathrm{d}t$ is the angular velocity of the body and is directed along the instantaneous axis of rotation as is the vector $\mathrm{d}\boldsymbol{\varphi}$ of an element of rotation of the body during the infinitely short time $\mathrm{d}t$, $\mathbf{r}$ is the radius vector from fixed point O, about which the body rotates, to point M, and ρ is the distance from point M to the instantaneous axis of rotation. Such a body can have three independent motions: it can rotate about each of three mutually perpendicular axes passing through fixed point O. It consequently has three degrees of freedom (1.2.7).

1.5.7

The *angular acceleration* vector $\boldsymbol{\epsilon}$ is introduced to indicate the rate of change of the angular velocity vector with time in nonuniform rotation of a body about a fixed axis or its rotation about a fixed point. Vector $\boldsymbol{\epsilon}$ is equal to the first derivative of the angular velocity $\boldsymbol{\omega}$ of the body with respect to the time t:

$$\boldsymbol{\epsilon} = \frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}t}$$

If the body rotates about a fixed axis, vector $\boldsymbol{\epsilon}$ is directed along this axis. It coincides in direction with $\boldsymbol{\omega}$ in accelerated rotation ($\mathrm{d}\omega/\mathrm{d}t > 0$) and is opposite to $\boldsymbol{\omega}$ in direction in decelerated rotation ($\mathrm{d}\omega/\mathrm{d}t < 0$). The projection of the angular acceleration on the fixed axis OZ of rotation is equal to

$$\epsilon_z = \frac{\mathrm{d}\omega_z}{\mathrm{d}t}$$

where ω_z is the projection of vector $\boldsymbol{\omega}$ on the same axis.

1.5.8

As distinct from the angular acceleration, the acceleration a of arbitrary point M of a rigid body rotating about fixed point O, or about a fixed axis passing through point O, is often called the *linear acceleration*. It equals

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}[\boldsymbol{\omega}\mathbf{r}] = \mathbf{a}_r + \mathbf{a}_{ax}$$

where $\mathbf{a}_r = [\boldsymbol{\epsilon}\mathbf{r}]$ is the *rotary acceleration* of the point, and $\mathbf{a}_{ax} = [\boldsymbol{\omega}[\boldsymbol{\omega}\mathbf{r}]]$ is its *axipetal acceleration*, which is directed toward the instantaneous axis of rotation. If the

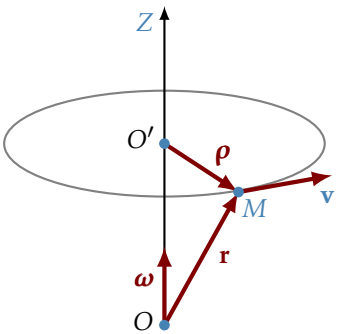


Figure 1.7: Relation between linear and rotational acceleration.

body rotates about the fixed axis OZ (Figure 1.7), the rotary acceleration of point M coincides with its tangential acceleration $\mathbf{a}_\tau$ (1.4.5), and its axipetal acceleration coincides with its normal acceleration (1.4.6):

$$\mathbf{a}_\tau = [\boldsymbol{\epsilon}\mathbf{r}] = [\boldsymbol{\epsilon}\boldsymbol{\rho}] \text{ and } \mathbf{a}_n = -\boldsymbol{\omega}^2\boldsymbol{\rho}$$

1.5.9

Any complex motion of a rigid body can be broken down into two simple motions: translational motion at the velocity $\mathbf{v}_A$ of a certain arbitrarily chosen point A of the body and rotation about an instantaneous axis passing through this point. The angular velocity $\boldsymbol{\omega}$ of rotation is independent of the choice of point A. The velocity of arbitrary point M of the body is

$$\mathbf{v} = \mathbf{v}_A + [\boldsymbol{\omega}(\mathbf{r} - \mathbf{r}_A)]$$

where $\mathbf{r}$ and $\mathbf{r}_A$ are the radius vectors of points M and A.

In the dynamics of a rigid body it usually proves convenient to deal with its complex motion as the combination of two simultaneous motions: translational motion at the velocity of the body's centre of mass (2.3.3) and rotation about the centre of mass.

The simplest case of complex motion of a body is *plane*, or *plane-parallel*, *motion*, in which all the points of the body travel in parallel planes. This is the motion, for instance, of a homogeneous circular cylinder rolling down an inclined plane. In plane motion the direction of the instantaneous axis of rotation of the body about point A remains constant and vectors $\boldsymbol{\omega}$ and $\mathbf{v}_A$ are perpendicular to each other.

2 Newton's Laws

2.1 Newton's First Law. Inertial Frames of Reference

2.1.1

As the *first law of dynamics* Newton took a law that had already been established by Galileo: a particle persisting its state of rest or uniform motion in a straight line until the action of other bodies compel it to change that state.

Newton's first law indicates that no external action is required to maintain a state of rest or of uniform motion in a straight line. This manifests a special dynamic property of bodies, called their *inertia*. Newton's first law is consequently called the *law of inertia*, and the motion of a body not subject to the action of other bodies is said to be *inertial motion*.

2.1.2

Mechanical motion is relative: its nature may differ for different frames of reference (1.2.1) that are moving in relation to each other. An astronaut, for instance, on board a man-made satellite orbiting the earth, may be at rest in a reference frame fixed in the satellite. At the same time, he is travelling together with the satellite with respect to the earth along an elliptic orbit, i.e. not with uniform motion and not in a straight line. It is thus obvious that Newton's first law does not hold for any frame of reference. A ball, for example, lying on the smooth floor of a stateroom in a ship travelling with uniform motion at sea in a straight line, may suddenly start moving along the floor without being subject to the action of any other bodies. This will occur when the velocity of the ship, i.e. its speed and/or direction, begins to change.

A frame of reference with respect to which a particle, not subject to external action, remains at rest or moves uniformly in a straight line is called an *inertial frame of reference*. The content of Newton's first law reduces, in essence, to two statements: in the first place, that all material bodies possess the property of inertia, and secondly, that inertial reference frames exist.

2.1.3

Any two inertial reference frames can move only with translational motion with respect to each other, and only uniformly in a straight line. It has been experimentally established that the *heliocentric frame of reference* is practically inertial. The origin of its coordinate system is located at the centre of mass (2.3.3) of the solar system (ap-

proximately at the centre of the sun) and the three axes extend in the directions of three distant stars selected, for example, so that these coordinate axes are mutually perpendicular.

A *laboratory frame of reference*, whose coordinate axes are rigidly fixed in the earth, is noninertial chiefly owing to the earth's diurnal (daily) rotation. But the earth rotates so slowly that the maximum normal acceleration (1.4.6) of points on its surface due to diurnal rotation does not exceed $0.034\,\mathrm{m/s^2}$. Therefore, in the great majority of practical problems, the laboratory reference frame can be regarded as being approximately inertial.

2.1.4

Inertial reference frames are of prime importance in all branches of physics, not only mechanics. This is due to the fact that according to Einstein's principle of relativity (5.1.2) the mathematical expression of any physical law should have the same form in ail inertial reference frames. Hence, in the following we shall make use of only inertial frames of reference, without stipulation each time. The laws of motion of a particle with respect to a noninertial reference frame is dealt with in Chapter 7.

2.2 Force

2.2.1

Force is a vector quantity which is a measure of the mechanical action exerted on the body being considered by other

bodies. Mechanical interaction can be accomplished either be tween bodies in direct contact (for instance, in friction, and in the pressure of bodies against one another), or between remote bodies. A special form of matter, connecting particles of matter into unified systems and transmitting the action of certain particles on others at finite velocities, is called a *physical field*, or simply *field*. Interaction between remote bodies is accomplished by the gravitational or electromagnetic fields they set up (for example, the attraction of the planets to the sun, and the inter action of charged bodies or current-carrying conductors). Mechanical action exerted on a given body by other bodies can be manifested in two ways. In the first place, it can lead to a change in the state of mechanical motion of the body being considered, and secondly, to its deformation. Both of these elects can serve as the basis for measuring forces. For example, the measurement of forces by means of a spring dynamometer is based on Hooke's law (**??**) for axial tension.

Using the concept of force, we usually speak in mechanics of the motion and deformation of a body by the action of the forces applied to it. Here, of course, each force always corresponds to a certain body exerting this force on the body being considered.

Force $\mathbf{F}$ is completely specified by giving its magnitude, direction in space and point of application. The straight line along which a force is exerted is called its *line of action*.

A field acting on a particle with the force $\mathbf{F}$ is said to be *steady*, or *stationary*, if it does not change in the course of time t, i.e. if force $\mathbf{F}$ is explicitly independent of time at any point in the held: $\partial\mathbf{F}/\partial t \equiv 0$. For a held

to be steady it is necessary that the bodies which set it up are at rest with respect to the reference frame used in considering the held.

2.2.2

The simultaneous action of several forces, $\mathbf{F_1}, \mathbf{F_2}, \mathbf{F_3}, \ldots \mathbf{F_n}$ (Figure 2.1 (a)) on a particle M is equivalent to the action of a single force, called the *resultant force* and equal to the vector sum of the component forces. Thus

$$\mathbf{F} = \sum_{i=1}^{n} \mathbf{F_i}$$

It is the closing line of the polygon of forces $\mathbf{F_1}, \mathbf{F_2}, \mathbf{F_3}, \ldots \mathbf{F_n}$ (Figure 2.1 (b)).

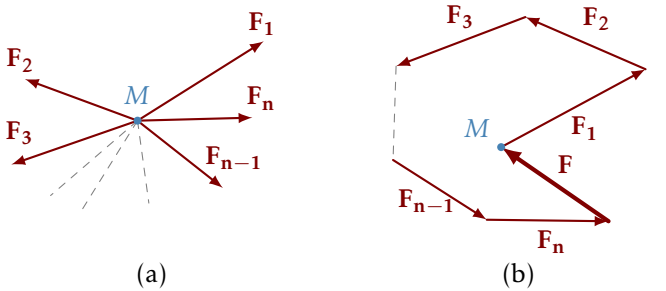


Figure 2.1: Resultant force is equal to vector sum of the component forces.

If a body is perfectly rigid, the action of a force on it does not change when the point of application is displaced along the line of action, but within the limits of

the body. In other words, forces applied to perfectly rigid bodies can be regarded as non-localized vectors.

2.2.3

A body is said to be *free* if no limitations are imposed on its position and motion in space. An airplane, for example, Hying in the air is a free body, as is a submarine travelling in the depths of the sea. Most bodies that we deal with are not free; certain limitations are imposed on their possible positions and motion. These limitations are called *constraints* in mechanics. A ball, for example, hung on a nonextensible thread, can not move away from its point of suspension by more than the length of the thread; a streetcar (tram) must travel only along its tracks. Constraints are realized due to the action on the body being considered exerted by other bodies attached to or contacting this body (for instance, the thread on the ball tied to it, the track on the streetcar, etc.).

In investigating the behaviour of constrained bodies or systems of bodies in mechanics, the *principle of releasability* is applied. It states that a restrained body (or system of bodies) can be treated as a free body (or system) if the bodies imposing the constraints are replaced by the corresponding forces. These forces are called *reaction forces* of the constraints, or *constraint reactions*. All other forces exerted on the body are called *active forces*. Thus, the motion of the ball hung by the thread can be treated as the motion of a free ball subject, in addition to all the active forces applied to it (for instance, gravity), to the reaction of the thread as well.

In contrast to the active forces, which should he given

in each specific problem, the constraint reactions are not known beforehand. They are to be determined in solving the problem. Their values should be such that under the combined effects of the active forces and constraint reactions, the motion of the "released" body completely agrees with the limitations imposed by the constraints on the constrained body being considered. There are no other distinctions between constraint reactions and active forces.

2.2.4

Bodies not included in the mechanical system being in vestigated are called *external bodies*. Forces exerted on the system by external bodies are called *external forces*. Correspondingly, forces of interaction between the parts or particles of the system being considered are called *internal forces*.

A mechanical system is said to be *closed*, or *isolated*, if it does not interact with external bodies. No external forces act on any of the bodies of a closed system.

2.3 Mass. Momentum

2.3.1

In classical (Newtonian) mechanics, the *mass of a particle* (material point) is a positive scalar quantity which is a measure of the inertia of this particle. Under the action of a force the velocity of a particle changes *gradually*, not instantaneously, i.e. it acquires an acceleration *finite* in magnitude and inversely proportional to the mass of the

particle. To compare the masses, m_1 and m_2, of two particles, it is sufficient to measure the magnitudes a_1 and a_2 of the accelerations acquired by these particles when the same force is applied to them: $m_2/m_1 = a_1/a_2$. The mass of a body is ordinarily determined by weighing by beam scales.

Classical (Newtonian) mechanics contends that:

(a) the mass of a particle is independent of its state of motion, being its unalterable characteristic;

(b) mass is an additive quantity, i.e. the mass of a system (for instance, a body) is equal to the sum of the masses of all the particles making up the system;

(c) the mass of an isolated (closed) system (2.2.4) does not change with any processes occurring within the system (*law of conservation of mass*).

These propositions of Newtonian mechanics were reconsidered and revised in relativistic mechanics (5.6.1, 5.6.2, 5.7.3 and 5.7.6).

2.3.2

The density ρ of a body at a given point M is the ratio of the mass dm of an infinitely small element of the body, which includes point M, to the quantity dV, which is the volume of this element. Thus

$$\rho = \frac{dm}{dV}$$

The size of the element being considered should be so small that any variation of density within its limits can

be neglected. On the other hand, this size should he a great many times larger than intermolecular distances.

A body is said to be *homogeneous* if the density is constant at all of its points. The mass of a homogeneous body is the product of its density by its volume: $m = \rho V$.

The mass of a nonhomogeneous body is

$$m = \int_{(V)} \rho dV$$

where ρ is a function of the coordinates, and integrating is carried out over the whole volume of the body. The average density $\langle \rho \rangle$ of a nonhomogeneous body is the ratio of its mass to its volume: $\langle \rho \rangle = m/V$.

2.3.3

The centre of inertia, or centre of mass, of a system of particles is point C, whose radius vector $\mathbf{r}_c$ equals

$$\mathbf{r}_C = \frac{1}{m} \sum_{i=1}^{n} m_i \mathbf{r}_i$$

where m_i and $\mathbf{r}_i$ are the mass and radius vector of the ith particle, n is the total number of particles in the system, and $m = \sum_{i=1}^{n} m_i$ is the mass of the whole system.

The velocity of the centre of mass is

$$\mathbf{v}_C = \frac{\mathrm{d}\mathbf{r}_C}{\mathrm{d}t} = \frac{1}{m} \sum_{i=1}^{n} m_i \mathbf{v}_i$$

2.3.4

The vector quantity $\mathbf{p}_i$, equal to the product of the mass m_i of a particle by its velocity $\mathbf{v}_i$, is called the *momentum of this particle*. The momentum of a system of particles is the vector $\mathbf{p}$, which is the geometric sum of the momenta of all the particles making up the system:

$$\mathbf{p} = \sum_{i=1}^{n} \mathbf{p}_i$$

The momentum of a system is equal to the product of the mass of the whole system by the velocity of its centre of mass: $\mathbf{p} = m\mathbf{v}_C$.

2.4 Newton's Second Law

2.4.1

The basic law of the dynamics of a particle is Newton's second law, which discusses how the mechanical motion of a particle is changed by the effect of forces applied to it. *Newton's second law* states: the rate of change in the momentum p of a particle is equal to the force $\mathbf{F}$ acting on it. Thus

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = \mathbf{F} \text{ or } \frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{v}) = \mathbf{F}$$

where m and $\mathbf{v}$ are the mass and velocity of the particle.

If several forces act simultaneously on a particle, the force $\mathbf{F}$ in Newton's second law is understood to he the geometric sum of all acting forces, both active forces and constraint reactions (2.2.3), i.e. the resultant force (2.2.2).

2.4.2

The vector quantity $\mathbf{F}\mathrm{d}t$ is called an element of *impulse of force* $\mathbf{F}$ during the infinitely short time $\mathrm{d}t$ of its action. The impulse of force $\mathbf{F}$ acting in the finite time interval from $t = t_1$ to $t = t_2$ is equal to the definite integral $\int_{t_1}^{t_2} \mathbf{F}\mathrm{d}t$, where $\mathbf{F}$, in the general case, depends upon the time t.

According to Newton's second law, the change in the momentum of a particle is equal to the impulse of the force acting on it:

$$\mathrm{d}\mathbf{p} = \mathbf{F}\mathrm{d}t = \Delta\mathbf{p} = \mathbf{p}_2 - \mathbf{p}_1 = \int_{t_1}^{t_2} \mathbf{F}\mathrm{d}t$$

where $\mathbf{p}_2 = \mathbf{p}(t_2)$ and $\mathbf{p}_1 = \mathbf{p}(t_1)$ are the values of the momentum of a particle at the end $t = t_2$ and beginning $t = t_1$ of Ihe time interval being considered.

2.4.3

Since in Newtonian mechanics the mass m of a particle is independent of its state of motion, $\mathrm{d}m/\mathrm{d}t = 0$. Hence, the mathematical expression of Newton's second law can also be presented in the form

$$\mathbf{a} = \frac{\mathbf{F}}{m}$$

where $\mathbf{a} = \mathrm{d}\mathbf{v}/\mathrm{d}t = \mathrm{d}^2\mathbf{r}/\mathrm{d}t^2$ is the acceleration of the particle and $\mathbf{r}$ is its radius vector. The corresponding statement of *Newton's second law reads*: the acceleration of a particle coincides in direction with the force acting on it and is equal to the ratio of this force to the mass of

the particle. The tangential and normal accelerations of a particle (1.4.4, 1.4.5 and 1.4.6) are determined by the corresponding components of force **F**:

$$\mathbf{a}_\tau = \frac{\mathbf{F}_\tau}{m},\ a_\tau = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{F_\tau}{m}$$

and

$$\mathbf{a}_n = \frac{\mathbf{F}_n}{m},\ a_n = \frac{v^2}{R} = \frac{F_n}{m}$$

where v is Ihe magnitude of the particle's velocity vector, and R is the radius of curvature of its path. The force F_n, imparting normal acceleration to a particle, is directed toward the centre of curvature (1.2.4) of the particle's path and is therefore called the *centripetal force*.

2.4.4

If a particle is subject to the simultaneous action of sev eral forces, $\mathbf{F_1}, \mathbf{F_2}, \mathbf{F_3}, \ldots \mathbf{F_n}$, its acceleration is

$$\mathbf{a} = \frac{1}{m}\sum_{l=1}^{n}\mathbf{F}_l = \sum_{l=1}^{n}\mathbf{a}_l$$

where $\mathbf{a}_l = \mathbf{F}_l/m$. Consequently, each of the forces acting simultaneously on a particle imparts to it the acceleration it would have if there were no other forces (*principle of superposition, or independence, of forces*).

The equation

$$m\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} = \mathbf{F} = \sum_{l=1}^{n}\mathbf{F}_l$$

is called the *differential equation of motion of a particle*.

In projections on the axes of a rectangular Cartesian coordinate system, this equation is of the form

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = F_x,\ m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = F_y,\ m\frac{\mathrm{d}^2z}{\mathrm{d}t^2} = F_z$$

where x, y and z are the coordinates of the moving particle.

2.5 Newton's Third Law. Motion of the Centre of Mass

2.5.1

The mechanical action exerted by bodies on one another is of the nature of their *interaction*. This is what *Newton's third law* deals with: two particles exert forces on each other which are equal in magnitude and opposite in direction along a line connecting these particles.

If $\mathbf{F}_{ik}$ is a force exerted on the ith particle by the kth one, and $\mathbf{F}_{ki}$ is, the force exerted on the kth particle by the ith one, then, according to Newton's third law,

$$\mathbf{F}_{ki} = -\mathbf{F}_{ik}$$

Forces $\mathbf{F}_{ik}$ and $\mathbf{F}_{ki}$ are applied to different particles and can be mutually counterbalanced only when particles i and k belong to the same perfectly rigid body.

2.5.2

Newton's third law is an essential supplement to the first and second laws. It permits us to go over from the dynamics of a separate particle to that of an arbitrary mechanical system (system of particles). It follows from Newton's third law that in any mechanical system the geometric sum of all the internal forces (2.2.4) equals zero:

$$\sum_{i=1}^{n} \sum_{k=1}^{n} \mathbf{F}_{ik} = 0$$

where n is the number of particles in the system, and $\mathbf{F}_{ii} = 0$. The vector $\mathbf{F}^{\text{ext}}$, equal to the vector sum of all the external forces (2.2.4) acting on the system, is called the *principal vector of external forces*:

$$\mathbf{F}^{\text{ext}} = \sum_{i=1}^{n} \mathbf{F}_i^{\text{ext}}$$

where $\mathbf{F}_i^{\text{ext}}$ is the resultant external force applied to the ith particle.

2.5.3

It follows from Newton's first and second laws that the first derivative with respect to the time t of the momentum $\mathbf{p}$ of a mechanical system (2.3.4) is equal to the principal vector of all external forces applied to the system. Thus

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = \mathbf{F}^{\text{ext}}$$

This equation expresses the *law of the change in the momentum of a system*.

Since $\mathbf{p} = m\mathbf{v}_c$, where m is the mass of a system and $\mathbf{v}_c$ is the velocity of its centre of inertia, or mass, the *law of motion of the centre of mass* of a mechanical system is of the form

$$\frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{v}_c) = \mathbf{F}^{\mathrm{ext}} \text{ or } m\mathbf{a}_c = \mathbf{F}^{\mathrm{ext}}$$

where $\mathbf{a}_c = \mathrm{d}\mathbf{v}_c/\mathrm{d}t$ is the acceleration of the centre of mass. Thus, the centre of mass (centre of inertia) of a mechanical system travels like a particle whose mass is equal to the mass of the whole system and which is subject to a force equal to the principal vector of the external forces applied to the system. If the system being considered is a rigid body that travels with translational motion (1.5.1), the velocities $\mathbf{v}_i$of all the points of the body and $\mathbf{v}_c$ of its centre of mass are the same and are equal to velocity $\mathbf{v}$ of the body. Correspondingly, the acceleration of the body $\mathbf{a} = \mathbf{a}_c$ and the *basic equation in the dynamics of translational motion of a rigid body* is of the form $m\mathbf{a} = \mathbf{F}^{\mathrm{ext}}$.

2.6 Motion of a Body of Variable Mass

2.6.1

In Newtonian mechanics the mass of a body can be changed only if particles of matter are detached from or added to it. An example of such a body is a rocket. In its flight the mass of the rocket is gradually reduced because the gaseous products of combustion in the rocket engine are ejected through the nozzles.

The equation of translational motion of a body of variable

mass (Meshchersky equation) is

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{F}^{\mathrm{ext}} + (\mathbf{v}_1 - \mathbf{v})\frac{\mathrm{d}m}{\mathrm{d}t}$$

where m and $\mathbf{v}$ are the mass and velocity of the body at the instant of time being considered, $\mathbf{F}^{\mathrm{ext}}$ is the principal vector of the external forces (2.5.2) acting on the body, and $\mathbf{v}_1$ is the velocity of the detracted particles *after separation* from the rocket (if $\mathrm{d}m/\mathrm{d}t < 0$) or the added particles *before being joined* (if $\mathrm{d}m/\mathrm{d}t > 0$).

2.6.2

The second member on the right-hand side of the Meshchersky equation represents the additional force acting on a body of variable mass. It is called the *reactive force*:

$$\mathbf{F}_{\mathrm{r}} = (\mathbf{v}_1 - \mathbf{v})\frac{\mathrm{d}m}{\mathrm{d}t} = \mathbf{u}\frac{\mathrm{d}m}{\mathrm{d}t}$$

where $\mathbf{u} = \mathbf{v}_1 - \mathbf{v}$ is the relative velocity of the particles being separated from or joined to the body, i.e. their velocity with respect to a reference frame in translational motion together with the body.

The reactive force characterizes the mechanical action exerted on the body by the particles separated from or joined to it (for instance, the effect of the ejected exhaust gases on the rocket).

2.6.3

The equation of motion of a rocket not subject to the action of external forces is

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{u}\frac{\mathrm{d}m}{\mathrm{d}t}$$

If the initial velocity of the rocket is zero the rocket flies in a straight line in the direction opposite to that of the relative velocity $\mathbf{u}$ of the exhaust gases ejected from the engine nozzle. Here

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = -u\frac{\mathrm{d}m}{\mathrm{d}t}$$

and at $\mathbf{u} = \text{const}$ the relation between the velocity of the rocket and its mass is expressed by *Tsiolkovsky's formula*

$$v = u\ln\frac{m_0}{m}$$

where m_0 is the initial, or launching, mass of the rocket.

2.6.4

The maximum velocity that a rocket can reach in the absence of external forces is called its *characteristic velocity*. This velocity is acquired at the moment the rocket engine stops because it has used up all the fuel and oxidizer that were on board,

$$v_{\max} = u\ln\frac{m_0}{m_0 - m_\mathrm{f}}$$

where m_f is the initial mass of the fuel and oxidizer. The effects of the earth's gravity and air resistance lead to an

appreciable reduction in the maximum velocity actually acquired by the rocket, during the operation of its engine, as compared to the characteristic velocity.

2.6.5

The *characteristic velocity of a multistage* (combination) *rocket* is

$$v_{\max} = \sum_{i=1}^{n} u_i \ln \frac{m_{0i}}{m_{0i} - m_{fi}}$$

where n is the total number of stages of the rocket, m_{fi} is the mass of the fuel and oxidizer intended for the operation of the ith stage engine, u_i is the exhaust gas velocity (nozzle velocity) of the ith stage engine, and m_{0i} is the launching mass of the multistage rocket, including all the stages from the ith to the nth. The velocity increase of a multistage rocket, compared to that of a single-stage one having the same launching mass and the same fuel and oxidizer capacity, is due to the reduction in the mass of the rocket as the first, second and subsequent stages are consecutively jettisoned when their fuel has been exhausted.

2.7 Law of Conservation of Momentum

2.7.1

The *law of conservation of momentum* is: the momentum **p** of a closed, or isolated, system does not change in the

course of time, i.e.

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} \equiv 0 \text{ and } \mathbf{p} = \text{const}$$

In contrast to Newton's laws, the law, or principle, of conservation of momentum is valid not only within the framework of classical mechanics. It belongs to the most fundamental of physical laws because it is concerned with a definite property of the symmetry of space: its homogeneity. The *homogeneity of space* is manifested in the fact that the physical properties of a closed system and the laws of its motion are independent of the choice of position of the origin of the inertial reference frame, i.e. these properties and laws are unchanged by a parallel translation of the closed system as a whole in space. According to modern concepts, fields can also have momentum, not only particles and bodies. Light, for example, exerts a pressure on the surface of a body that reflects or absorbs it because the electromagnetic field of the light wave has momentum.

2.7.2

In application to systems described by classical (Newtonian) mechanics, the principle of conservation of momentum can be regarded as a consequence of Newton's laws. For a closed mechanical system the principal vector of external forces

$\mathbf{F}^{\text{ext}} \equiv 0$ and the principle of conservation of momentum follows from (2.5.3):

$$\mathbf{p} = \sum i = 1 n m_i \mathbf{v}_i = \text{const}$$

where m_i and $\mathbf{v}_i$ are the mass and velocity of the ith particle of the system, which consists of n particles.

Consequently, there is also no change in the projections of the momentum of a closed system onto the axes of Cartesian coordinates of an inertial reference frame:

$$p_x = \sum_{i=1}^{n} m_i v_{ix} = \text{const}$$

$$p_y = \sum_{i=1}^{n} m_i v_{iy} = \text{const}$$

$$p_z = \sum_{i=1}^{n} m_i v_{iz} = \text{const}$$

The momentum of the system is $\mathbf{p} = m\mathbf{v}_c$, where m is the mass of the system and $\mathbf{v}_c$ is the velocity of its centre of mass (2.3.4). It follows, therefore, from the principle of conservation of momentum that in any processes occurring within a closed system the velocity of its centre of mass does not change: $\mathbf{v}_c = \text{const}$.

2.7.3

If a system is not closed, hut the external forces acting on it are such that their principal vector is identically zero ($\mathbf{F}^{\text{ext}} \equiv 0$), then, according to Newton's laws (2.5.3), the momentum of the system does not change with time: $\mathbf{p} = \text{const}$.

As a rule, $\mathbf{F}^{\text{ext}} \not\equiv 0$ and $\mathbf{p} \neq \text{const}$. But if the projection of the principal vector of external forces onto any fixed axis is identically zero, the projection onto the same axis of the momentum vector of the system does not change

with time. Thus, $p_x = \text{const}$ under the condition that $F_x^{\text{ext}} \equiv 0$. If, for instance, no other external forces, except gravity, are acting on a system, the horizontal component of the momentum of the system, perpendicular to the direction of the gravitational force, remains constant.

2.7.4

In some processes (e.g. collisions or in firing shells, etc.), the momenta of the parts of the system undergo great changes in relatively short time intervals. This is due to internal forces of interaction, of short duration, but of quite considerable magnitude, developed within the system. In comparison with these forces, all external forces acting continuously on the system (for instance, gravity) are small. In such processes, the action of the system of external forces can usually be ignored, i.e. it can be assumed that the momentum of the system as a whole is not changed in the process being considered.

2.8 Galilean Transformations. Mechanical Principle of Relativity

2.8.1

A *Galilean transformation* is a transformation of coordinates and time that is applied in Newtonian mechanics in going over from one inertial frame of reference $K(x, y, z, t)$ to another $K'(x, y', z', t')$ which travels with translational motion with respect to K at the constant velocity $\mathbf{V}$. The Galilean transformation is based on axioms

concerning the absolute nature of time intervals and of lengths. The first axiom contends that the passage of time (time interval between any two events) is the same in all inertial frames of reference. According to the second axiom, the dimensions of a body are independent of its velocity of motion with respect to the frame of reference. If the like axes of Cartesian coordinates of in-

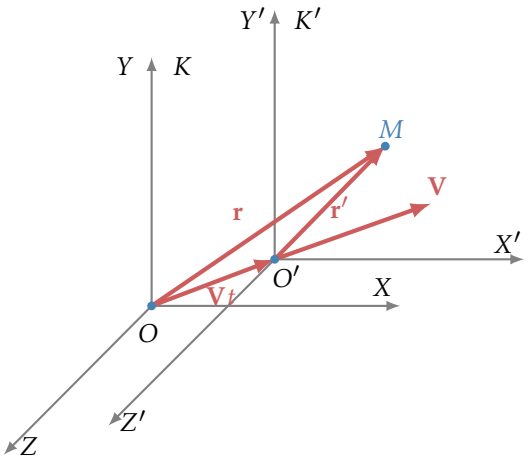


Figure 2.2: A general Galilean transformation between two frames of reference K and K' moving with a constant velocity $\mathbf{V}$.

ertial reference frames K and K' are drawn parallel to one another pairwise and if at the initial instant of time ($t = t' = 0$) the origins O and O' of coordinates coincide with each other (Figure 2.2), the Galilean transfor-

mations are of the form:

$$x' = x - V_x t,\ y' = y - V_y t,\ z' = z - V_z t \text{ and } t' = t \text{ or}$$
$$\mathbf{r}' = \mathbf{r} - \mathbf{V}t \text{ and } t' = t,$$

where x, y, z and x', y', z' are the coordinates of point M in the frames of reference K (at the instant of time t) and K' (at the instant of time $t' = t$), $\mathbf{r}$ and $\mathbf{r}'$ are the radius vectors of point M in the same frames of reference, and V_x, V_y and V_z are the projections of the velocity $\mathbf{V}$ of reference frame K' onto the coordinate axes of reference frame K.

The coordinate axes are usually arranged so that frame of reference K' travels along the positive direction of axis OX (Figure 2.3). Under this condition the Galilean transformations are of their simplest form:

$$x = x - Vt,\ y' = y,\ z' = z \text{ and } t' = t$$

2.8.2

A consequence of the Galilean transformations is the following law for transforming the velocity of arbitrary point M (Figure **??**) in going over from one inertial frame of reference K (the velocity of the point is $\mathbf{v} = \mathrm{d}\mathbf{r}/\mathrm{d}t$) to another K' (in which the velocity of the same point is $\mathbf{v}' = \mathrm{d}\mathbf{r}'/\mathrm{d}t$):

$$\mathbf{v}' = \mathbf{v} - \mathbf{V}$$

The projections of the velocity on like coordinate axes are transformed accordingly:

$$v'_{x'} = v_x - V_x,\ v'_{y'} = v_y - V_y \text{ and } v'_{z'} = v_z - V_z$$

In particular, when reference frame K' travels along the positive direction of axis OX (Figure 2.3)

$$v'_{x'} = v_x - V,\ v'_{y'} = v_y \text{ and } v'_{z'} = v_z$$

The accelerations of point M in frames of reference $K'(\mathbf{a} = \mathrm{d}\mathbf{v}/\mathrm{d}t)$ and $K'(\mathbf{a}' = \mathrm{d}\mathbf{v}'/\mathrm{d}t)$ are the same: $\mathbf{a}' = \mathbf{a}$.

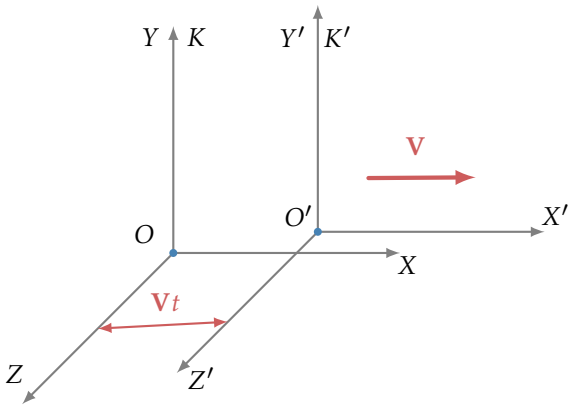


Figure 2.3: Galilean transformation between two frames of reference K and K' when K' travels along direction of axis OX.

Hence, the acceleration of a particle is independent of the choice of the inertial frame of reference or, in other words, it is invariant with respect to Galilean transformations.

2.8.3

The forces of interaction between particles depend only upon their relative position and their velocity with respect to one another. The relative position of any two particles, 2 and 1, is represented by a vector equal to the difference between the radius vectors of the two particles, i.e. to the vector $\mathbf{r}_{21} = \mathbf{r}_2 - \mathbf{r}_1$ in reference frame K and to the vector $\mathbf{r}'_{21} = \mathbf{r}'_2 - \mathbf{r}'_1$(in reference frame K'. It follows from the Galilean transformations that $\mathbf{r}_{21} = \mathbf{r}'_{21} - \mathbf{r}_{21}$. Therefore, the distance between particles 1 and 2 is the same in reference frames K and K':

$$\mathbf{r}'_{21} = \mathbf{r}_{21} = \text{ or}$$
$$(x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2$$
$$= (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$$

The velocity of particle 2 with respect to particle 1 is equal to the difference of the velocities of these two particles: $\mathbf{v}_2 - \mathbf{v}_1$(in reference frame K) and $\mathbf{v}'_2 - \mathbf{v}'_1$ (in reference frame K'). It follows from the Galilean transformations that $\mathbf{v}'_2 - \mathbf{v}'_1 = \mathbf{v}_2 - \mathbf{v}_1$.

Thus, the relative position and the velocity of relative motion of any two particles are independent of the choice of inertial reference frame; they are invariant under a Galilean transformation. Accordingly, the forces acting on a particle are also invariant under a Galilean transformation: $\mathbf{F}' = \mathbf{F}$.

2.8.4

Equations expressing Newton's laws (2.4.3) and (2.5.1) are invariant with respect to Galilean transformations,

i.e. their form is unchanged in transforming the coordinates and time from one inertial frame of reference (K) to another (K'):

$$m\mathbf{a} = \mathbf{F} \text{ and } \mathbf{F}_{ki} = -\mathbf{F}_{ik}$$

(in reference frame K)

$$m'\mathbf{a}' = \mathbf{F}' \text{ and } \mathbf{F}'_{ki} = -\mathbf{F}'_{ik}$$

(in reference frame K')

where $m' = m$ is the mass of the particle being considered and is the same in all frames of reference.

Thus, the *mechanical principle of relativity (Galileo's principle of relativity)* is valid in classical mechanics. It contends that the laws of mechanics are the same in all inertial frames of reference. This implies that under the same conditions all mechanical processes proceed in the same way in different frames of reference. Consequently, it is impossible to establish, by means of any mechanical experiments conducted within a closed system of bodies, whether the system is at rest or is travelling at uniform velocity in a straight line (with respect to any inertial frame of reference whatsoever).

The mechanical principle of relativity points to the fact that all inertial frames of reference are equally valid in mechanics. It is impossible to single out any special, or "main", inertial frame of reference with respect to which the motion of bodies could be regarded as "absolute motion".

2.8.5

The principle of relativity was generalized to include all physical phenomena by Albert Einstein in his special theory of relativity (5.1.2). It was found that coordinates and time in various inertial frames of reference are related by the Lorentz transformations (5.3.2) rather than the Galilean transformations. But at low velocities of relative motion of the frames of reference (in comparison to the velocity of light in free space), the Lorentz transformations become the Galilean transformations.

3 Work And Mechanical Energy

3.1 Energy, Work and Power

3.1.1

Energy is a scalar physical quantity, which is a common measure of the various forms of motion of matter dealt with in physics. The energy of a system quantitatively characterizes this system with respect to the conversions of motion that are possible within it. These conversions take place due to the interaction between parts of the system, both with one another and with external bodies (the environment). The concept of various forms of energy has been introduced in physics to analyze the qualitatively different forms of motion and the interactions corresponding to them. These include: mechanical (3.4.1), internal (9.1.2), electromagnetic (**??**), nuclear (**??**) and various other forms of energy.

3.1.2

The mechanical motion of a body is changed by forces exerted on the body by other bodies. Used in mechanics for a quantitative description of such a process of energy exchange between interacting bodies is the concept of the work done by the forces applied to the body being considered. An *element of work* done by force $\mathbf{F}$ in the infinitely small displacement $d\mathbf{r}$ is the scalar quantity

$$\delta W = \mathbf{F}\,\mathrm{d}\mathbf{r} = \mathbf{F}\mathbf{v}\,\mathrm{d}t$$

where $\mathbf{r}$ and $\mathbf{v} = \mathrm{d}\mathbf{r}/\mathrm{d}t$ are the radius vector and velocity of the point of the body at which the force is applied, and $\mathrm{d}t$ is the infinitely short time interval during which force $\mathbf{F}$ performs the work δW (the meaning of the notation δW is given in (3.1.8)). In rectangular Cartesian coordinates

$$\begin{aligned}\delta W &= F_x\mathrm{d}x + F_y\mathrm{d}y + F_z\mathrm{d}z \\ &= (F_x v_x + F_y v_y + F_z v_z)\mathrm{d}t\end{aligned}$$

where x, y and z are the coordinates of the point of application of the force, and F_x, F_y, F_z and v_x, v_y, v_z are the projections of vectors $\mathbf{F}$ and $\mathbf{v}$ on the coordinate axes.

3.1.3

The expression for an element of work can also be presented in the form

$$\delta W = F\,\mathrm{d}s\cos\alpha = F_\tau \mathrm{d}s$$

where $\mathrm{d}s = |\mathrm{d}\mathbf{r}|$ is an element of the length of the path of the point of force application during the infinitely short

time interval $\mathrm{d}t$, α is the angle between vectors $\mathbf{F}$ and $\mathrm{d}\mathbf{r}$, and $F_\tau = F\cos\alpha$ is the projection of the force in the direction of the displacement $\mathrm{d}\mathbf{r}$. A force normal to the path of its point of application performs no work.

Force $\mathbf{F}$ is called the *motive force* if $F_\tau > 0$, so that $\delta W > 0$. If $F_\tau < 0$ (so that $\delta W < 0$), force $\mathbf{F}$ is called the *retarding force* (*force of resistance*).

3.1.4

If a mechanical system is subject to the simultaneous action of forces $\mathbf{F}_1, \mathbf{F}_2 \ldots \mathbf{F}_n$, the work δW done by them during the infinitely short time $\mathrm{d}t$ is equal to the algebraic sum of the work done during the same time $\mathrm{d}t$ by each of the forces separately. Thus

$$\delta W = \sum_{i=1}^{n} \delta W_i = \sum_{i=1}^{n} \mathbf{F}_i \mathrm{d}\mathbf{r}_i = \sum_{i=1}^{n} \mathbf{F}_i \mathbf{v}_i \mathrm{d}t$$

where $\mathbf{r}_i$ and $\mathbf{v}_i$ are the radius vector and velocity of the point of application of force $\mathbf{F}_i$.

For example, for a particle $\mathbf{r}_i = \mathbf{r}$ is the radius vector of the particle and $\mathbf{v}_i = \mathbf{v}$ is its velocity. Accordingly, $\delta W = \mathbf{F}\mathrm{d}\mathbf{r} = \mathbf{F}\mathbf{v}\mathrm{d}t$, where $\mathbf{F} = \sum_{i=1}^{n} \mathbf{F}_i$ is the resultant force (2.2.2). It follows from Newton's second law (2.4.1) that for a particle

$$\delta W = \mathbf{v}\mathrm{d}\mathbf{p},$$

where $\mathbf{p} = m\mathbf{v}$ is the momentum of the particle, and m is its mass.

In the translational motion of a perfectly rigid body $\mathrm{d}\mathbf{r}_i == \mathrm{d}\mathbf{r}_C$ and $\mathbf{v}_i = \mathbf{v}_C$, where $\mathbf{r}_C$ and $\mathbf{v}_C$ are the radius

vector and velocity of the centre of mass (2.3.3) of the body.

The work done by internal forces in any motion of a perfectly rigid body equals zero. Hence, in the translational motion of such a body $\delta W = \mathbf{F}^{\text{ext}} d\mathbf{r}_C = \mathbf{F}^{\text{ext}} \mathbf{v}_C dt$, where $\mathbf{F}^{\text{ext}}$ is the principal vector of external forces (2.5.2). It follows from the law of motion of the centre of mass (2.5.3) that

$$\delta W = \mathbf{v}_C d\mathbf{p},$$

where $\mathbf{p} = m\mathbf{v}_C$ is the momentum of a rigid body of mass m travelling at the velocity $\mathbf{v} = \mathbf{v}_C$.

3.1.5

The work W done by force $\mathbf{F}$ along a finite portion of the path L of its point of application is equal to the algebraic sum of the work done over all the infinitely small portions of this path, i.e. it is expressed by the line integral

$$W = \int_{(L)} \mathbf{F} d\mathbf{r} = \int_0^s F_\tau ds$$

where s is the length of arc measured off along the path from the beginning of the portion of the path being considered, and F_τ the projection of the force onto the direction of displacement $d\mathbf{r}$ of the point of force application.

To calculate this integral it is necessary to know the dependence of F_τ on s along the given path L. If this dependence is represented graphically (Figure 3.2), the required work W is proportional to the hatched area in Figure 3.2.

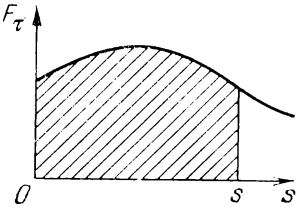


Figure 3.1: Graphical representation of the resultant work.

3.1.6

Potential-field (conservative) forces are ones whose work depends only on the initial and final positions of their points of application; it depends neither on the shape of the path of these points nor on their laws of motion along this path.

For example, the forces of interaction between the parts (i.e. particles) of a system are said to be potential (conservative) forces if they depend only upon the configuration of the system, i.e. on the relative position of all its particles. The work done by these forces in moving the system from one arbitrary position to another is independent of the kind of displacement, but is completely determined by the initial and final configurations of the system. Examples are the forces of electrostatic and gravitational interaction.

A *steady (stationary)* field (2.2.1) is said to be *potential*, or conservative, if force F, which it exerts on a particle located in the field, is a potential, or conservative force. This implies that force F depends only upon the position of the particle in the field, and that the amounts of work

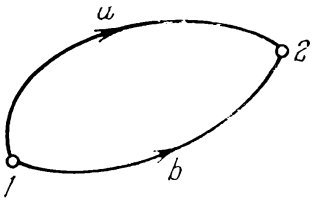


Figure 3.2: Graphical representation of the resultant work.

done by force F in displacing the particle from one arbitrary position (1) to another (2) (Figure **??**) along any two paths, for instance, $1a2$ (work W_{1a2}) and $1b2$ (work W_{1b2}), are the same:

$$W_{1a2} = W_{1b2} = \int_1^2 \mathbf{F} \frac{\mathrm{d}}{\mathrm{d}\mathbf{r}}$$

Accordingly, the work done by a conservative force in displacing its point of application along any closed path L (for example, $1a2b1$) is equal to zero:

$$\oint_{(L)} \mathbf{F}\,\mathrm{d}\mathbf{r} \equiv 0$$

In the general case, the external bodies that set up the field being considered can be in motion with respect to an inertial frame of reference, so that their field is not stationary, i.e. force $\mathbf{F}$ depends explicitly on time:

$$\frac{\partial \mathbf{F}}{\partial t} \not\equiv 0$$

A *nonstationary field* is a *conservative* one if the work done by force $\mathbf{F}$ upon instantaneous displacement of its point

of application along any closed path L equals zero:

$$\oint_{(L)} \mathbf{F}\,d\mathbf{r} \equiv 0$$

Here $\mathbf{F}$ depends, not only on the coordinates of the point of application, but on the time as well. To calculate this integral, how ever, time is assumed to be a fixed parameter.

3.1.7

Among the nonconservative forces are dissipative and gyroscopic forces. *Dissipative forces* are ones whose total work done in displacing a closed system in any way is always negative. Such, for instance, are the forces of sliding friction and the forces retarding the motion of bodies in liquids and gases. In contrast to conservative forces, dissipative forces depend, not only on the relative positions of the interacting bodies, but on their relative velocities as well.

Gyroscopic forces are ones that depend upon the velocity of the particle on which they act. They are directed perpendicular to this velocity. An example is the Lorentz force (**??**) exerted by a magnetic field on a charged particle travelling in this field. The work done by gyroscopic forces is always equal to zero, regardless of how the particle is displaced.

A mechanical system (system of particles) is said to be *conservative* if all the internal forces in it are conservative ones and all the external forces are both conservative and stationary. Systems that do not satisfy these conditions are said to be *non-conservative*.

3.1.8

An element of work done by force **F**, exerted on a particle by a stationary conservative field, can be represented in the form of an exact differential of a scalar function $\Phi(x, y, z)$ of the coordinates. This is called the force function of the field:

$$\mathbf{F}\,\mathrm{d}r = \mathrm{d}\Phi \quad \text{or}$$

$$F_x\,\mathrm{d}x + F_y\,\mathrm{d}y + F_z\,\mathrm{d}z = \frac{\partial\Phi}{\partial x}\,\mathrm{d}x + \frac{\partial\Phi}{\partial y}\,\mathrm{d}y + \frac{\partial\Phi}{\partial z}\,\mathrm{d}z$$

Hence

$$F_x = \frac{\partial\Phi}{\partial x}, \quad F_x = \frac{\partial\Phi}{\partial x}, \quad F_x = \frac{\partial\Phi}{\partial x} \quad \text{and}$$

$$\mathbf{F} = \nabla\Phi$$

These last relations are also valid for a non-stationary conservative field, whose force function depends, not only on the coordinates, but on time as well: $\Phi = \Phi(x, y, z, t)$. But here

$$\mathbf{F}\,\mathrm{d}\mathbf{r} = \mathrm{d}\Phi - \frac{\partial\Phi}{\partial t}\,\mathrm{d}t$$

An element of work done by a nonconservative force cannot be represented in the form of an exact differential of any function whatsoever of the coordinates. This is precisely why an element of work done by an arbitrary force is denoted by δW.

3.1.9

The concept of power is used in mechanics to characterize the amount of work done in unit time. *Power (instantaneous power)* is the scalar physical quantity P. It

is equal to the ratio of the element of work δW to the infinitely short time interval $\mathrm{d}t$ during which it is performed:

$$P = \frac{\delta W}{\mathrm{d}t}$$

If **F** is the force doing the work δW, the power is the scalar product of force **F** by the velocity v of its point of application:

$$P = \mathbf{F}\mathbf{v} = F_\tau v$$

In the general case, power may vary with time.

The *average power* in the time interval from t to $t + \Delta t$ is the physical quantity $\langle P \rangle$, which is equal to the ratio of the work W done during this time interval to the length of this interval Δt:

$$\langle P \rangle = \frac{W}{\Delta t}$$

3.2 Kinetic Energy

3.2.1

The *kinetic energy* of a body is the energy of its mechanical motion. A change in the kinetic energy E_k of a particle due to the action of force F is equal to the work done by this force:

$$\mathrm{d}E_k = \delta W = \mathbf{v}\,\mathrm{d}\mathbf{p}$$

where $\mathbf{p} = m\mathbf{v}$ is the momentum of the particle, and m and **v** are its mass and velocity. In Newtonian mechanics $m = \text{const}$ and the expression for the kinetic energy of a particle is of the form

$$E_k = \frac{m\mathbf{v}^2}{2} = \frac{mv^2}{2}$$

Kinetic energy in relativistic mechanics is discussed in (5.7.1).

3.2.2

The kinetic energy of a mechanical system is equal to the sum of the kinetic energies of all the parts of the system. For a system consisting of n particles, for instance,

$$E_k = \sum_{i=1}^{n} \frac{m_i \mathbf{v}_i^2}{2} = \sum_{i=1}^{n} \frac{m_i v_i^2}{2}$$

where m_i and $\mathbf{v}_i$ are the mass and velocity of the i-th particle in the system.

The kinetic energy of a body is

$$E_k = \frac{1}{2} \int_{(V)} \rho \mathbf{v}^2 \mathrm{d}V = \frac{1}{2} \int_{(V)} \rho v^2 \mathrm{d}V$$

where $\mathbf{v}$ is the velocity of points of the infinitely small element of volume $\mathrm{d}V$ of the body, whose density is ρ and mass is $\mathrm{d}m = \rho \mathrm{d}V$. Integrating is carried out throughout the whole volume V of the body. If a perfectly rigid body of mass m travels with translational motion at the velocity $\mathbf{v}$, its kinetic energy is $E_k = mv^2/2$. The kinetic energy of rotating bodies is discussed in (4.3.3) and (4.3.5).

3.2.3

The change in the kinetic energy of a mechanical system is equal to the algebraic sum of the amounts of work done by all the external and internal forces acting on the system (2.2.4). Thus

$$\mathrm{d}E_k = \delta W^{\mathrm{ext}} + \delta W^{\mathrm{int}}$$

For example, for a system consisting of n particles

$$\mathrm{d}E_k = \sum_{i=1}^{n} \mathbf{F}_i^{\mathrm{ext}} \mathrm{d}\mathbf{r}_i = \sum_{i=1}^{n} \sum_{k=1}^{n} \mathbf{F}_{ik} \mathrm{d}\mathbf{r}_i$$

where $\mathbf{r}_i$ is the radius vector of the i-th particle, $\mathbf{F}_i^{\mathrm{ext}}$ is the resultant of the external forces acting on this particle, and $\mathbf{F}_{ii} = 0$.

If the system is not deformed, then the work done by the internal forces

$$\delta W^{\mathrm{int}} = 0, \text{ and } \mathrm{d}E_k = \delta W^{\mathrm{ext}}$$

For example, the change in the kinetic energy of a perfectly rigid body in translational motion is

$$\mathrm{d}E_k = \mathbf{F}^{\mathrm{ext}} \mathrm{d}\mathbf{r}$$

where $\mathbf{F}^{\mathrm{ext}}$ is the principal vector of external forces (2.5.2), and $\mathrm{d}\mathbf{r}$ is the vector of an element of displacement of the body.

3.2.4

The kinetic energy of a mechanical system depends upon the choice of the frame of reference. If in the inertial reference frame K the kinetic energy of the system equals E_k and in the reference frame K', travelling with translational motion at the velocity $\mathbf{V}$ with respect to K, it is equal to E_k', then

$$E_k = E_k' + \frac{mV^2}{2} + \mathbf{p}'\mathbf{V}$$

where m is the mass of the system, $\mathbf{p}' = m\mathbf{v}'c$ is the momentum of the system in its motion with respect to reference frame K', and $\mathbf{v}'_C$ is the velocity of the system's centre of mass with respect to K'. This equation is valid both for the case when $\mathbf{V} = \text{const}$, i.e. when K' is an inertial frame of reference, as well as for $\mathrm{d}\mathbf{V}/\mathrm{d}t \neq 0$.

In particular, if the frame of reference K' travels with translational motion with respect to reference frame K at the velocity $\mathbf{v}_C$ of the centre of mass of the system, i.e. $\mathbf{V} = \mathbf{v}_C$, then $\mathbf{v}'_C = 0$ and

$$E_k = \frac{mv_C^2}{2} + E'_k$$

This equation is an expression of *Koenig's theorem*: the kinetic energy of a mechanical system is equal to the sum of the kinetic energy that a particle would have if its mass was that of the whole system and if it travelled at the velocity of the system's centre of mass, plus the kinetic energy of the same system in its motion with respect to a translational reference frame (i.e. one travelling with translational motion) having its origin at the centre of mass.

It follows from Koenig's theorem that the kinetic energy of a perfectly rigid body is equal to the kinetic energy of translation al motion of this body at the velocity of its centre of mass plus the kinetic energy of rotation of the body about its centre of mass.

3.3 Potential Energy

3.3.1

Potential energy is the part of the energy of a mechanical system that depends only on its configuration, i.e. on the relative arrangement of all the particles of the system and their positions in the external potential (conservative) field (3.1.6). The decrease in the potential energy when the system is moved from arbitrary position **1** to another arbitrary position **2** is measured by the work W_{12} done in this displacement by the conservative forces (internal and external) acting on the system. Thus

$$E_p(1) - E_p(2) = W_{12}$$

where $E_p(1)$ and $E_p(2)$ are the values of the potential energy of the system in its initial and final positions. Accordingly, the work done by conservative forces upon an infinitely small change in the configuration of the system is $\delta W = -\mathrm{d}E_p$.

Note. It is assumed here that the external conservative forces are *stationary (steady)*, i.e. they can change with time only as the consequence of a change of the position of the system being considered with respect to the frame of reference. Otherwise

$$\mathrm{d}E_p = -\delta W + \frac{\partial E_p}{\partial t}\mathrm{d}t$$

In the simplest case, when the system is a particle located in a conservative field, the relation between force **F**, acting on the particle, and the potential energy E_p of this particle in the held is of the form

$$F_x = -\frac{\partial E_p}{\partial x},\ F_y = -\frac{\partial E_p}{\partial y},\ F_z = -\frac{\partial E_p}{\partial z}$$

and $F = -\operatorname{grad} E_p$.

The potential energy E_p of a particle is related to the force function (3.1.8) of the corresponding conservative field by the equation

$$dE_p = -d\Phi \text{ or } E_p(x, y, z, t) = -\Phi(x, y, z, t) + C$$

where C is the integration constant.

3.3.2

The equations in (3.3.1) enable the dependence of the potential energy of a system on its configuration to be determined only to an arbitrary constant term that has no effect on the *change* in energy. To obtain a unique dependence of the potential energy of the system on its configuration, a so-called *zero configuration* is chosen, in which the potential energy of the system is conditionally assumed to equal zero. Thus, the potential energy of a system in an arbitrary state is equal to the work done by all the conservative forces acting on the system in transforming it from the state being considered to the state corresponding to the zero configuration.

3.3.3

Example 1. *Potential energy of a particle in a uniform (homogeneous) force field.* Assume that force $\mathbf{F}$, exerted on the particle by the field, is directed along axis OZ, i.e. $\mathbf{F} = F_z\mathbf{k}$, where $\mathbf{k}$ is the unit vector of axis OZ, and the projection F_z of force $\mathbf{F}$ onto axis OZ is independent of the coordinates of the particle. Then

$$dE_p = -\mathbf{F}d\mathbf{r} = -F_z dz \text{ and } E_p(z) = -F_z + E_p(0)$$

where $E_p(0)$ is the value of the potential energy of the particle at the level $z = 0$.

In particular, the potential energy of a particle of mass m located in a uniform gravity field at the earth's surface (axis OZ is directed vertically upward, $F_z = -mg$, where g is the free fall acceleration) is equal to

$$E_p(z) = mgz + E_p(0).$$

3.3.4

Example 2. *Potential energy of a particle in a central force field.* In a conservative field of central forces the particle is subject to the action of forces $\mathbf{F}$ which are directed everywhere along straight lines passing through a fixed point, called the *centre of force*, and depend only on the distance r to this point:

$$\mathbf{F} = F_r(r)\frac{\mathbf{r}}{r}$$

where $\mathbf{r}$ is the radius vector from the centre of force to the point being considered in the field, and $F_r(r)$ is the projection of force $\mathbf{F}$ onto the direction of vector $\mathbf{r}$; it depends only on the distance r. If the particle is attracted to the centre of force, $F_r(r) = -|F| < 0$, but if it is repelled from the centre of force, then $F_r(r) = |F| > 0$. An element of work done by force $\mathbf{F}$ is $\delta W = \mathbf{F}\mathrm{d}\mathbf{r} = F_r(r)\mathrm{d}r$. The potential energy of the particle is

$$E_p(r) = \int_r^\infty F_r(r)\mathrm{d}r + E_p(\infty)$$

Usually the energy of a particle, located at an infinite distance from the centre of force, is taken as the refer-

ence point for potential energy, i.e. it is assumed that $E_p(\infty) = 0$. Thus

$$E_p(r) = \int_r^{\infty} F_r(r)\mathrm{d}r$$

Examples of a central force field, in which the force is inversely proportional to the square of the distance to the centre of force $(F_r(r) \propto r^{-2})$, are the gravitational fields of a particle or a homogeneous sphere, the electrostatic field of a point charge, as well as that of a hollow sphere and a solid sphere uniformly charged along the surface and throughout the volume, respectively.

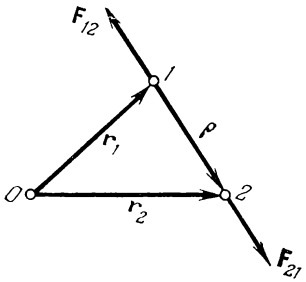


Figure 3.3: Potential energy of a system of two particles with central forces.

3.3.5

Example 3. *Potential energy of a system of two particles between which central forces* act, i.e. forces depending upon the distance between the particles and directed along the

straight line joining the particles. Shown in Figure 3.3 are mutually repulsive forces $\mathbf{F}_{12}$ and $\mathbf{F}_{21} = -\mathbf{F}_{12}$:

$$\mathbf{F}_{21} = F_\rho(\rho)\frac{\boldsymbol{\rho}}{\rho}$$

where $\boldsymbol{\rho} = \mathbf{r}_2 - \mathbf{r}_1$ is the radius vector from particle 1 to particle 2, and $F_\rho(\rho)$ is the projection of force $\mathbf{F}_{21}$ on the direction of vector $\boldsymbol{\rho}$ and depends only on the distance ρ between the particles. An infinitely small change in the potential energy of the system

$$\mathrm{d}E_p = -(\mathbf{F}_{12}\mathrm{d}\mathbf{r}_1 + \mathbf{F}_{21}\mathrm{d}\mathbf{r}_2) = -\mathbf{F}_{21}\mathrm{d}\boldsymbol{\rho} = -F_\rho(\rho)\mathrm{d}\rho$$

If we assume that $E_p \to 0$ as $\rho \to \infty$, then

$$E_p(\rho) = \int_\rho^\infty F_\rho(\rho)\mathrm{d}\rho$$

This energy is often called the *mutual potential energy of two particles.*

3.3.6

Example 4. *Potential energy of an elastic body (for instance, a spring) in axial tension or compression.* When an elastic body is deformed, conservative internal forces (elastic forces) are developed in the body that oppose deformation. According to Hooke's law, the force $\mathbf{F}_{\text{el}}$ exerted by the deformed body A (Figure 3.4) on body B, causing the deformation, is proportional to the amount of deformation (stretch or compression):

$$\mathbf{F}_{\text{el}} = -kx\mathbf{i}.$$

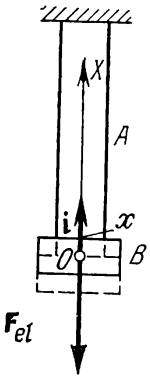


Figure 3.4: Potential energy of an elastic body in axial tension or compression.

Here x_i is the displacement vector of body B that characterizes the deformation of body A (in the undeformed state $x = 0$; in compression $x > 0$ and intension $x < 0$),and $k > 0$ is a factor characterizing the elastic properties of body A.

The potential energy of an elastically deformed body (in the absence of deformation, i.e. at $x = 0$, this energy is assumed equal to zero)

$$E_p = \frac{kx^2}{2}.$$

3.4 Law of Conservation of Mechanical Energy

3.4.1

The *mechanical energy*, or *total mechanical energy*, is the energy of mechanical motion and interaction. The mechanical energy E of a system of particles is equal to the sum of their kinetic energy E_k and the potential energy E_p of interaction of these particles with one another and with external bodies:

$$E = E_k + E_p$$

An elementary increment of the mechanical energy of a system during the infinitely short time $\mathrm{d}t$ is equal to

$$\mathrm{d}E = \delta W_{\mathrm{nonc}} + \frac{\partial E_p}{\partial t}\mathrm{d}t$$

where δW_{nonc} is the algebraic sum of the elements of work done in the time interval $\mathrm{d}t$ by all the internal and external *nonconservative* forces. The term $(\partial E_p/\partial t)\mathrm{d}t$ represents the change during the time interval $\mathrm{d}t$ in the potential energy of the system and, consequently, its total mechanical energy due to nonstationary external conservative forces (3.3.1).

3.4.2

If the system is *conservative* (3.1.7), $\delta W_{\mathrm{nonc}} \equiv 0$ and $\partial E_p/\partial t = 0$, because, in this case, there are no nonconservative forces and the external conservative forces are stationary. Consequently, the mechanical energy of such a system $E = \mathrm{const}$, i.e. the following law, called the *law of conservation of mechanical energy*, is valid: in the motion of a conservative system, its mechanical energy remains unchanged.

In particular, this law holds for closed conservative systems: the mechanical energy of a closed system does not change with time if all the internal forces acting in this system are conservative ones.

The law of conservation of mechanical energy is associated with the *uniformity of time*. This property of time is manifested in the fact that the laws of motion of a closed system (or a system located in a stationary external field) are independent of the choice of the zero time reference. For example, in the free fall of a body in a stationary conservative gravity field at the earth's surface, the velocity of the body and the length of path travelled depend only on the length of time in free fall and initial velocity of the body, but not on the specific instant of time the body began to fall.

3.4.3

The mechanical energy of a closed nonconservative system is changed owing to the work done by all the nonconservative internal forces:

$$\mathrm{d}E = \delta W_{\mathrm{nonc}}$$

Gyroscopic forces (3.1.7) do no work and make no contribution to δW_{nonc}, i.e. the existence of such forces in a system does not lead to changes in its mechanical energy.

The action of dissipative forces (3.1.7), for instance, the force of friction, leads to a gradual reduction in the mechanical energy of a closed system. This process is called the *dissipation of energy*. Consequently, a system whose mechanical energy continually decreases with time

is called a *dissipative system*. In the dissipation of energy, the mechanical energy of the system is converted into other forms of energy (for example, into the energy of chaotic motion of molecules). The conversion of mechanical energy is accomplished in full agreement with a universal law of nature: the law of conservation of energy (9.2.7).

According to this law, energy can be converted from one form to another and redistributed within the system, but its total amount in a closed system must remain constant. It follows from the law of conservation and conversion of energy that the changes in energy of an open system, occurring in interaction of the system with the external medium (external bodies and fields), must be numerically equal and opposite in sign to the changes in the energy of the external medium. In other words, the change in the energy of a system upon its interaction with the external medium should be equal to the energy the system receives or yields up in the process being considered.

3.4.4

Forces of resistance and friction act in all real mechanical systems. Hence, these systems are nonconservative. In certain cases, however, they can be assumed approximately conservative, and the law of conservation of mechanical energy can be applied to them. Such an approach is feasible if, in the process being considered, the work W_{nonc} done by all the nonconservative forces acting on the system is negligibly small in comparison to the mechanical energy E of the system, i.e. $|W_{\text{nonc}}/E| \ll 1$, so

that $|\Delta E/E| \ll 1$, where $\Delta E = W_{\text{nonc}}$ is the change in the mechanical energy of the system.

3.4.5

The state of *mechanical equilibrium of a system* is one that can be disturbed only as a result of external force action applied to the system. In this equilibrium state all the particles of the system are at rest so that the kinetic energy of the system equals zero. The state of mechanical equilibrium is said to be *stable* if a small external action applied to the system leads to only a small change in its state. In this case, forces are developed in the system that tend to restore the equilibrium state. A state of mechanical equilibrium is said to be *unstable* if it can be disturbed by even an infinitely small external action on the system, which does not return again to the equilibrium state. In this case, forces are developed in the system that tend to further disturb the equilibrium state.

The law of conservation of mechanical energy enables the conditions of equilibrium of conservative systems to be formulated:

in states of stable equilibrium the potential energy curve of the system has minima, whereas in states of unstable equilibrium, it has maxima.

3.4.6

On the basis of the law of conservation of mechanical energy, the region of possible configurations of a conservative system (3.3.1) can be found. The kinetic energy of the system $E_k \gg 0$. Hence, at a given value E of the

mechanical energy of the system, the latter can only be in states that conform to the condition: $E_p \ll E$. Figure 3.5 illustrates the simplest case in which a particle has one-dimensional motion along axis OX in an external stationary conservative field. The potential energy of the particle is a function of only a single coordinate x, i.e. $E_p = E_p(x)$. The curve of this relation, shown in Figure 3.5, is called a *potential energy curve*.

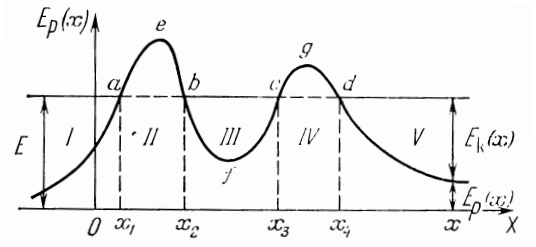


Figure 3.5: Potential energy curve for the motion of a particle in an external stationary conservative field.

For a fixed value of the mechanical energy E of the particle, as shown in Figure 3.5, the particle can remain, in its travel, in one of the following three regions:

$$x \leqslant x_1 \text{ (region I)},$$
$$x_2 \leqslant x \leqslant x_3 \text{ (region III), and}$$
$$x \geqslant x_4 \text{ (region V)}$$

These regions are separated from one another by the regions II and IV, which are the so-called *potential barriers* *aeb* and *cgd*, within whose limits the particle cannot exist. At the boundaries of the potential barriers (at points

a, *b*, *c* and *d*), the particle reverses its direction of motion; in region I the particle can travel without limit to the left of boundary *a* of the barrier, and in region V, without limit to the right of boundary *d* of the barrier. In region III, the particle oscillates between points *b* and *c*; it is in a so-called *potential well b f c*.

3.5 Perfectly Elastic and Inelastic Collisions

3.5.1

An impact in which a considerable change in the velocities of the colliding bodies occurs in an extremely short time interval is called a *collision*. Examples of collisions are a ham mer striking a billet being forged that lies on the anvil, and a hammer striking the head of a nail being driven.

The *line of impact* is the common normal to the surfaces of the colliding bodies at their point of contact. If the centres of mass (2.3.3) of the colliding bodies lie on the line of impact at the instant of contact, the collision is said to be *central*. An example is the collision of two billiard balls. The collision, or impact, is said to be *head-on*, or *straight-line*, if the velocities of the centres of mass of the colliding bodies are parallel, before the collision, to the line of impact. Otherwise, it is called an *oblique impact*, or *glancing collision*.

3.5.2

In a collision the bodies are deformed and instantaneous, but quite substantial, forces are developed. They are called *impact forces*. For a system of colliding bodies these forces are internal ones[1], i.e. they do not change the total momentum of the system. External forces, constantly exerted on the system (for instance, the gravity forces of the bodies) are usually very small compared to the impact forces. Therefore, even though the impulses of impact forces (2.4.2) in the duration τ of the collision are commensurable with the momenta (2.3.4) of the colliding bodies, the resultant impulse of all the constantly acting external forces during the same time interval τ is small compared to the momenta of the bodies. Accordingly, the work done on the system by the external forces during the time inter val t is small compared to the mechanical energy of the system. Hence, a system of bodies can be assumed, in the collision pro cess, to be an approximately closed system (2.2.4). This means that in calculating the results of the collision, use can be made of the laws of the conservation of momentum (2.7.1), of angular momentum (4.4.1) and of energy (9.2.7). If the bodies deform in the collision like perfectly elastic ones, the impact forces are conservative and the law of conservation of mechanical energy (3.4.2) is valid in the system.

[1] It is assumed that the colliding bodies are either free (2.2.3) or the constraints imposed on them are such that no impact constraint reactions are developed.

3.5.3

The collision of two bodies is said to be *perfectly inelastic* if, after impact, they travel together as a single body. Sufficiently close to a perfectly inelastic collision are, for example, the impact of the pile-driving hammer with the end of the pile being driven, and the firing of a bullet into a wagon filled with sand, when the bullet remains embedded in the sand. Various processes occur in the colliding bodies in an inelastic collision (plastic deformation of the bodies, friction, etc.). As a result, the kinetic energy of the system is partly transformed into its internal energy (9.1.2).

If.two bodies of masses m_1 and m_2, travelling with translational motion at the velocities $\mathbf{v}_1$ and $\mathbf{v}_2$, undergo a *perfectly inelastic head-on central collision*, they continue their translational motion together at the velocity

$$\mathbf{u} = \frac{m_1\mathbf{v}_1 + m_2\mathbf{v}_2}{m_1 + m_2}$$

Note. In the case of a perfectly inelastic collision that is neither head-on nor central, the preceding equation can be used to find the velocity of the centre of the mass of the bodies joined together in the collision. As the result of such a collision, how ever, the system may begin to rotate about its centre of mass in accordance with the law of conservation of angular momentum (4.4.1).

3.5.4

The change in the kinetic energy of a system of two colliding bodies in a perfectly inelastic head-on central colli-

sion is

$$\Delta E_k = \frac{m_1 + m_2}{2}\mathbf{u}^2 - \frac{m_1}{2}\mathbf{v}_1^2 - \frac{m_2}{2}\mathbf{v}_2^2$$
$$= -\frac{m_1 m_2}{2(m_1 + m_2)}(\mathbf{v}_1^2 - \mathbf{v}_2^2) < 0$$

In particular, if the second, body is at rest before the collision (for instance, a pile being driven by a pile driver, or a forging lying on the anvil), the relative reduction in the kinetic energy of the system in a perfectly inelastic head-on central collision is

$$-\frac{\Delta E_k}{E_{k_1}} = \frac{m_2}{m_1 + m_2}$$

In engineering, a perfectly inelastic head-on central collision (impact) is made use of to change the shape of bodies (smith and die forging, rivetting, etc.) or to move a body in a medium with high resistance to motion (driving nails, piles, etc.). It proves expedient in the first case to have the ratio $-\Delta E_k/E_{k_1}$ as near as possible to unity, i.e. it is necessary that $m_2 \gg m_1$ (the mass of the forged item and the anvil should be many times greater than that of the hammer). In the second case, on the contrary, it is necessary that the loss in kinetic energy in the collision (impact) be as low as possible, i.e. that $m_1 \gg m_2$ (the mass of the hammer should be many times greater than that of the nail being driven).

3.5.5

A collision is said to be *perfectly elastic* if it does not lead to any change in the mechanical energy of the system, i.e. if the bodies are perfectly elastic.

Example 1. A perfectly elastic head-on central collision of two bodies (for instance, balls) of masses m_1 and m_2, which travel before the collision with translational motion at the velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ along axis OX, passing through their centres of mass (Figure 3.6 (a)). The velocities of the bodies after the collision, $\mathbf{u}_1$ and $\mathbf{u}_2$ (Figure 3.6 (b)), can be found from the laws of conservation of momentum and mechanical energy:

$$m_1\mathbf{u}_1 + m_2\mathbf{u}_2 = m_1\mathbf{v}_1 + m_2\mathbf{v}_2,$$
$$m_1\mathbf{u}_1^2 + m_2\mathbf{u}_2^2 = m_1\mathbf{v}_1^2 + m_2\mathbf{v}_2^2$$

Velocities $\mathbf{u}_1$ and $\mathbf{u}_2$ are directed along axis OX and their

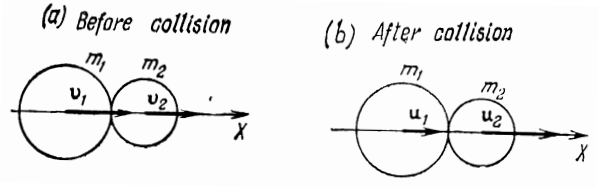


Figure 3.6: A perfectly elastic head-on central collision of two bodies.

projections on this axis are

$$u_{1x} = \frac{(m_1 - m_2)v_{1x} + 2m_2v_{2x}}{m_1 + m_2}$$

and

$$u_{2x} = \frac{2m_1v_{1x} + (m_2 - m_1)v_{2x}}{m_1 + m_2}$$

In particular, if the masses of the bodies are the same, the bodies exchange velocities in a collision: $u_{1x} = v_{2x}$ and $u_{2x} = v_{1x}$.

If the mass of the second body is many times greater than that of the first body, then $u_{1x} \approx 2v_{2x} - v_{1x}$ and $u_{2x} \approx v_{2x}$.

3.5.6

Example 2. *Perfectly elastic oblique central collision.* If the bodies are smooth the impulse of friction forces in the collision can be neglected. In this case the tangential components of the velocities of the bodies are not changed, i.e. components perpendicular to the line of impact: $u_{1\tau} = v_{1\tau}$ and $u_{2\tau} = v_{2\tau}$. The normal components, directed along the line of impact, change in the same way as in a head-on collision:

$$u_{1n} = \frac{(m_1 - m_2)v_{1n} + 2m_2 v_{2n}}{m_1 + m_2}$$

and

$$u_{2n} = \frac{2m_1 v_{1n} + (m_2 - m_1)v_{2n}}{m_1 + m_2}$$

In particular, in the perfectly elastic oblique collision of a smooth ball with a fixed flat wall($m_2 \gg m_1$ and $u_2 = v_2 = 0$): $u_{1\tau} = v_{2\tau}$ and $u_{1n} = -v_{1n}$, i.e. the ball rebounds from the wall according to the law of mirror reflection: the angle of reflection is equal to the angle of incidence. The magnitude of the velocity is conserved: $u_1 = v_1$. The vector $\Delta\mathbf{p}_1$ of the change in momentum of the ball in the collision is directed perpendicular to the wall:

$$\Delta\mathbf{p}_1 = m_1(\mathbf{u}_1 - \mathbf{v}_1) = -2m_1\mathbf{v}_{1n}$$

The impulse of impact force acting on the wall is equal to $2m_1\mathbf{v}_{1n}$.

4 Dynamics of Rotary Motion

4.1 Moment of Force and Angular Momentum

4.1.1

The moment of force, or force moment is a concept introduced in mechanics to characterize the external mechanical action exerted on a body that changes its rotary motion. Distinction is made between the moment of force about a fixed point and about a fixed axis.

The *moment of the force* **F** *about a fixed point* O (the *pole*) is the vector quantity **M**, equal to the vector product of the radius vector **r**, from point O to point A of force application (Figure 4.1), by the force vector **F**:

$$\mathbf{M} = [\mathbf{r}\,\mathbf{F}]$$

The magnitude of the moment of force is $M = Fr\sin\alpha = Fl$, where α is the angle between vectors **r** and F, and

$l = r\sin\alpha$ is the length of the perpendicular OB (Figure 4.1) dropped from point O to the line of action of the force. The quantity l is called the *arm of the force* with respect to point O. If the point of application of force $\mathbf{F}$ is displaced along its line of action, the moment $\mathbf{M}$ of this force about the same fixed point O is not changed. If the line of action of the force passes through point O, the moment of the force about point O equals zero.

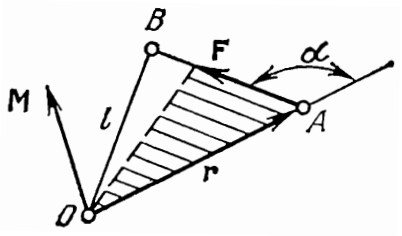


Figure 4.1: The moment of force **M**.

4.1.2

The *principal*, or *resultant*, *moment* of a system of forces *about fixed point* O (*pole*) is the vector $\mathbf{M}$, equal to the geometric sum of the moments about point O of all n forces of the system:

$$\mathbf{M} = \sum_{i=1}^{n} [\mathbf{r}_i \mathbf{F}_i]$$

where $\mathbf{r}_i$ is the radius vector from point O to the point of application of force $\mathbf{F}_i$.

It follows from Newton's third law (Section 2.5.1) that the moments of the internal forces of interaction between the particles of a system about pole O are compensated for (cancelled out) pair wise: $[\mathbf{r}_i\mathbf{F}_{ik}] = -[\mathbf{r}_k\mathbf{F}_{ki}]$. Consequently, in calculating the resultant moment of forces, only the *external* forces exerted on the mechanical system being considered need be taken into account.

4.1.3

The *moment of force* $\mathbf{F}$ *about a fixed axis* a is the scalar quantity M_a, equal to the projection on this axis of vector $\mathbf{M}$, which is the moment of force $\mathbf{F}$ about arbitrary point O of axis a.

The value of moment M_a is independent of the choice of the position of point O along axis a.

Note. Sometimes the moment of a force about fixed axis a is understood to mean the vector quantity $\mathbf{M}_a = M_a\mathbf{i}_a$, where $\mathbf{i}_a$ is the unit vector of axis a. The vector $\mathbf{M}_a$ is the component of vector $\mathbf{M}$ of the moment of force about pole O, this component being directed along axis a.

If the line of action of the force intersects or is parallel to axis a, the moment of force about this axis is equal to zero.

Let A be the point of application of force $\mathbf{F}$, and O_1 be the foot of the perpendicular dropped from point A to the axis OZ being considered (Figure 4.2). It proves convenient to resolve force $\mathbf{F}$ into three mutually perpendicular components: axial component F_z, parallel to axis OZ; radial component $\mathbf{F}_n$, directed along vector $\boldsymbol{\rho} = \vec{O_1A}$, and tangential component $\mathbf{F}_\tau$ directed perpendicular

to the axis a and perpendicular to vector $\boldsymbol{\rho}$.

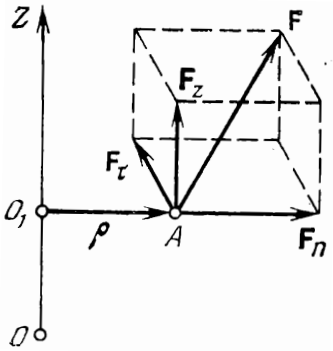


Figure 4.2: Resolving the components of the moment of force **M**.

The moment of force **F** about axis OZ is

$$M_z = [\boldsymbol{\rho}\mathbf{F}_\tau]_z \text{ and } \mathbf{M}_z = [\boldsymbol{\rho}\mathbf{F}_\tau]$$

Since vectors $\boldsymbol{\rho}$ and $\mathbf{F}_\tau$ are perpendicular to each other

$$|\mathbf{M}_z| = |M_z| = \rho|\mathbf{F}_\tau|$$

The *principal*, or *resultant*, *moment* of a system of forces *about a fixed axis a* is equal to the algebraic sum of the moments of all the forces of the system about this axis.

4.1.4

The *angular momentum*, or *moment of momentum*, *of a particle about fixed point O* (*pole*) is the vector **L**. It is equal to the vector product of radius vector **r**, from point O to the

location of the particle, by vector **p** of its momentum:

$$\mathbf{L} = [\mathbf{r}\,\mathbf{p}] = [\mathbf{r}m\mathbf{v}],$$

where m and $\mathbf{v}$ are the mass and velocity of the particle.

The *angular momentum of a system about fixed point O* is denoted by vector **L** and equals the vector sum of the angular momenta of all the particles of the system about the same point O:

$$\mathbf{L} = \sum_{i=1}^{n} [\mathbf{r}_i\,\mathbf{p}_i] = \sum_{i=1}^{n} [\mathbf{r}_i m_i \mathbf{v}_i],$$

where m_i, $\mathbf{r}_i$ and $\mathbf{v}_i$ are the mass, radius vector and velocity of the i-th particle, and n is the total number of particles.

The *angular momentum of a system about fixed axis a* is the scalar quantity L_a, equal to the projection on axis a of vector **L**, the angular momentum of the system about some point O lying on this axis:

$$L_a = \sum_{i=1}^{n} [\mathbf{r}_i m_i \mathbf{v}_i]_a$$

The numerical value L_a of the angular momentum is independent of the choice of the position of point O along axis a.

Note. Sometimes the angular momentum of a system about a fixed axis a is understood to mean the vector quantity $\mathbf{L}_a = L_a\mathbf{i}_a$ where $\mathbf{i}_a$ is the unit vector of axis a.

4.1.5

The *angular momentum of a body rotating about a fixed point O* at the angular velocity is equal to:

$$\mathbf{L} = \int_{(m)} [\mathbf{rv}] \mathrm{d}m = \int_{(m)} [\mathbf{r}[\boldsymbol{\omega}\mathbf{r}]] \mathrm{d}m$$

where $\mathbf{r}$ is the radius vector from point O to an infinitely small (dement of the body of mass $\mathrm{d}m$, and $\mathbf{v} = [\boldsymbol{\omega}\mathbf{r}]$ is the velocity of the element of the body. Since $[\mathbf{r}[\boldsymbol{\omega}\mathbf{r}]] = r^2\boldsymbol{\omega} - (\boldsymbol{\omega}\mathbf{r})\mathbf{r}$, vectors $\mathbf{L}$ and $\boldsymbol{\omega}$ do not coincide in direction in the general case:

$$\mathbf{L} = \boldsymbol{\omega} \int_{(m)} r^2 \mathrm{d}m - \int_{(m)} (\boldsymbol{\omega}\mathbf{r})\mathbf{r} \mathrm{d}m$$

The angular momentum of a body hinged at point O and its angular velocity coincide in direction if the body rotates about one of its principal axes of inertia (4.2.4) passing through point O

$$\mathbf{L} = \mathbf{I}\boldsymbol{\omega}$$

where $\mathbf{I}$ is the moment of inertia (4.2.1) of the body about this principal axis.

4.1.6

The resultant moments $\mathbf{M}$ and $\mathbf{M}^*$ of a system of forces about two different fixed points O and O^* are related by the equation

$$\mathbf{M} = \mathbf{M}^* + [\mathbf{r}^*\mathbf{F}]$$

where $\mathbf{r}^*$ is the radius vector from origin O to point O^*, and $\mathbf{F}$ is the principal vector of the system of forces being

considered. If $\mathbf{F} = 0$ the resultant moment of the system of forces is the same about any fixed point: $\mathbf{M}^* = \mathbf{M}$. This, precisely, is the property possessed by a *force couple*, or simply *couple*, i.e. a system of two forces, equal in magnitude and directed along two parallel straight lines in opposite directions. The shortest (perpendicular) distance d between the lines of action of the forces is called the *arm of the couple*. The moment, or torque, of a force couple is directed perpendicular to the plane of the forces, and its magnitude is equal to $M = Fd$, where F is the magnitude of each of the forces of the couple.

The resultant moment $\mathbf{M}_C$ of all the forces acting on a mechanical system about the centre of mass (2.3.3) of this system is related to the resultant moment $\mathbf{M}$ of the same system of forces about fixed point O by the equation

$$\mathbf{M} = \mathbf{M}_C + [\mathbf{r}_C \mathbf{F}]$$

where $\mathbf{r}_C$ is a radius vector from origin O to point C, and $\mathbf{F}$ is the principal vector of the system of forces.

4.1.7

The value of the angular momentum of a mechanical system about its centre of mass C is the same for the absolute motion of the particles at the velocities (i.e. with respect to a fixed inertial frame of reference) and for the relative motion at the velocities $\mathbf{v}'_i = \mathbf{v}_i - \mathbf{v}_C$ (i.e. with respect to a frame of reference in translational motion with the origin of coordinates at point C):

$$\sum_{i=1}^{n} [\mathbf{r}_i m_i \mathbf{v}_i] = \sum_{i=1}^{n} [\mathbf{r}'_i m_i \mathbf{v}'_i] = \mathbf{L}_C$$

where $\mathbf{r}_i' = -\mathbf{r}_C$ is the radius vector of the i-th particle in a frame of reference travelling together with the centre of mass.

The relation between the values of the angular momentum $\mathbf{L}$ of a mechanical system about fixed point O and $\mathbf{L}_C$ about the system's centre of mass is of the form

$$\mathbf{L} = \mathbf{L}_C + [\mathbf{r}_C \mathbf{p}]$$

where $\mathbf{p} = \sum_{i=1}^{n} m_i \mathbf{v}_i$ is the momentum of the system in its absolute motion.

4.2 Moment of Inertia

4.2.1

The *moment of inertia of a mechanical system* about fixed axis a is the physical quantity denoted by I_a and equal to the sum of the products of the masses of all n particles of the system by the squares of their normal distances from the axis:

$$I_a = \sum_{i=1}^{n} m_i \rho_i^2$$

where m_i and ρ_i are the mass of the i-th particle and its normal distance from the axis.

The moment of inertia of a body about fixed axis a is

$$I_a = \int_{(m)} \rho^2 \mathrm{d}m = \int_{(V)} \rho^2 D \mathrm{d}V$$

where $\mathrm{d}m = D\mathrm{d}V$ is the mass of an infinitely small element of volume $\mathrm{d}V$ of the body, D is the density and ρ is the normal distance from the element $\mathrm{d}V$ to axis a.

If the body is homogeneous, i.e. its density is the same through out, then

$$I_a = D \int_{(V)} \rho^2 \mathrm{d}V$$

The moment of inertia I_a is a measure of inertness of a body in its rotary motion about fixed axis a (4.3.4), in the same way that its mass is a measure of inertness of the body in its translational motion.

4.2.2

The moment of inertia of a given body about some axis depends, not only on the mass, shape and size of the body, but also on the position of the body with respect to this axis. Ac cording to *Steiner's theorem of parallel axes*, the moment of inertia I of a body about an arbitrary axis is equal to the sum of its moment of inertia I_C about a parallel axis through the centre of mass (or of inertia) of the body plus the product of the mass m of the body by the square of the normal distance d between the two axes:

$$I = I_C + md^2$$

4.2.3

The moments of inertia of certain homogeneous regularly shaped bodies about certain axes are listed in Table 4.1.

4.2.4

The *products of inertia of a body* about the axes of a rectangular Cartesian coordinate system are the quantities:

$$I_{xy} = \int_{(m)} xy\,\mathrm{d}m = \int_{(V)} xy\,D\mathrm{d}V$$
$$I_{xz} = \int_{(m)} xz\,\mathrm{d}m = \int_{(V)} xz\,D\mathrm{d}V$$
$$I_{yz} = \int_{(m)} yz\,\mathrm{d}m = \int_{(V)} yz\,D\mathrm{d}V$$

where x, y and z are the coordinates of an infinitely small element of the body of volume $\mathrm{d}V$, density D and mass m.

Axis OX is called the *principal axis of inertia of the body* if the products of inertia I_{xy} and I_{xz} are simultaneously equal to zero. Three principal axes of inertia can be passed through each point of the body. The axes are mutually perpendicular.

The moments of inertia of a body about the three principal axes of inertia, passed through an arbitrary point O of the body, are called the *principal moments of inertia of the body*.

The principal axes of inertia, passing through the centre of mass of the body, are called the *principal central axes of inertia of the body*, and the moments of inertia about these axes are called the *principal central moments of inertia of the body*. An axis of symmetry of a homogeneous body is always one of its principal central axes of inertia.

4.3 The Fundamental Law in the Dynamics of Rotary Motion

4.3.1

It follows from Newton's laws that the first derivative with respect to time t of the angular momentum $\mathbf{L}$ of a mechanical system about any fixed point 0 is equal to the resultant moment M^{ext} of all the external forces, applied to the system, about the same point O:

$$\frac{d\mathbf{L}}{dt} = M^{\text{ext}}$$

This equation expresses the *law of the variation of angular momentum of the system*. It is valid, in particular, for a rigid body, hinged at point O and rotating about it. In this case, the equation expresses the *fundamental law of dynamics of a rigid body rotating about a fixed point*.

In terms of the projections on the axes of a fixed rectangular Cartesian coordinate system with its origin at point O, the law for the variation of the angular momentum is written in the form:

$$\frac{dL_x}{dt} = M_x^{\text{ext}}, \quad \frac{dL_y}{dt} = M_y^{\text{ext}}, \quad \frac{dL_z}{dt} = M_z^{\text{ext}}$$

Here L_x, L_y, L_z, M^{ext_x}, M_x^{ext}, M_y^{ext} M_z^{ext} are the angular momenta of the system and the resultant moments of the external forces about the corresponding coordinate axes.

4.3.2

Example. The *regular precession of a gyroscope* due to its gravitational force. A *gyroscope (symmetrical gyroscope)*

is a rigid body spinning (rotating rapidly) about its axis of symmetry that can change its position in space. A gyroscope has three degrees of freedom (1.5.6) if it is mounted at fixed point O, which lies on its axis and is called the *centre of suspension of the gyroscope*. If the cen-

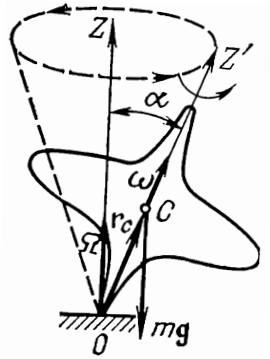


Figure 4.3: Analysing the rotary motion of a heavy gyroscope or a top.

tre of suspension coincides with the centre of gravity C (7.3.2) of the gyroscope, the gyroscope is said to be *balanced*: the force of gravity does not affect its state of rotation. Otherwise, it is said to be a *heavy gyroscope*, or top (Figure 4.3). Owing to the action of the moment of gravitational force about point O

$$M^{\text{ext}} = [\mathbf{r}_C m\mathbf{g}]$$

a heavy gyroscope rotates about this point so that its axis OZ' rotates uniformly about vertical axis OZ describing a conical surface (shown by dashed lines in Figure 4.3).

Such motion of the gyroscope is called *regular precession*. If the angular velocity of precession $\Omega \ll \omega$ (where ω is the spin angular velocity or, simply, spin velocity of the gyroscope about its axis of symmetry OZ'), it can be assumed with some approximation that the angular momentum $\mathbf{L}$ of the gyroscope about point O is directed along axis OZ' of the gyroscope and equals

$$\mathbf{L} = I\boldsymbol{\omega}$$

where I is the moment of inertia of the gyroscope about axis OZ'. Hence

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = [\mathbf{r}_C m\mathbf{g}] = \left[\frac{\mathbf{r}_C}{I\omega_{z'}}\mathbf{L}m\mathbf{g}\right] = [\boldsymbol{\Omega}\mathbf{L}]$$

where $\boldsymbol{\Omega} = -m\mathbf{r}_C g/I\omega_{z'}$, is the angular velocity of precession, and $\omega_{z'} = \omega$ in the case shown in Figure 4.3. The greater the spin angular velocity of the gyroscope the slower its precession.

4.3.3

The kinetic energy of a rigid body rotating about a fixed point at the angular velocity $\boldsymbol{\omega}$ is

$$E_k = \frac{I\boldsymbol{\omega}^2}{2}$$

where I is the moment of inertia of the body about the instantaneous axis of rotation (1.5.6).

An element of work done in the infinitely short time interval $\mathrm{d}t$ by force $\mathbf{F}$ acting on the body is

$$\delta W = \mathbf{M}\boldsymbol{\omega}\mathrm{d}t = \mathbf{M}\mathrm{d}\boldsymbol{\varphi} = M_\omega \mathrm{d}\varphi$$

where $\mathbf{M} = [\mathbf{rF}]$ is the moment of force $\mathbf{F}$ about point O ($\mathbf{r}$ is the radius vector from point O to the point of application of force $\mathbf{F}$), $\mathrm{d}\varphi = \omega \mathrm{d}t$ and $\mathrm{d}\boldsymbol{\varphi} = \boldsymbol{\omega}\mathrm{d}t$ are the angle of rotation and the vector of an element of rotation of the body in the time interval $\mathrm{d}t$, and M_ω is the moment of force $\mathbf{F}$ about an instantaneous axis of rotation ($\mathbf{M}_\omega$ equal to the projection of vector $\mathbf{M}$ on the direction of vector ω).

The change in the kinetic energy of a rotating rigid body during the time interval $\mathrm{d}t$ is equal to the work done by the external forces:

$$\mathrm{d}E_k = M_\omega^{\mathrm{ext}} \mathrm{d}\varphi$$

where is the resultant moment (4.1.3) of external forces about an instantaneous axis of rotation of the body.

4.3.4

If a rigid body rotates about fixed axis OZ at the angular velocity ω, its angular momentum about this axis is

$$L_z = I_z \omega_z \text{ and } \mathbf{L}_z = I_z \boldsymbol{\omega}$$

Here I_z is the moment of inertia of the body about axis OZ and does not change with time ($I_z = \text{const}$), and $|\omega_z| = \omega > 0$ ($\omega_z = \omega$ if vector $\boldsymbol{\omega}$ and the unit vector of axis OZ coincide in direction, otherwise $\omega_z = -\omega$).

The *fundamental law in the dynamics of a rigid body rotating about fixed axis OZ is*

$$I_z = \frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}t} = \mathbf{M}_z^{\mathrm{ext}} \text{ or } \boldsymbol{\varepsilon} = \frac{1}{I_z}\mathbf{M}_z^{\mathrm{ext}}$$

where $\boldsymbol{\varepsilon} = \mathrm{d}\boldsymbol{\omega}/\mathrm{d}t$ is the angular acceleration of the body.

It is evident from the preceding equation that the moment of inertia of a rigid body about some fixed axis is a measure of inertness of the body in its rotation about the given axis: the greater the moment of inertia of the body, the less the angular acceleration it acquires under the action of the same moment of external forces.

4.3.5

The kinetic energy of a rigid body rotating about fixed axis OZ at the angular velocity ω is

$$E_k = \frac{1}{2} I_z \omega^2$$

An element of work done during the infinitely short time $\mathrm{d}t$ by force $\mathbf{F}$ applied to the body is

$$\delta W = \mathbf{M}_z \boldsymbol{\omega} \mathrm{d}t = M_z \mathrm{d}\varphi$$

where M_z is the moment of force $\mathbf{F}$ about the axis OZ of rotation (the unit vector of axis OZ coincides in direction with vector $\boldsymbol{\omega}$).

The change in the kinetic energy of a rigid body during the time interval $\mathrm{d}t$ is equal to the work done by the external forces during this interval:

$$\mathrm{d}E_k = M_z^{\mathrm{ext}} \mathrm{d}\varphi$$

where M_z^{ext} is the resultant moment of external forces about the axis of rotation of the body.

4.3.6

The motion of a free rigid body satisfies the following two differential equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{v}_C) = \mathbf{F}^{\mathrm{ext}} \text{ and } \frac{\mathrm{d}\mathbf{L}_C}{\mathrm{d}t} = \mathbf{M}_C^{\mathrm{ext}}$$

Here m is the mass of the body, $\mathbf{v}_C$ is the velocity of its centre of mass C, $\mathbf{F}^{\mathrm{ext}}$ is the principal vector of external forces applied to the body (2.5.2), $\mathbf{M}_C^{\mathrm{ext}}$ is the resultant moment of external forces about point C (4.1.6), and $\mathbf{L}_C$ is the angular momentum about the same point C (4.1.7).

The first equation describes translational motion of the free body at the velocity of its centre of mass (2.5.3). The second equation follows from the law of the variation of angular momentum (4.3.1) and describes the rotation of a rigid body about its centre of mass (1.5.9).

4.3.7

The kinetic energy of a free rigid body can be deter mined on the basis of Koenig's theorem (3.2.4):

$$E_k = \frac{mv_C^2}{2} + \frac{I_C\omega^2}{2}$$

where I_C is the moment of inertia of the body about an instantaneous axis of rotation passing through its centre of mass C, and ω is the angular velocity of the body. In the general case, the instantaneous axis is displaced in the body and the moment of inertia I_C varies with time. The value of I_C is constant if the body is in plane motion (1.5.9).

Example. The kinetic energy of a homogeneous circular cylinder rolling down an inclined plane without slipping. The cylinder has plane motion because all of its points travel in vertical planes that are parallel to one another. The cylinder travels with translational motion at the velocity $\mathbf{v}_C$ directed along the inclined plane, and rotates about its axis ($I_C = mR^2/2$, where m and R are the mass and radius of the cylinder) at the angular velocity ω. From the condition that the cylinder rolls without slipping it follows that the instantaneous velocities of the points of contact between the cylinder and inclined plane equal zero, i.e. $\omega = v_C/R$. Hence, the kinetic energy of the rolling cylinder is

$$E_k = \frac{mv_C^2}{2} + \frac{I_C\omega^2}{2} = \frac{3}{4}mv_C^2.$$

4.4 Law of Conservation of Angular Momentum

4.4.1

The *law of the conservation of angular momentum* can be formulated as follows: the angular momentum of a closed system (2.2.4) about any fixed point remains constant with time, i.e.

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} \equiv 0 \text{ and } \mathbf{L} = \text{const}$$

Accordingly, the angular momentum of a closed system about its centre of mass (4.1.7) remains constant with

time:

$$\frac{d\mathbf{L_C}}{dt} \equiv 0 \text{ and } \mathbf{L_C} = \text{const}$$

Like the laws of conservation of momentum and energy, the law of conservation of angular momentum is of significance far beyond the scope of classical mechanics. It belongs to the most Fundamental of physical laws because it is associated with a definite symmetry property of space: its isotropy. The *isotropy of space* is manifested in the fact that the physical properties and laws of motion of a closed system are independent of the direction chosen for the coordinate axes in an inertial frame of reference. This means that these properties and laws do not change when the closed system is turned as a whole through any angle in space.

According to modern concepts, not only particles and bodies can have an angular momentum, hut fields as well. Elementary particles and systems built from them (for instance, atomic nuclei) can have an angular momentum that is not related to their motion in space and is called their spin (Table **??**).

4.4.2

With respect to systems that can be described by classical (Newtonian) mechanics, the law of conservation of angular momentum can be regarded as a consequence of Newton's laws. For a closed mechanical system, the resultant moment of external forces about any fixed point (as well as about the centre of mass of the system) is identically zero: $\mathbf{M}^{\text{ext}} \equiv 0$ (consequently, $\mathbf{M}_C^{\text{ext}} \equiv 0$, see (4.1.6), where $\mathbf{F} = \mathbf{F}^{\text{ext}} \equiv 0$) and the law of conservation of angu-

lar momentum follows from (4.3.1):

$$\mathbf{L} = \sum_{i=1}^{n} [\mathbf{r}_i m_i \mathbf{v}_i] = \text{const}$$

where $m_i, \mathbf{r}_i$ and $\mathbf{v}_i$ are the mass, radius vector and velocity of the i-th particle of the system, which consists of n such particles.

Accordingly (see (4.1.7) and (2.5.3)),

$$\mathbf{L}_C = \sum_{i=1}^{n} [\mathbf{r}'_i m_i \mathbf{v}'_i] = \sum_{i=1}^{n} [\mathbf{r}_i m_i \mathbf{v}_i] = \text{const}$$

where $\mathbf{r}'_i = \mathbf{r}_i - \mathbf{r}_C$, and $\mathbf{v}'_i = \mathbf{v}_i - \mathbf{v}_C$ and $\mathbf{r}_C, \mathbf{v}_C$ are the radius vector and velocity of the system's centre of mass.

4.4.3

If a system is not closed, but the external forces acting on it are such that their resultant moment about fixed point O is identically zero ($\mathbf{M}^{\text{ext}} \equiv 0$), then, according to Newton's laws (4.3.1), the angular momentum of the system about the same point O remains constant with time: $\mathbf{L} = \text{const}$. This condition is practically complied with, for example, by a balanced gyroscope (4.3.2) with three degrees of freedom, if the moment of the friction force in its suspension is sufficiently small.

If the stand of the gyroscope, holding its centre of suspension at rest, is turned in any way, the axis of the gyroscope retains its orientation with respect to a fixed inertial frame of reference.[1]

[1] It is assumed that vector $\mathbf{L}$ is directed along the axis of the gyro-

Usually $\mathbf{M}^{\text{ext}} \not\equiv 0$ and $\mathbf{L} \neq \text{const}$. But if the resultant moment of external forces about any fixed axis passing through point O is identically zero, the angular momentum of the system about this axis remains constant with time. For instance, if $M_z^{\text{ext}} \equiv 0$, then $L_z = \text{const}$.

If the system rotates about fixed axis OZ and the resultant moment of external forces about this axis $M_z^{\text{ext}} \equiv 0$, the angular momentum of the system about the axis of rotation remains constant with time:

$$I_z \omega = \text{const}$$

where ω and I_z are the angular velocity and moment of inertia of the system.

If the action of internal forces, as well as external forces that comply with the condition $M_z^{\text{ext}} \equiv 0$, leads to deformation of the system, changing its moment of inertia I_z, the angular velocity ω correspondingly increases or decreases.

4.4.4

Free axes of a body are ones about which a free rigid body (2.2.3) can rotate at constant angular velocity $\boldsymbol{\omega}$ in the absence of all external action. Such rotation of a body is said to be *free*, or *inertial rotation*. The free axes of a body coincide with its principal central axes of inertia (4.2.4). In the general case, the values I_1, I_2 and I_3 of the principal central moments of inertia (4.2.4) differ. The

scope. Otherwise, a free gyroscope has a precessional motion: its axis describes a circular conical surface whose vertex is at the centre of suspension and whose axis is directed along the vector $\mathbf{L} = \text{const}$.

free rotation of such a body (for ex ample, a homogeneous rectangular parallelepiped with edges of different length) is accomplished practically only about two free axes, corresponding to the extreme values of the principal central moments of inertia – the maximum and minimum values. The rotation of the body about its third principal central axis, corresponding to the intermediate value of the moment of inertia of the body, proves to be unstable: even very small external action is capable of leading to substantial deviation of the instantaneous axis of rotation from its initial direction in the body.

If the values of two principal central moments of inertia of a body are equal: $I_1 = I_2 \neq I_3$, stable free rotation of such a body (for instance, a homogeneous circular cylinder) is possible only about the free axis corresponding to the third, differing value I_3 of the moment of inertia. For a homogeneous circular cylinder such a free axis is its axis of symmetry. But if a long thin cylinder is rotated by means of a string secured to one end, stable rotation of the cylinder can be accomplished about the free axis corresponding to the maximum value of its moment of inertia. This free axis is perpendicular to the axis of symmetry of the cylinder.

Body	**Position of axis a**	**Moment of inertia I_a**
Hollow thin-walled circular cylinder of radius R and mass m	Axis of the cylinder	mR^2
Solid circular cylinder (or disk) of radius R and mass m	Axis of the cylinder	$mR^2/2$
Solid sphere of radius R and mass m	Axis through the centre of the sphere	$2mR^2/5$
Hollow thin-walled sphere of radius R and mass m	Axis through the centre of the sphere	$2mR^2/3$
Thin straight rod of length l and mass m	Axis perpendicular to the rod and through its centre along its length	$ml^2/12$
The same rod	Axis perpendicular to the rod and through one end of it	$ml^2/3$

Table 4.1: The moments of inertia of certain homogeneous regularly shaped bodies about certain axes.

5 Fundamentals of the Special Theory of Relativity

5.1 Postulates of the Special Theory of Relativity

5.1.1

The *special theory of relativity* (sometimes called the *restricted relativity theory*) is the modern physical theory of space and time. The special relativity theory and quantum mechanics (**??**) serve as the theoretical basis for all modern physics and engineering (for example, nuclear physics and nuclear technology). The special relativity theory is often called *relativistic theory*, and specific phenomena described by this theory are called *relativistic effects*. Relativistic effects are manifested, as a rule, at velocities of bodies close to that of light in free space ($c = 3 \times 10^8$ m/s) and called *relativistic velocities*. Relati-

vistic mechanics refers to the mechanics of motion at relativistic velocities and is based on the special theory of relativity.

As in classical Newtonian mechanics it is assumed in special relativity theory that time is uniform (3.4.2) and that space is homogeneous (2.7.1) and isotropic (4.4.1).

5.1.2

The special theory of relativity is based on two main principles accepted as the initial postulates.

The *first postulate* is a generalization of Galileo's mechanical principle of relativity (2.8.4) to cover any physical processes.

This postulate, called the *principle of relativity*, or *Einstein's relativity principle*, states: all physical phenomena proceed in the same way, under the same conditions, in all inertial frames of reference (2.1.2). In other words, the principle of relativity contends that physical laws are independent (invariant) with respect to the chosen inertial frame of reference; equations expressing these laws have the same form in all inertial frames of reference.

Consequently, it is impossible to ascertain, by means of any physical experiments whatsoever, conducted in a closed system of bodies (2.2.4), whether the system is at rest or is travelling at uniform velocity in a straight line with respect to some inertial frame of reference. All inertial frames of reference are equally justified in physics. It is impossible, on the basis of physical experiments, to select from the host of inertial reference frames any "preferred" or "absolute" frame of reference possessing any qualitative distinctions from other inertial reference

frames.

5.1.3

The *second postulate* expresses the *principle of the invariance of the velocity of light*: the velocity of light in free space is independent of the motion of the source of light. It is the same in all directions and in all inertial frames of reference, being one of the most basic physical constants. Experiments show that c, the velocity of light in free space, is the limiting velocity in nature. The velocity of any particles and bodies, as well as the velocity of propagation of any interactions or signals, cannot exceed c.

These specific laws governing the process of light propagation in free space enable this real physical process to be made use of in establishing a procedure for the *chronometry of a frame of reference*, i.e. for the synchronization of clocks located at various points in space and travelling together with the frame of reference being considered (5.2.3).

5.1.4

The postulates of the special theory of relativity contradict the concepts of the properties of space and time that are accepted in classical mechanics and on which the Galilean transformations are based (2.8.1). In particular, this refers to the contention, considered to be "self-evident" in Newtonian mechanics, that the passage of time is identical in all inertial frames of reference. From this it follows that the time interval between any two

events is of an absolute nature. For instance, if two events occur simultaneously according to the clock in one inertial frame of reference, then, according to classical concepts, they also occur simultaneously according to the clock of any other inertial frame.

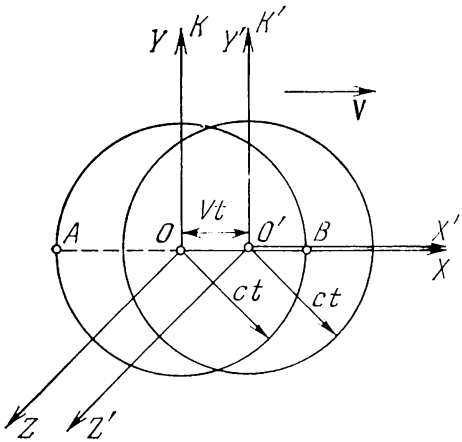


Figure 5.1: Consequences of constancy of speed of light as a postulate in theory of realtivity.

The above-mentioned contradiction can be illustrated by the following example (Figure 5.1). We have two inertial frames of reference: fixed frame K and frame K' which travels along the OX axis at constant velocity V. Assume that a flash of light is produced at point O at the instant of the zero time reference in both frames K and K' ($t = t' = 0$), when their origins O and O' of coordinates coincide. By the instant of time $t > 0$, light, propagating in free space at the velocity c, reaches points at the

surface of a hollow sphere in the frame K. This sphere has its centre at point O and its radius equals ct.

It can be assumed that in frame K' the light flash occurred at the instant of time $t' = O$ at point O'. Hence, according to the postulates of the special theory of relativity, by the instant of time $t' = t$, light in frame K' reaches points at the surface of a hollow sphere of the same radius ct as in the frame K but with its centre at point O'. At this instant, however, point O' is not at point O, but at a distance of Vt from this point. Hence, any attempt to combine the postulates of the special theory of relativity with classical conceptions of absolute time that elapses identically in all frames of reference leads to an absurdity: the light flash must *simultaneously* reach points in space that belong to two different hollow spheres.

5.2 Simultaneity of Events. Synchronization of Clocks

5.2.1

The concept of the simultaneity of two or several events is extensively employed in making various physical measurements. For example, to determine the length l of a rod, arranged along the axis OX of reference frame K and travelling with respect to this frame, it is necessary to note the values of the coordinates $x_2(t)$and $x_1(t)$ of the ends of the rod *simultaneously*, i.e. at the same instant of time t. Then

$$l = |x_2(t) - x_1(t)|$$

The instant of time at which some event occurs (for instance, the launching or touchdown times of a spaceship) is determined by noting a clock's reading that is *simultaneous* with the event being considered. This can be simply done by means of a clock located at the place where the event occurs. Hence, there should be many clocks located at various points in space in each frame of reference. Obviously, these clocks should be in agreement with one another, i.e. run *synchronously*: their readings should be identical at each instant of time t.

5.2.2

The synchronism of clocks located adjacently, i.e. at the same place in space, can be checked by the coincidence of their readings at each arbitrary instant of time. The synchronism of clocks that are separated in space at the points A and B, distant from each other, could be checked in a similar manner if we had at our disposal standard time signals that could propagate *instantaneously* from A to B. Experiments indicate, however, that this method is unfeasible because the velocity of any signal cannot exceed the velocity of light in free space.

We can, however, proceed in the following manner. We can convey the clock from B to A, check whether it is synchronous with the clock located at A, and then convey the clock carefully back to B. Then we can check whether the clock brought back to B runs at the same rate as the clock left at A by transmitting time signals from A to B after definite equal time intervals according to the clock at A. But it cannot be established by this procedure whether a shift in the zero time reference has occurred in

the clock reading while it was being conveyed back to B, i.e. whether the clock brought back to B is now fast or slow with respect to the clock at A by a constant amount Δt.

5.2.3

The problem of the synchronism of clocks separated in space at points A and B can be solved only by an unambiguous agreement (or definition) about when these clocks are to be regarded as synchronous. Einstein made use of a real physical process – the propagation of light in free space – as the basis for such a definition. He proceeded from the facts that the velocity of light in free space is the maximum possible velocity at which signals can be transmitted in nature and that it is identical in all directions and in all inertial frames of reference.

Assume that according to the clock at point A the light signal is transmitted from this point at the instant of time t_1 and, after being reflected at point B, returns to A at the instant of time t_3. Then, by definition, the clock at B is in synchronism with that at A if they run at the same rate and if, at the instant the light signal reaches B, the clock located there indicates the time $t_2 = (t_1 + t_3)/2$.

5.2.4

According to the special theory of relativity, the passage of time may differ in different inertial frames of reference. Consequently, the time interval between any two definite events is relative: it can change in going over from one inertial reference frame to another. In particu-

lar, the simultaneity of two events occurring at different points separated in space is also relative. Events occurring simultaneously in one inertial frame of reference are not at all simultaneous in other inertial frames that are travelling with respect to the first frame. In some frames of reference one of these events occurs before the other event; in other frames it occurs after the other event.

Thus, in the example illustrated in Figure 5.1 (5.1.4), the arrival of the light flash at points A and B are events that occur simultaneously in the fixed frame of reference A. In the travelling frame of reference K' these events are not simultaneous. The light flash reaches A, receding from the light source O', later than it reaches B, approaching the source O'.

Events connected by a cause-and-effect relationship, i.e. by causality, cannot occur simultaneously in any frame of reference whatsoever, because any effect is due to some process initiated by the cause. At the same time, no process (physical, chemical, biological, etc.) can proceed instantaneously. Therefore, relativity does not contradict causality in any way. In any inertial frame of reference an event that is an effect always occurs later than the event causing this effect.

5.3 Lorentz's Transformations

5.3.1

It follows from the postulates of the special theory of relativity, as well as from the homogeneity (2.7.1) and isotropy (4.4.1) of space and the uniformity of time (3.4.2), that the relations between the coordinates and times of

the same event in two inertial frames of reference are expressed by the *Lorentz transformations*, rather than the Galilean transformations (2.8.1) used in classical (Newtonian) mechanics. According to the principle of relativity and the above-mentioned properties of the symmetry of space and time, the Lorentz transformations should he linear.

5.3.2

The Lorentz transformations are in their simplest form if the like axes of the Cartesian coordinates of the fixed (K) and moving (K') inertial frames of reference are pairwise parallel to each other and if frame K' travels with respect to frame K at the constant velocity V along axis OX (Figure 2.3, see (2.8.1)). If, in addition, the zero time reference in both frames ($t = 0$ and $t' = 0$) is taken as the instant when the origins O and O' of coordinates of the two systems coincide, then the Lorentz transformations are of the form

$$x' = \frac{x - Vt}{\sqrt{1 - \frac{V^2}{c^2}}}, \qquad x = \frac{x' + Vt}{\sqrt{1 - \frac{V^2}{c^2}}}$$

$$y = y', \qquad y' = y$$

$$z = z', \qquad z' = z$$

$$t = \frac{t - Vx/c^2}{\sqrt{1 - \frac{V^2}{c^2}}}, \qquad t' = \frac{t + Vx'/c^2}{\sqrt{1 - \frac{V^2}{c^2}}}$$

where c is the velocity of light in free space.

5.3.3

The Lorentz transformations show that not only the spatial coordinates of an event being considered are changed in going over from one inertial frame of reference to another, hut the corresponding instants of time as well. But a definite interrelation exists between the spatial coordinates x', y' and z' of an event and the time instant t' it occurs in an arbitrary inertial frame of reference K'. Hence, the quantity $\left[(x')^2 + (y')^2 + (z')^2 - c^2(t')^2\right]$ is independent of the velocity V of frame K', i.e. it is identical in all inertial frames of reference:

$$(x')^2 + (y')^2 + (z')^2 - c^2(t')^2 = (x)^2 + (y)^2 + (z)^2 - c^2(t)^2$$

The coordinate x' and time t' cannot be imaginary. It follows, therefore, from the Lorentz transformations that the velocity of relative motion of any two inertial frames of reference cannot exceed the velocity of light in free space ($V \leqslant c$).

5.3.4

According to Einstein's principle of relativity (5.1.2), physical laws must comply with the condition of *relativistic invariance (Lorentz invariance)*. This requirement implies that equations expressing physical laws retain their form in going over from one inertial frame of reference to another in accordance with the Lorentz transformations.

The Lorentz transformations reduce to the Galilean transformations (2.8.1) at $V \ll c$ or, more precisely, in the limit as $V/c \to 0$, i.e. as $c \to \infty$. In other words, the Galilean transformations and classical (Newtonian) mechanics based on them are founded on the assumption

of instantaneous propagation of interaction. Such an approximate approach is permissible only in considering the laws of mechanical motion of bodies at velocities a great many times less than the velocity of light in free space.

5.4 Relativity of Lengths and Time Intervals. Time Interval Between Two Events

5.4.1

It follows from the Lorentz transformations (5.3.2) that the linear dimension of a body travelling with respect to an inertial frame of reference is reduced in the direction of motion. This change in the longitudinal size of a body in motionis called the *Lorentz-FitzGerald contraction*. Assume l_0 to he the length of a rod at rest in reference frame K'. If the rod is located along axis $O'X'$ (Figure 5.2), then

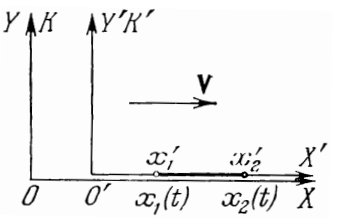


Figure 5.2: Length contraction in relative motion.

$l_0 = x'_2 - x'_1$, where x'_2 and x'_1 are the coordinates of the

ends of the rod. The length l of the same rod in reference frame K, with respect to which it travels along axis OX at the velocity V, is equal to the difference in the coordinates of the ends of the rod, measured at the same instant of time t:

$$\begin{aligned} l &= x_2(t) - x_1(t) \\ &= (x_2' - x_1')\sqrt{1 - \frac{V^2}{c^2}} \\ &= l_0\sqrt{1 - \frac{V^2}{c^2}} \end{aligned}$$

Transverse dimensions of a body are independent of its velocity and are identical in all inertial frames of reference:

$$y_2 - y_1 = y_2' - y_1', \text{ and } z_2 - z_1 = z_2' - z_1'$$

Thus, the linear dimensions of a body are relative. They are maximal in the inertial frame of reference with respect to which the body is at rest. These are said to be the body's *proper dimensions*.

5.4.2

The Lorentz-FitzGerald contraction is a kinematic effect of the special theory of relativity. It is not due to the effect of any kind of longitudinal forces compressing the body along its direction of motion. This contraction becomes appreciable only at velocities approaching that of light in free space. It follows from the contraction formula that bodies cannot travel at velocities $V \geqslant c$ because at $V = c$ the longitudinal dimension of the body

(in the direction of motion) becomes zero, and at $V > c$ it would have to become imaginary.

5.4.3

It is evident from the Lorentz transformations that in the theory of relativity we can specify a definite "instant of time" only with respect to a single definite inertial frame of reference. Thus, for instance, a single "instant of time" in reference frame K (a single definite value of time t in this frame) corresponds to many values of time t' in frame K', depending upon the values of coordinate x:

$$t' = \frac{t - Vx/c^2}{\sqrt{1 - \dfrac{V^2}{c^2}}}$$

Likewise, a single "instant of time" in frame K', i.e. a single definite value of time t', corresponds to many values of time t in frame K, depending upon the values of coordinate x':

$$t = \frac{t' + Vx'/c^2}{\sqrt{1 - \dfrac{V^2}{c^2}}}.$$

5.4.4

Another important consequence of the Lorentz transformations is the relativity of the time interval between any two events (for example, the beginning and end of some process), i.e. the dependence of this time interval on the choice of the inertial frame of reference. Assume that two events, 1 and 2, occur in the moving inertial frame

of reference K' at the same point A, fixed in frame K' (i.e. $x'_2 = x'_1$), at the instants of time t'_1 and t'_2. Thus, the time interval between the events is $\tau_0 = t'_2 - t'_1$.

With respect to fixed inertial frame of reference K, point A travels with the same velocity V as does frame K'. Hence in frame K events 1 and 2 occur at different points in space, having the coordinates x_1 and x_2, and $x_2 - x_1 = V\tau$, where $\tau = t_2 - t_1$ is the time interval between events 1 and 2 according to the clock in frame K. It follows from the Lorentz transformations that

$$\tau = t_2 - t_1 = \frac{t'_2 - t'_1}{\sqrt{1 - \dfrac{V^2}{c^2}}} = \frac{\tau_0}{\sqrt{1 - \dfrac{V^2}{c^2}}}$$

Thus, the time interval between two events is minimal in the inertial frame of reference with respect to which both events occur at the same point. The time measured by a clock travel ling together with some object is said to be the *proper time* of the object.

5.4.5

The consequence of the Lorentz transformations that is dealt with in the preceding subsection is evidence of the existence of the relativistic effect of the slowing of time, called *time dilation*, in a moving inertial frame of reference as compared to a fixed one. A clock travelling at the velocity V with respect to a given inertial frame runs slower by a factor of $\sqrt{1 - V^2/c^2}$ than a fixed clock.

Correspondingly, in agreement with the relativity principle, all physical processes in a moving frame of reference proceed at a slower rate than in a fixed frame.

The time dilation effect becomes appreciable only at extremely high velocities V, close to that of light in free space. This effect has been demonstrated, for example, by experiments conducted with muons (**??**). The muon is an unstable elementary particle. The mean proper lifetime of muons (according to a clock in the inertial frame of reference with respect to which they are at rest) $\tau_0 = 2.2 \times 10^{-6}$ s. Muons are created in the upper layers of title atmosphere by the action of primary cosmic rays, and they travel with respect to the earth at velocities V close to c. If there was no relativistic effect of time dilation, a muon would be capable of travelling an average distance, with respect to an observer on the earth, of $\tau_0 c = 660$ m through the atmosphere. In other words, muons could not reach the earth's surface. Actually, they are registered by instruments installed at the surface of the earth, because the mean lifetime of a travelling muon, according to the clock of the observer on the earth, $\tau = \tau_0/\sqrt{1 - V^2/c^2} \gg \tau_0$, and the path travelled by a muon during this time, $\tau V \gg 660$ m.

5.4.6

The relativistic effect of time dilation in a spaceship travelling with respect to the earth makes it feasible to undertake space flights of almost unlimited length and journeys "into the future". According to the relativity principle, all processes in a spaceship, including the aging process of the astronauts, proceed in agreement with the

same laws as on the earth. But in the spaceship time must be measured by a clock travelling together with it at the velocity V with respect to the earth.

If V is close to c the clock in the ship runs considerably slower than earth clocks (at the space-launch complex), by a factor of $1/\sqrt{1 - V^2/c^2}$. For example, at $\beta = V/c = 0.99999$ the rates of the spaceship and earth clocks differ by a factor of 224. Consequently, in a time interval of $\tau_0 = 10$ years in such a spaceship, one can make a space flight taking $\tau = 2240$ years according to earth clocks, aging only 10 years during the flight. In this flight the spaceship can travel a tremendous distance from the earth: $l = V\tau = \beta c\tau = 2239.98$ light years[1]. The closer V is to c, the greater the distance l travelled by the spaceship with respect to the earth in the same interval τ_0 of proper time in the spaceship, i.e. the longer the space flight that the astronauts can make during their lifetime.

If an astronaut completes a space flight at a velocity V close to c by returning to the earth, he will find that people remaining on the earth (including his twin brother) has aged more than he has during the flight. If V differs from c by a sufficiently small amount, when $(1 - V^2/c^2)^{-1/2} \gg 1$, the astronaut can, during the flight, outlive all his contemporaries and find him self, upon his return, among future generations.

5.4.7

It would seem, on the face of it, that, on the basis of the relativity principle, we should arrive at a conclusion ex-

[1] A light year is the distance travelled by light in free space in one year: 1 light year = 9.4605×10^{15} m.

actly the opposite of that given in the preceding subsection. The clock on the earth, travelling at the velocity $-\mathbf{V}$ with respect to the spaceship, should run slower than the clock in the ship. Therefore, the time required for the flight should be longer for the astronaut, and not for the people left on the earth. Hence, the twin that made the flight should age more than his brother.

We have thus found that the difference between the readings of the clocks at the space-launch complex and in the ship is positive according to (5.4.6), while, on the other hand, it is negative. This absurd result is called the *clock*, or *twin*, *paradox*. Actually, no clock paradox exists. It arose due to improper application of the relativity principle. This principle contends that not all frames of reference are equally justified, but only all *inertial* frames of reference. But the frame of reference of the spaceship, in contrast to that of the earth, is not an inertial one all throughout the flight. During launching, turning around and in braking orbit when approaching the earth, the spaceship travels with acceleration (or deceleration).

Consequently, the problem concerning the running of the clock at the launching complex which is always at rest with respect to a single inertial frame, and that of the clock in the spaceship, is, in principle, not a symmetrical one. Hence, the earth and ship frames of reference are not equally justified in the given problem. The reasoning in (5.4.6) is the correct one because it is based on the use of an inertial (the earth) frame of reference. Accordingly, the reasoning given in (5.4.7), leading to the clock paradox, is incorrect. In the second case, the general theory of relativity should be applied, rather than the special the-

ory. When this is done, it is found that, from the point of view of the astronaut as well, his clock runs slower than the clock at the launching complex.

5.4.8

The *interval*, or *space-time interval*, between two events, measured in the inertial frame of reference K' is the quantity

$$s'_{12} = \sqrt{c^2(t'_{12})^2 - (l'_{12})^2}$$

where $t'_{12} = t'_2 - t'_1$ is the time interval between the two events being considered (according to the clock in frame K'), and $l'_{12} = \sqrt{(x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2}$ is the distance between the points at which events **1** and **2** occur, also measured in frame of reference K'.

It follows from the Lorentz transformations that the interval between two given events, **1** and **2**, is invariant with respect to the choice of the inertial frame of reference, i.e. it does not change in transferring from moving inertial frame K' to fixed frame K:

$$s'_{12} = s_{12} = \text{inv, where } s_{12} = \sqrt{c^2 t_{12}^2 - l_{12}^2}$$

If $s_{12}^2 > 0$, i.e. if s_{12} is a real number, the interval s_{12} is said to be *timelike*.

The interval s_{12} is said to be *spacelike* if $s_{12}^2 < 0$, i.e. s_{12} is an imaginary number.

5.4.9

It follows from the invariance of the interval with respect to the choice of the inertial frame K' that in all frames K'

the values of t'_{12} and l'_{12} for the given two events, 1 and 2, satisfy the equation of a hyperbola:

$$c^2(t'_{12})^2 - (l'_{12})^2 = s^2_{12}$$

If $s^2_{12} > 0$ the relation between t'_{12} and l'_{12} in various inertial frames of reference K' travelling with respect to fixed frame K at any possible velocities ($0 \geqslant V < c$), can be represented graphically as the two branches I and II of a hyperbola (Figure 5.3). Consequently, the sign of

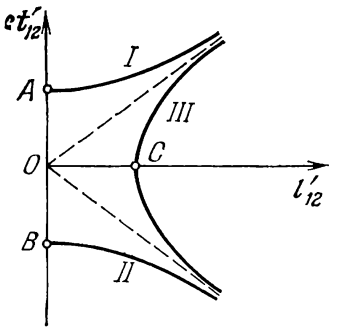


Figure 5.3: The invariance of interval between two inertial frames.

the time interval between events 1 and 2, related by a timelike interval, is absolute. It is independent of the choice of the frame of reference: in all frames K' the second event always occurs either after the first, i.e. $t'_{12} > 0$ (branch I) or always before the first, i.e. $t'_{12} < 0$ (branch II). The distance l'_{12} is relative; an inertial frame K' can be chosen in which $l'_{12} = 0$, i.e. events 1 and 2 occur at the same place (points A and B on the branches I and II of the hyperbola).

Two events related by causality should always correspond to a timelike interval or, in any case, one equal to zero ($s_{12} = 0$). This is due to the fact that the signal by means of which event 1 (the cause) leads to event 2 (the effect) cannot be propagated in space at a velocity exceeding that of light in free space:

$$l'_{12} \leqslant c(t'_2 - t'_1)$$

5.4.10

In the case of events related by a spacelike interval ($s_{12}^2 < 0$), the sign of t'_{12} is relative: $t'_{12} > 0$ (upper part of hyperbola III in Figure 5.3) in certain inertial frames K' whereas in others $t'_{12} < 0$ (lower part of hyperbola III). Point C corresponds to the frame K' in which $t'_{12} = 0$, i.e. events 1 and 2 occur simultaneously.

5.5 Transformation of Velocities and Accelerations in Relativistic Kinematics

5.5.1

The velocities $\mathbf{v}$ and $\mathbf{v}'$ of a particle in two inertial frames of reference K and K' are

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = v_x\mathbf{i} + \mathbf{v}_y\mathbf{j} + \mathbf{v}_z\mathbf{k}, \text{ and}$$

$$\mathbf{v}' = \frac{\mathrm{d}\mathbf{r}'}{\mathrm{d}t'} = v'_{x'}\mathbf{i}' + \mathbf{v}'_{y'}\mathbf{j}' + \mathbf{v}'_{z'}\mathbf{k}'$$

where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $\mathbf{r}' = x'\mathbf{i}' + y'\mathbf{j}' + z'\mathbf{k}'$ are the radius vectors of the particle in frames K and K'. The projections of velocities $\mathbf{v}$ and $\mathbf{v}'$ on the axes of Cartesian coordinates of frames K and K' are:

$$v_x = \frac{dx}{dt}, \qquad v_y = \frac{dy}{dt}, \qquad v_z = \frac{dz}{dt}$$
$$v'_{x'} = \frac{dx'}{dt'}, \qquad v'_{y'} = \frac{dy'}{dt'}, \qquad v'_{z'} = \frac{dz'}{dt'}$$

If the like axes of the Cartesian coordinates of frames K' and K are pairwise parallel, and frame K' travels with respect to frame K at the constant velocity V along axis OX (Figure 2.3, see (2.8.1)) and, in addition, if at the zero time references in frames K and K' ($t = 0$ and $t' = 0$), the origins O and O' of coordinates of these frames coincide, the Lorentz transformations in the form given in (5.3.2) are valid. It follows from these transformations that the relations between the projections of the velocities of the particle on the axes of the Cartesian coordinates of frames K and K' are of the form:

$$v_x = \frac{dx}{dt}, \qquad v_y = \frac{dy}{dt}, \qquad v_z = \frac{dz}{dt}$$
$$v'_{x'} = \frac{dx'}{dt'}, \qquad v'_{y'} = \frac{dy'}{dt'}, \qquad v'_{z'} = \frac{dz'}{dt'}$$

These equations express the *law of the composition of velocities in relativistic kinematics*. In the limit as $c \to \infty$ these equations lead to the ordinary law for the composition of velocities in classical mechanics (2.8.2):

$$v'_{x'} = v_x - V, \quad v'_{y'} = v_y, \quad v'_{z'} = v_z, \text{ and}$$
$$\mathbf{v}' = \mathbf{v} - \mathbf{V}$$

5.5.2

The relations between the squares of the magnitudes of vectors **v** and **v′** are

$$v^2 = c^2\left[1 - \frac{\left[1-(v'/c)^2\right]\left(1-(V/c)^2\right)}{\left(1+Vv'_{x'}/c^2\right)^2}\right]$$

$$(v')^2 = c^2\left[1 - \frac{\left[1-(v/c)^2\right]\left(1-(V/c)^2\right)}{\left(1-Vv_x/c^2\right)^2}\right]$$

In particular, if $v' = c$, then $v = c$ and vice versa. Hence, if the velocity of a particle with respect to some inertial frame of reference is equal to the velocity of light in free space, then it must have the same magnitude with respect to any other inertial frame regardless of the velocity of relative motion of these two frames. In other words, the sum of two velocities, of which one is equal to c, is always equal to c. This law, detected in the motion of such elementary particles as photons (**??**) and the neutrino (Table **??**), demonstrates the limiting nature of the velocity of light in free space (5.1.3).

5.5.3

It is evident from the equations in the preceding subsection that a particle, travelling with respect to some inertial frame of reference at a velocity less than c, has a velocity with respect to any other inertial frame that is also less than c (for instance, if $v < c$, then $v' < c$, and vice versa). From this it follows, among other things, that no matter how close to c the velocities of two particles are, their relative velocity is always less than c. Assume, for

example, that two particles are travelling toward each other along the axis OX of frame K at the respective velocities: $\mathbf{v}_1 = 0.8c\mathbf{i}$ and $\mathbf{v}_2 = -0.8c\mathbf{i}$. The velocity $\mathbf{u}_{21}$ of the second particle with respect to the first is not equal, as is assumed in classical mechanics, to the vector difference $\mathbf{v}_2 - \mathbf{v}_1 = -1.6c\mathbf{i}$ if only because the magnitude of this velocity exceeds c. The required velocity is equal to the velocity of the second particle with respect to inertial frame K', travelling together with the first particle ($\mathbf{V} = 0.8c\mathbf{i}$), i.e. $\mathbf{u}_{21} = \mathbf{v}'_2$. It follows from the equation in (5.5.1) that

$$v'_{2x'} = \frac{v_{2x} - V}{1 - Vv_{2x}/c^2} = -\frac{1.6c}{1 + 0.64} = 0.976c,$$
$$v'_{2y'} = v'_{2z'} = 0$$

i.e. $\mathbf{u}_{21} = -0.976c\mathbf{i}'$ and $|\mathbf{u}_{21}| < c$.

5.5.4

The projections of the acceleration of a particle on the axes of the Cartesian coordinates of two inertial frames of reference K and K' (5.5.1) are related by the following

equations:

$$a'_{x'} = \frac{\mathrm{d}v'_{x'}}{\mathrm{d}t'} = a_x \left(\frac{\sqrt{1-(V/c)^2}}{1-\dfrac{Vv_x}{c^2}} \right)^3,$$

$$a'_{y'} = \frac{\mathrm{d}v'_{y'}}{\mathrm{d}t'} = \left[\left(1-\frac{Vv_x}{c^2} \right) a_y + \frac{Vv_y}{c^2} a_x \right] \frac{1-(V/c)^2}{\left(1-\dfrac{Vv_x}{c^2} \right)^3},$$

$$a'_{z'} = \frac{\mathrm{d}v'_{z'}}{\mathrm{d}t'} = \left[\left(1-\frac{Vv_x}{c^2} \right) a_z + \frac{Vv_z}{c^2} a_x \right] \frac{1-(V/c)^2}{\left(1-\dfrac{Vv_x}{c^2} \right)^3},$$

$$a_x = \frac{\mathrm{d}v_x}{\mathrm{d}t} = a'_{x'} \left(\frac{\sqrt{1-(V/c)^2}}{1+\dfrac{Vv'_{x'}}{c^2}} \right)^3,$$

$$a_y = \frac{\mathrm{d}v_y}{\mathrm{d}t} = \left[\left(1+\frac{Vv'_{x'}}{c^2} \right) a'_{y'} + \frac{Vv'_{y'}}{c^2} a'_{x'} \right] \frac{1-(V/c)^2}{\left(1+\dfrac{Vv'_{x'}}{c^2} \right)^3},$$

$$a_z = \frac{\mathrm{d}v_z}{\mathrm{d}t} = \left[\left(1-\frac{Vv'_{x'}}{c^2} \right) a'_{z'} + \frac{Vv'_{z'}}{c^2} a'_{x'} \right] \frac{1-(V/c)^2}{\left(1+\dfrac{Vv'_{x'}}{c^2} \right)^3},$$

5.6 Basic Law of Relativistic Dynamics

5.6.1

In contrast to classical mechanics, the mass of a particle is not constant in relativistic mechanics. It depends upon the velocity **v** of the particle. Its value m differs in two inertial frames of reference moving with respect to each other. The *dependence of mass on velocity* is expressed by the equation:

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

where m_0 is the *rest mass* of the particle, i.e. its mass measured in the inertial frame of reference with respect to which the particle is at rest, and c is the velocity of light in free space. The mass m is frequently called the *relativistic mass*.

The effect of the velocity of a particle on its mass becomes appreciable only at values of v close to c. For example, $m/m_0 = 1.005$ at $v/c = 0.1$ and $m/m_0 = 2.29$ at $v/c = 0.9$. It is evident from the law for the dependence of m on v that particles with a rest mass $m_0 \neq 0$ cannot travel at velocities greater than or equal to c (i.e. $v < c$). At the same time, particles whose rest mass equals zero [photons (**??**) and the neutrino (Table **??**)] cannot have a velocity other than c.

5.6.2

The momentum $\mathbf{p} = m\mathbf{v}$ of a particle is a nonlinear function of its velocity:

$$p = \frac{m_0\mathbf{v}}{\sqrt{1 - v^2/c^2}}$$

Vector $\mathbf{p}$ is sometimes called the *relativistic momentum* of the particle (to distinguish it from the value $m_0\mathbf{v}$ of its momentum in classical mechanics). It is obvious that at $v \ll c$, the momentum $\mathbf{p} = m\mathbf{v} \approx m_0\mathbf{v}$.

By virtue of the homogeneity of space (2.7.1), the *law of conservation of relativistic momentum* is valid in relativistic mechanics: the momentum of a closed system (2.2.4) does not change in the course of time.

Following from this law is the *law of conservation of relativistic mass*: in any processes that take place in a closed system, its total relativistic mass remains unchanged.

5.6.3

The *basic law of relativistic dynamics* is: the rate of change in the momentum of a particle is equal to the force $\mathbf{F}$ acting on it, i.e.

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = \mathbf{F}, \text{ or } \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{m_0\mathbf{v}}{\sqrt{1 - v^2/c^2}}\right) = \mathbf{F}$$

Note. If the particle is subject to the simultaneous action of several forces, $\mathbf{F}$ is understood to be the resultant force (2.2.2).

5.6.4

An element of work done by force $\mathbf{F}$ in the infinitely small displacement $\mathrm{d}\mathbf{r}$ of its point of application is

$$\delta W = (\mathbf{F}\,\mathrm{d}\mathbf{r}) = (\mathbf{F}\mathbf{v})\,\mathrm{d}t$$

Combining the basic law of relativistic dynamics (5.6.3) and the dependence of mass on the velocity (5.6.1) we obtain

$$\mathbf{F} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} + \mathbf{v}\frac{\mathrm{d}m}{\mathrm{d}t} \text{ and } \frac{\mathrm{d}m}{\mathrm{d}t} = \frac{mv}{c^2 - v^2}\frac{\mathrm{d}v}{\mathrm{d}t}$$

Hence

$$\begin{aligned}\delta W = (\mathbf{F}\mathbf{v})\,\mathrm{d}t &= m(\mathbf{v}\,\mathrm{d}\mathbf{v}) + v^2\,\mathrm{d}m \\ &= mv\,\mathrm{d}v + v^2\,\mathrm{d}m = c^2\,\mathrm{d}m\end{aligned}$$

5.6.5

The acceleration imparted to a particle by the force $\mathbf{F}$ is

$$\mathbf{a} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\mathbf{F}}{m} - \frac{\mathbf{v}}{m}\frac{\mathrm{d}m}{\mathrm{d}t} = \frac{1}{m}\left[\mathbf{F} - \frac{\mathbf{v}}{c^2}(\mathbf{F}\mathbf{v})\right]$$

Hence, in contrast to classical mechanics, the acceleration of a particle in relativistic mechanics does not, in general, coincide in direction with the force imparting the acceleration. Vector $\mathbf{a}$ is collinear with force $\mathbf{F}$ in only two cases:

(a) when force $\mathbf{F}$ is perpendicular to the velocity $\mathbf{v}$ of the particle (transverse force) so that $(\mathbf{F}\mathbf{v}) = 0$ and

$$\mathbf{a} = \frac{\mathbf{F}}{m} = \frac{\mathbf{F}}{m_0}\sqrt{1 - \frac{v^2}{c^2}}$$

(b) when force $\mathbf{F}$ is parallel to the velocity vector $\mathbf{v}$ of the particle (longitudinal force) so that $\mathbf{v}(\mathbf{F}\mathbf{v}) = v^2\mathbf{F}$ and

$$\mathbf{a} = \frac{\mathbf{F}}{m}\left(1 - \frac{v^2}{c^2}\right) = \frac{\mathbf{F}}{m_0}\left(1 - \frac{v^2}{c^2}\right)^{3/2}$$

A longitudinal force imparts to a particle an acceleration less by a factor of $1 - v^2/c^2$ than that imparted by a transverse force of the same magnitude. This is due to the fact that a transverse force changes the velocity of a particle only in direction (the magnitude v of the velocity and the relativistic mass are not changed), whereas a longitudinal force changes the magnitude of the velocity of a particle as well as its mass.

5.7 Mass-Energy Relation

5.7.1

The increment of the kinetic energy E_k of a particle is equal to the work done by the force $\mathbf{F}$ acting on the particle (5.6.4):

$$\mathrm{d}E_k = \delta W = c^2 \mathrm{d}m$$

where $\mathrm{d}m$ is the corresponding increment of relativistic mass of the particle. It follows that

$$E_k = (m - m_0)c^2 = m_0 c^2 \left[\frac{1}{\sqrt{1 - v^2/c^2}} - 1\right]$$

where m_0 is the rest mass of the particle. Expanding $(1 - v^2/c^2)^{-1/2}$ into a Maclaurin series we obtain

$$E_k = m_0 c^2 \left[\frac{1}{2} \left(\frac{v}{c} \right)^2 + \frac{3}{8} \left(\frac{v}{c} \right)^4 + \dots \right]$$

At $v \ll c$ this equation reduces to the ordinary expression for the kinetic energy in classical mechanics:

$$E_k = \frac{m_0 v^2}{2} = \frac{m v^2}{2}$$

5.7.2

It follows from the first equation in the preceding subsection that an increase in the kinetic energy of a body should be accompanied by a corresponding increase in its relativistic mass m:

$$\mathrm{d}m = \frac{1}{c^2} \mathrm{d}E_k$$

Changes in other kinds of energy of a body are also associated with an increase in its relativistic mass. If, for example, in heating a body at rest its internal energy (9.1.2) is increased by the amount $\mathrm{d}U$, the mass m of the body, equal to its rest mass m_0, is increased by

$$\mathrm{d}m = \mathrm{d}m_0 = \frac{1}{c^2} \mathrm{d}U$$

In the general case, a change in the total energy E of the body by the amount $\mathrm{d}E$ is accompanied by a change in its relativistic mass by the amount

$$\mathrm{d}m = \frac{1}{c^2} \mathrm{d}E$$

Accordingly, the following universal relation exists between E and m:

$$E = mc^2 = \frac{m_0 c^2}{\sqrt{1 - \dfrac{v^2}{c^2}}}$$

this is known as the *mass-energy relation*: the total energy of a body (or system) is equal to the product of the relativistic mass of the body (or system) by the square of the velocity of light in free space.

By virtue of the uniformity of time (3.4.2),the *law of the conservation of energy* is valid in relativistic mechanics as it is in classical mechanics: the total energy of a closed system (2.2.4) does not change in the course of time.

It follows from the mass-energy relation that the laws of the conservation of relativistic mass (5.6.2) and of the total energy are not independent laws.

5.7.3

The total energy of a particle or a system of particles (for instance, of an atomic nucleus, atom, molecule or body) at rest is equal to

$$E_0 = m_0 c^2$$

where m_0 is the rest mass; E_0 is called the *rest energy* of the particle or system. The values of m_0 and E_0 are independent of the choice of the inertial frame of reference. These quantities are constant characteristics of structureless (elementary) particles, similar, for example, to their electric charge and spin (Table **??**). The rest mass and rest energy of a system of particles depend

upon the composition of the system and its internal state. For example, the rest mass of an excited nucleus (or atom) is greater than the rest mass of the same nucleus (or atom) in the ground state.

5.7.4

The total energy E of a particle and its momentum $\mathbf{p}$ are related by the equations:

$$\mathbf{p} = \frac{E}{c^2}\mathbf{v} \text{ or } \frac{E^2}{c^2} - p^2 = m_0^2 c^2 \text{ or } E = \sqrt{p^2 c^2 + m_0^2 c^4}$$

The values of the total energy, relativistic mass and momentum of a given particle, in contrast to its rest mass m_0, are relative, i.e. they are different in two inertial frames of reference K (E, m and $\mathbf{p}$) and K' (E', m' and $\mathbf{p}'$). But the difference between the square of the total energy of a particle divided by c^2 and the square of the momentum of the particle, like the interval between two events (5.4.8), does not depend upon the choice of the inertial frame of reference:

$$\frac{(E')^2}{c^2} - (p')^2 = \frac{E^2}{c^2} - p^2 = m_0^2 c^2$$

5.7.5

In going over from one inertial frame K to another K' that is travelling with respect to the first at the velocity $\mathbf{V} = \text{const}$ along the OX axis (Figure 2.3, see (2.8.1)), the projections of the momentum of the particle on the coordinate axes and its total energy are transformed as

follows:

$$p'_{x'} = \frac{p_x - \frac{VE}{c^2}}{\sqrt{1 - V^2/c^2}}, \qquad p_x = \frac{p'_{x'} - \frac{VE'}{c^2}}{\sqrt{1 - V^2/c^2}}$$
$$p'_{y'} = p_y, \qquad p_y = p'_{y'}$$
$$p'_{z'} = p_z, \qquad p_z = p'_{z'}$$
$$E' = \frac{E - Vp_x}{\sqrt{1 - V^2/c^2}}, \qquad E = \frac{E' + Vp'_{x'}}{\sqrt{1 - V^2/c^2}}$$

5.7.6

It does not at all follow from the laws of conservation of relativistic mass (5.6.2) and of total energy (5.7.2) that the rest mass and rest energy of a closed system cannot change. For instance, the sum of the rest masses of free protons and neutrons is always greater than the rest mass of the atomic nucleus they form.

A quantity introduced to characterize systems held together with a margin of safety (for example, atomic nuclei, atoms and molecules) is the concept of the binding energy. The *binding energy of a system* is measured as the least amount of work that must be done to separate the system into its component parts (for instance, an atom into its nucleus and electrons). The binding energy of a system is

$$E_b = \sum_{i=1}^{n} m_{0i}c^2 - M_0c^2$$

where m_0 is the rest mass of a system of n particles, and

m_{0i} is the rest mass of the i-th. particle in the free state. The quantity

$$\Delta m = \sum_{i=1}^{n} m_{0i} - M_0 = \frac{E_b}{c^2}$$

is sometimes called the *mass defect of the system*.

6 Gravitation

6.1 Law of Universal

6.1.1

Isaac Newton's law of universal gravitation states that the force of mutual attraction between any two particles is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. Such forces are called *gravitational forces*. If m_1 and m_2 are the masses of the particles being considered and $\mathbf{r}_1$ and $\mathbf{r}_2$ are the radius vectors of these particles, then, according to the law of universal gravitation, the force $\mathbf{F}_{12}$ attracting the first particle toward the second particle, and the force $\mathbf{F}_{21}$ attracting the second particle toward the first particle are equal in magnitude and opposite in direction; they are equal to:

$$\mathbf{F}_{12} = -G\frac{m_1 m_2}{r_{12}^3}\mathbf{r}_{12} \text{ and } \mathbf{F}_{21} = -G\frac{m_1 m_2}{r_{21}^3}\mathbf{r}_{21}$$

Here $\mathbf{r}_{12} = \mathbf{r}_1 - \mathbf{r}_2$ and $\mathbf{r}_{21} = \mathbf{r}_2 - \mathbf{r}_1$ are radius vectors from the second particle to the first and from the first to

the second, respectively, and $r_{12} = r_{21} = |\mathbf{r}_{12}| = |\mathbf{r}_{21}|$ is the distance between the particles. The proportionality factor G is called the *constant of universal gravitation*, or the *gravitational constant*. In value the gravitational constant is equal to the force of mutual attraction of two particles of unit mass with a distance of unit length between them. It has been found from experiments that

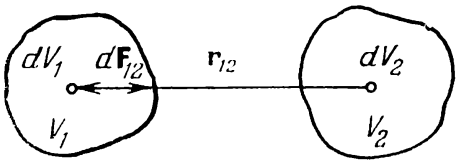


Figure 6.1: The gravitational interaction between two bodies of arbitrary size and shape.

$$G = (6.6720 \pm 0.0041) \times 10^{-11}\,\mathrm{N m^2/kg^2}$$

6.1.2

The gravitational interaction of two bodies of arbitrary size and shape (Figure 6.1) is described by the equation

$$\mathbf{F}_{12} = -G \int_{(V_1} \rho_1 \mathrm{d}V_1 \int_{(V_2} \frac{\rho_2}{r_{12}^3} \mathbf{r}_{12} \mathrm{d}V_2$$

where $\mathbf{r}_{12}$ is the radius vector from the infinitely small element $\mathrm{d}V_2$ of volume of the second body to the infinitely small element $\mathrm{d}V_2$ of volume of the first body, ρ_1 and ρ_2 are the densities of the two elements, and integration is carried out throughout the whole volume of both bodies.

The calculations of force $\mathbf{F}_{12}$ are substantially simplified in two cases:

(a) when there is a spherically symmetrical distribution of the masses in the two bodies, i.e. both bodies are of spherical shape and the density of each body depends only upon the distance to its centre (in particular, the bodies can be homogeneous);

(b) when one of the bodies is of negligibly small size with respect to the other body in which the distribution of its mass is spherically symmetrical.

In these cases

$$\mathbf{F}_{12} = -G\frac{m_1 m_2}{r_{12}^3}\mathbf{r}_{12}$$

where m_1 and m_2 are the masses of the bodies, and $\mathbf{r}_{12}$ is the radius vector connecting the centres of mass (2.3.3) of the second and first bodies.

6.1.3

As a first approximation, it can be assumed that the earth is of spherical shape and the distribution of its mass is spherical ly symmetrical. Hence the gravitational force $\mathbf{F}$, exerted by the earth on a body of mass m, is directed toward the centre of the earth and its magnitude is

$$F = G\frac{mM_\oplus}{r^2}$$

where $M_\oplus$ is the mass of the earth and r is the distance from the body to the earth's centre (the size of any body on earth is negligibly small compared to the radius of the earth).

6.1.4

In application to such microscopic objects as elementary particles, gravitational interaction is practically of no importance whatsoever. It is found to be extremely weak in comparison to all other kinds of interaction: strong, electromagnetic and weak interaction ((**??**) through (**??**)). For example, the electric force of mutual repulsion of two electrons exceeds their gravitational force by over 10^{42} times! Even for ordinary macroscopic objects on the earth the forces of gravitational inter action are extremely small. Two homogeneous spheres, for instance, with a mass of 1000 kg each and a distance of one metre between their centres are attracted to each other with a force of only 7×10^{-5} N.

But, at the same time, gravitational forces are the determining factors in the motion of objects that are investigated in astronomy and astronautics (spaceships, planets and their satellites, planetary systems, stars, etc.). This is due, primarily, to the huge size of astronomical bodies and, in the second place, to the small electromagnetic forces of interaction of such bodies, which, as a whole, are practically electroneutral.

6.2 Gravitational Field

6.2.1

Gravitational interaction between bodies is due to the *gravitational field* set up by the bodies. A distinctive feature of a gravitational field is that a particle placed in this field is subject to a force proportional to the mass of

the particle.

The force characteristic of a gravitational field is its *intensity*, the vector quantity $\mathbf{g}_f$, which is equal to the ratio of the force $\mathbf{F}$, exerted on the particle by the field in which it is placed, to the mass m of the particle:

$$\mathbf{g}_f = \frac{\mathbf{F}}{m}$$

The intensity of a gravitational field is independent of the mass m of the particle. It is a function of the coordinates (x, y, z) of points of the field being considered. In the case of an unsteady field the intensity also depends upon the time t.

A gravitational field is steady, or stationary (2.2.1), if the body that sets it up is fixed in the frame of reference chosen to describe the field. The intensity of a steady gravitational field depends only upon the coordinates: $\mathbf{g}_f = \mathbf{g}_f(x, y, z)$.

It follows from Newton's second law (2.4.3) that the forces of ;i gravitational field impart an acceleration $\mathbf{a}$ to a free particle in the field. This acceleration equals the intensity of the field:

$$\mathbf{a} = \frac{\mathbf{F}}{m} = \mathbf{g}_f$$

6.2.2

It follows from the law of universal gravitation (6.1.1) that the intensity of the gravitational field of a fixed particle of mass M, located at the origin of coordinates, equals

$$\mathbf{g}_f = -G\frac{M}{r^3}\mathbf{r}$$

where **r** is the radius vector of the point being considered in the field.

A gravitational field is a conservative, or potential, one (3.1.6) because the force acting on a particle of mass m brought into the field is a central force (3.3.4):

$$\mathbf{F} = m\mathbf{g}_f = -G\frac{mM}{r^2}\frac{\mathbf{r}}{r}$$

Correspondingly, the potential energy of a particle in such a field is equal to (3.3.4):[1]

$$E_p = \int_r^\infty F_r \mathrm{d}r = -GmM\int_r^\infty \frac{\mathrm{d}r}{r^2} = -G\frac{mM}{r}$$

It is equally correct to regard the quantity E_p as the potential energy of a particle of mass M in a gravitational field set up by a particle of mass m or, finally, as the *mutual potential energy of two particles* due to their gravitational interaction.

6.2.3

A gravitational field complies with the *principle of superposition of fields*: in superposing several (n) gravitational fields their intensities at each point in space are added vectorially, i.e. the intensity of the resultant field is

$$\mathbf{g}_f = \sum_{i=1}^{n} \mathbf{g}_{fi}$$

[1] Here and throughout (Section 6.2) and (Section 6.3) an infinitely distant point is chosen as the reference point for the potential energy, i.e. it is assumed that $E_p(\infty) = 0$.

where $\mathbf{g}_{fi}$ is the intensity of only the i-th field at the point being considered in space.

The intensity of the gravitational field of an arbitrary system consisting of n fixed particles is

$$\mathbf{g}_f = -G\sum_{i=1}^{n}\frac{m_i}{\rho_i^3}\boldsymbol{\rho}_i$$

where $\boldsymbol{\rho}_i = \mathbf{r} - \mathbf{r}_i$ is the radius vector from the i-th particle, whose radius vector is equal to $\mathbf{r}_i$, to the point in the field specified by radius vector $\mathbf{r}$. Consequently, the potential energy of a particle of mass m in this gravitational field is

$$E_p = -Gm\sum_{i=1}^{n}\frac{m_i}{\rho_i}$$

In particular, if the gravitational field is set up by a body whose mass M is distributed in a spherically symmetrical manner (6.1.2), then outside this body

$$\mathbf{g}_f = -G\frac{M}{r^3}\mathbf{r} \quad \text{and} \quad E_p = -\frac{GmM}{r}$$

where $\mathbf{r}$ is a radius vector from the centre of the body to the point being considered in the field. These equations are valid, for example, for the gravitational field of the earth.

6.2.4

By virtue of the conservative nature (potentiality) of the gravitational field (3.1.6), its energy characteristic, the

potential, ran be introduced. The *potential of a gravitational field* is the scalar φ, equal to the ratio of the potential energy $E_p$of a particle, placed at the point being considered in the field to the mass m of the particle:

$$\varphi = \frac{E_p}{m}$$

The potential φ is independent of the mass m of the particle; it is a function of the coordinates of points of the gravitational, held. For example, the potential of a gravitational field set up by a fixed particle of mass M is

$$\varphi = -\frac{GM}{r}$$

where r is the distance from the source of the field to the point being considered.

The potential of a gravitational field set up by an arbitrary system of n fixed particles is

$$\varphi = -\sum_{i=1}^{n} G\frac{m_i}{\rho_i}$$

where ρ_i is the distance from a particle of mass m_i to the point being considered in the field. Thus, in superposing gravitational fields, their potentials are added together algebraically, i.e. the potential φ at any point of a resultant field is equal to the algebraic sum of the potentials at the same point for all the superposed fields taken separately;

$$\varphi = -\sum_{i=1}^{n} \varphi_i$$

Note. In applying this equation it is necessary that the same reference point be chosen for the potentials φ_i of all the fields being superposed: $\varphi_i(\infty) = 0$ (see the footnote to (6.2.2)).

6.2.5

An element of work done by the forces of a gravitational field in an infinitely small displacement $\mathrm{d}r$ of a particle of mass m in this field is:

$$\mathrm{d}W = (\mathbf{F}\,\mathrm{d}\mathbf{r}) = m(\mathbf{g}_f\,\mathrm{d}\mathbf{r})$$

But, on the other hand, the work $\mathrm{d}W$ is equal to the decrease in the potential energy of the particle in the gravitational field:

$$\mathrm{d}W = -\mathrm{d}E_p = -m\mathrm{d}\varphi$$

Consequently, the potential and intensity of a gravitational field are related by the equation:

$$\mathrm{d}\varphi = -(\mathbf{g}_f\,\mathrm{d}\mathbf{r}) = -(g_{fx}\mathrm{d}x + g_{fy}\mathrm{d}y + g_{fz}\mathrm{d}z)$$

where g_{fx}, g_{fy} and g_{fz} are the projections of vector $\mathbf{g}_f$ on the axes of rectangular Cartesian coordinates. Since

$$\mathrm{d}\varphi = \frac{\partial\varphi}{\partial x}\mathrm{d}x + \frac{\partial\varphi}{\partial y}\mathrm{d}y + \frac{\partial\varphi}{\partial z}\mathrm{d}z$$

then

$$\frac{\partial\varphi}{\partial x} = -g_{fx}, \quad \frac{\partial\varphi}{\partial y} = -g_{fy}, \quad \frac{\partial\varphi}{\partial z} = -g_{fz},$$

Hence

$$\mathbf{g}_f = -\left(\frac{\partial\varphi}{\partial x}\mathbf{i} + \frac{\partial\varphi}{\partial y}\mathbf{j} + \frac{\partial\varphi}{\partial z}\mathbf{k} + \right) = -\operatorname{grad}\varphi = -\nabla\varphi$$

i.e. the intensity of a gravitational field is numerically equal and opposite in direction to the gradient of the potential of the field.

The relation between (p and gf can also be written in the form

$$\mathrm{d}\varphi = -\mathbf{g}_f \mathrm{d}l \cos\alpha = -g_{fl}\mathrm{d}l \quad \text{or} \quad g_{fl} = -\frac{\mathrm{d}\varphi}{\mathrm{d}l}$$

where α is the angle between vectors $\mathbf{g}_f$ and $\mathrm{d}\mathbf{r}$, $\mathrm{d}l = |\mathrm{d}\mathbf{r}|$, and g_{fl} is the projection of vector $\mathbf{g}_f$ onto the direction of vector $\mathrm{d}\mathbf{r}$. Thus, the projection of the intensity vector of a gravitational field onto some direction is numerically equal and opposite in sign to the change in the field potential per unit length in the same direction.

6.2.6

The non-relativistic theory of gravitation discussed above end based on Newton's law of universal gravitation is only approximate. It describes with sufficient accuracy only comparatively weak gravitational fields, having a potential $|\varphi| \ll c^2$, where $c^2 = 9 \times 10^{16}\,\mathrm{m}^2/\mathrm{s}^2$ is the square of the velocity of light in free space. It is, in particular, applicable to the gravitational lieIds of the earth and sun because the absolute values of the potentials of these fields at the surfaces of the earth and sun are equal to 6.3×10^7 and $1.9 \times 10^{11}\,\mathrm{m}^2/\mathrm{s}^2$.

6.2.7

The *modern* (relativistic) *theory of gravitation*, which is m unified theory of space, time and gravitation, was formulated by Albert Einstein. He called it the *general theory of relativity*. The existence of a close relationship between space and time had already been indicated in his special theory of relativity. This relationship is evident in the Lorentz transformations (5.3.2) and the invariance of the interval between two events (5.4.8). It was found necessary, in describing physical processes, l.o employ a four-dimensional space-time continuum, in which the position of a point is specified by three space coordinates mid the time coordinate ct.

According to the relativistic theory of gravitation, the geometrical properties (metric) of the space-time continuum depend upon the distribution of gravitating masses and their motion in space. Bodies setting up a gravitational field "distort" real three-dimensional space and differently change the passage of time at various points in this space. This means that the bodies cause a deviation of the metric of space-time from that of the "flat" space-time described by Euclidean geometry and dealt with in the special theory of relativity. Therefore, the motion of a body in a gravitational field can be regarded as inertial motion, but in a "distorted" (non-Euclidean) space-time continuum. Accordingly, a particle, subject to the action of a gravitational field, travels neither in a straight line nor at uniform velocity in a real three-dimensional space.

It was shown in the relativistic theory of gravitation that the principle of superposition (6.2.3) does not hold for arbitrary gravitational fields. This principle, as well

as the whole non-relativistic theory of gravitation, is sufficiently accurate only in weak fields ($|\varphi| \ll c^2$) and for motion in these fields at low velocities $v \ll c$.

6.3 Kepler's Laws. Space Velocities

6.3.1

The motion of the planets of the solar system along their orbits around the sun complies with the three laws of Kepler. These laws can be derived from Newton's law of universal gravitation, dealing, as a first approximation, with the sun and planets as particles. In the central force field of gravitation of the sun, a planet of mass m is subject to the action of the gravitational force

$$\mathbf{F} = -G\frac{mM_\odot}{r^3}\mathbf{r}$$

where $M_\odot$ is the mass of the sun and $\mathbf{r}$ is the radius vector of the planet from the centre of forces O, which is taken as the origin of coordinates.

The moment of force $\mathbf{F}$ about the centre of forces is $\mathbf{M} = [\mathbf{r}\mathbf{F}] \equiv 0$. Hence, the angular momentum $\mathbf{L}$ of the planet about the same point O does not change with time (4.3.1):

$$\mathbf{L} = [\mathbf{r}\, m\mathbf{v}] = \text{const}$$

Consequently, the planet travels along a plane path, (orbit), whose plane is perpendicular to vector $\mathbf{L}$. According to (1.3.6), $\mathbf{L} = \left[\mathbf{r}\, m\mathbf{v}_\varphi\right]$, where $\mathbf{v}_\varphi$ the transverse velocity of the planet. Therefore, the orbital motion of the planet

satisfies the condition

$$r^2 \frac{\mathrm{d}\varphi}{\mathrm{d}t} = \frac{L}{m} = \text{const}$$

where r and φ are the polar coordinates of the planet.

A second condition is imposed by the law of conservation of mechanical energy: $E_k + E_p = E = \text{const}$. According to (1.3.6) and (6.2.2):

$$E_k = \frac{mv^2}{2} = \frac{m}{2}\left[\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + \left(r\frac{\mathrm{d}\varphi}{\mathrm{d}t}\right)^2\right]$$
$$= \frac{m}{2}\left[\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + \left(\frac{L}{mr}\right)^2\right]$$

and

$$E_p = -G\frac{mM_\odot}{r}$$

so that the second condition is of the form

$$\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + \left(\frac{L}{mr}\right)^2 - \frac{2GM_\odot}{r} = \frac{2E}{m}$$

6.3.2

The equation of the path of the planet (in the polar coordinates r and φ) is

$$r = \frac{p}{1 + e\cos\varphi}$$

where $p = L^2/Gm^2M_\odot$ and $e = \sqrt{(2EL^2/G^2m^3M_\odot^2) + 1}$. The total mechanical energy of the planet $E < 0$, so that $e < 1$ and the path (orbit) of the planet is an ellipse.

Kepler's first law states that all the planets of the solar system travel along elliptic orbits of which the sun occupies one focus. It follows from the first condition (6.3.1) that the areal velocity (1.3.6) of the planet is constant:

$$\sigma = \frac{1}{2} r^2 \frac{\mathrm{d}\varphi}{\mathrm{d}t} = \frac{L}{2m} = \text{const}$$

Kepler's second law states that the radius vector of a planet sweeps out equal areas in equal lengths of time.

6.3.3

According to Kepler's second law, the orbital period, or period of revolution, T of a planet about the sun is equal to the ratio of the area A of the orbit to the areal velocity σ:

$$T = \frac{A}{\sigma} = \frac{\pi ab}{\sigma} = \text{const}$$

where $a = p/(1 - e^2)$ and $b = a\sqrt{1 - e^2}$ are the major and minor semi-axes of the elliptic orbit. Thus

$$T^2 = \frac{\pi^2 p}{L^2/4m^2} a^3 = \frac{4\pi^2}{GM_\odot} a^3 = \text{const}$$

This equation is an expression of *Kepler's third law*: the squares of the periods of revolution of the planets about the sun are proportional to the cubes of the major semi-axes of the elliptic orbits of these planets.

6.3.4

The *orbital velocity* is the minimum velocity that it is necessary to impart to a body to convert it into an artificial

satellite of the earth. It is also called the *circular orbital velocity* because it is equal to the velocity of an artificial satellite in circular orbit about the earth in the absence of air friction of the atmosphere. The orbital velocity is

$$v_1 = \sqrt{\frac{GM_\oplus}{r}}$$

where $GM_\oplus$ is the mass of the earth, and r is the radius of the circular orbit. At the earth's surface $v_1 = 7.9\,\mathrm{km/s}$.

6.3.5

The *escape velocity* is the minimum velocity that it is necessary to impart to a body to enable it, without any additional forces, to overcome the earth's gravitational attraction and become an artificial satellite of the sun. It is also called the *parabolic velocity* because it launches the body on a parabolic orbit in the earth's gravitational field (in the absence of air friction of the atmosphere). The escape velocity

$$v_2 = \sqrt{\frac{2GM_\oplus}{r}}$$

where r is the radius from the launching site of the body to the centre of the earth. At the earth's surface $v_2 = 11.2\,\mathrm{km/s}$.

6.3.6

The *solar escape velocity* is the minimum velocity that it is necessary to impart to a spacecraft, launched at the

earth's surface, to overcome the sun's gravitational attraction and leave the solar system. This velocity is $v_1 = 16.7\,\text{km/s}$.

7 Motion in non-inertial Frames of Reference

7.1 Kinematics of Relative Motion

7.1.1

It is assumed in classical (Newtonian) mechanics that distances and time intervals remain unchanged in going over from one frame of reference to any other frame that is in motion in even the most arbitrary manner with respect to the first frame. For example, assume that K is an inertial frame with origin at point O^* and S is a *non-inertial* frame with its origin at point O (Figure 7.1).

In the general case, the motion of frame S with respect to frame K can be regarded as the sum of two motions: translational motion of point O at the velocity $\mathbf{v}_0$ and rotation about this point at the angular velocity $\mathbf{\Omega}$. Radius vectors $\mathbf{r}^*$ and $\mathbf{r}$ of arbitrary particle M, measured in frames K and S, are related by the equation

$$\mathbf{r}^* = \mathbf{r}_0^* + \mathbf{r}$$

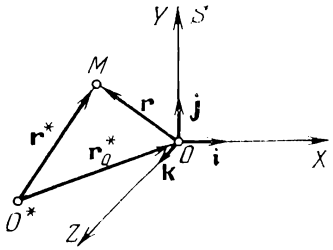


Figure 7.1: Inertial and non-inertial frames of reference.

where $\mathbf{r}_0^*$ is the radius vector of point O measured in frame of reference K.

7.1.2

The motion of particle M with respect to any inertial frame of reference K, conditionally assumed to be fixed, is called the *absolute motion* of particle M. The motion of the same particle with respect to the non-inertial frame of reference S is said to be *relative motion*.

The *relative velocity* $\mathbf{v}_r$ of particle M, i.e. its velocity with respect to frame S, equals

$$\mathbf{v}_r = \frac{\mathrm{d}x}{\mathrm{d}t}\mathbf{i} + \frac{\mathrm{d}y}{\mathrm{d}t}\mathbf{j} + \frac{\mathrm{d}z}{\mathrm{d}t}\mathbf{k}$$

where x, y and z are the Cartesian coordinates of particle M, and $\mathbf{i}$, $\mathbf{k}$ and $\mathbf{k}$ are the unit vectors of the coordinate axes in frame S.

The *absolute velocity* of particle M, i.e. its velocity $\mathbf{v}$ in the frame of reference K, equals

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}^*}{\mathrm{d}t} = \mathbf{v}_0 + x\frac{\mathrm{d}\mathbf{i}}{\mathrm{d}t} + y\frac{\mathrm{d}\mathbf{j}}{\mathrm{d}t} + z\frac{\mathrm{d}\mathbf{k}}{\mathrm{d}t} + \mathbf{v}_r$$

where $\mathbf{v}_0 = \mathrm{d}\mathbf{r}_0^*/\mathrm{d}t$ is the absolute velocity of point O. Since the unit vectors of moving frame S can change in frame of reference K only due to the rotation of frame S about point O at the angular velocity $\boldsymbol{\Omega}$,

$$\frac{\mathrm{d}\mathbf{i}}{\mathrm{d}t} = [\boldsymbol{\Omega}\mathbf{i}], \quad \frac{\mathrm{d}\mathbf{j}}{\mathrm{d}t} = [\boldsymbol{\Omega}\mathbf{j}], \quad \frac{\mathrm{d}\mathbf{k}}{\mathrm{d}t} = [\boldsymbol{\Omega}\mathbf{k}]$$

and

$$\mathbf{v} = \mathbf{v}_t + \mathbf{v}_r$$

where $\mathbf{v}_t = \mathbf{v}_0 + [\boldsymbol{\Omega}\mathbf{k}]$ is the *velocity of transportation* of particle M. It is also called the *bulk*, or *drift*, *velocity* and is equal to the absolute velocity of the point of moving frame of reference S (i.e. rigidly fixed in this frame) in which particle M is located at the given instant of time.

7.1.3

The *relative acceleration* $\mathbf{a}_r$ of particle M (its acceleration with respect to frame S) equals

$$\mathbf{a}_r = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}\mathbf{i} + \frac{\mathrm{d}^2y}{\mathrm{d}t^2}\mathbf{j} + \frac{\mathrm{d}^2z}{\mathrm{d}t^2}\mathbf{k}$$

The *absolute acceleration* of particle M, i.e. its acceleration with respect to frame K, equals

$$\mathbf{a} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{a}_t + \mathbf{a}_C + \mathbf{a}_r$$

Here

$$\mathbf{a}_t = \frac{\mathrm{d}\mathbf{v}_0}{\mathrm{d}t} + \left[\frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t}\mathbf{r}\right] + [\boldsymbol{\Omega}[\boldsymbol{\Omega}\mathbf{r}]]$$

is the *acceleration of transportation* of particle M. It is equal to the absolute acceleration of the point in moving frame S in which particle M is located at the given instant of time, and

$$\mathbf{a}_C = 2[\mathbf{\Omega}\mathbf{v}_r]$$

is the *Coriolis acceleration* of particle M. The Coriolis acceleration has its maximum value if the relative velocity $\mathbf{v}_r$ of the particle is perpendicular to the vector $\mathbf{\Omega}$ of angular velocity of the moving frame of reference. It is equal to zero if the angle between vectors $\mathbf{v}_r$ and $\mathbf{\Omega}$ equals zero or π, or if even one of these vectors equals zero.

7.2 Inertial Forces

7.2.1

Newton's laws of motion are not valid in non-inertial frames of reference. In particular, a particle can change the slate of its motion with respect to non-inertial frame S without any action being exerted on the particle by other bodies. A ball, for instance, suspended by a string from the ceiling of a railway coach of a train travelling at uniform velocity in a straight line, deviates backward when the motion of the train is accelerated and frontward when the motion is decelerated, i.e. it begins to move with respect to a non-inertial frame of reference attached to the coach. But no horizontal forces act on the ball when it deviates.

7.2.2

The fundamental law of dynamics for a particle in non-inertial frames of reference can be derived from Newton's second law and the relation between the absolute and relative accelerations of a particle. It follows from (7.1.3) that the product of the mass m of a particle by its relative acceleration equals

$$m\mathbf{a}_r = m\mathbf{a} - m\mathbf{a}_t - m\mathbf{a}_C$$

According to Newton's second law, formulated to apply to the absolute motion of a particle, i.e. its motion with respect to inertial frame K,

$$m\mathbf{a} = \mathbf{F}$$

where $\mathbf{F}$ is the vector sum of all forces acting on the particle. Hence, the *fundamental equation of dynamics of a particle's relative motion* is of the form:

$$m\mathbf{a}_r = \mathbf{F} - m\mathbf{a}_t - m\mathbf{a}_C$$

This equation can be reduced to a form similar to the fundamental law of dynamics for the absolute motion of a particle:

$$m\mathbf{a}_r = \mathbf{F} + \mathbf{I}_t + \mathbf{I}_C$$

The vector quantities $\mathbf{I}_t = -m\mathbf{a}_t$ and $\mathbf{I}_C = -m\mathbf{a}_C$ have the dimensionality of force and are called the *inertial force of transportation* and the *Coriolis inertial force*, respectively.

7.2.3

It follows from (7.1.3) that in the general case the inertial force of transportation is the sum of three terms:

$$\mathbf{I}_t = -m\frac{\mathrm{d}\mathbf{v}_0}{\mathrm{d}t} - m\left[\frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t}\mathbf{r}\right] - m[\boldsymbol{\Omega}[\boldsymbol{\Omega}\mathbf{r}]]$$

The last term on the right-hand side of this equation

$$\mathbf{I}_{cf} = -m[\boldsymbol{\Omega}[\boldsymbol{\Omega}\mathbf{r}]]$$

is called the *centrifugal force of inertia*, or simply the *centrifugal force*, because this vector is perpendicular to the instantaneous axis of rotation (to vector $\boldsymbol{\Omega}$) of the non-inertial frame S and its direction is away from this axis. Numerically the centrifugal force equals

$$I_{cf} = -m\Omega^2\rho$$

where ρ is the distance from a particle of mass m to the instantaneous axis of rotation of the frame of reference.

The inertial force of transportation coincides with the centrifugal force when the non-inertial frame of reference travels with translational motion at constant velocity ($\mathbf{v}_0 = \text{const}$) and rotates at constant angular velocity ($\boldsymbol{\Omega} = \text{const}$).

7.2.4

The Coriolis inertial force is

$$I_C = 2m[\mathbf{v}_r\boldsymbol{\Omega}]$$

This force acts on the particle only when the non-inertial frame of reference rotates and the particle moves with

respect to this frame. Thus, for example, particles of water in rivers of the Northern Hemisphere that flow meridionally are subject to the action of a Coriolis inertial force. This force is perpendicular to the velocity of flow of the river and leads to under mining the right bank in the direction of flow.

The Coriolis inertial force does no work in the relative motion of a particle because the force is perpendicular to the velocity of relative motion of the particle. Hence, the Coriolis inertial force is an example of the gyroscopic forces (3.1.7).

7.2.5

The action of inertial forces on a particle in a non-inertial frame of reference are real and can be measured in this frame by, for example, a spring dynamometer. But, in contrast to ordinary forces of interaction between bodies, it is impossible lo indicate for inertial forces the action of what specific bodies on the particle being considered they express. This feature of inertial forces is due to the fact that the very occurrence of the vector quantities $\mathbf{I}_t$ and $\mathbf{I}_C$ in the fundamental equation of the dynamics of relative motion results only from the non-inertial nature of the frame of reference used to describe the relative motion of the particle. The addition of the inertial forces $\mathbf{I}_t$ and $\mathbf{I}_C$ to the force $\mathbf{F}$, which characterizes the action on the particle of all the other bodies, enables the fundamental equation of the dynamics of relative motion to be written in a form that resembles the expression for Newton's second law in an inertial frame of reference.

There can be no closed systems of bodies in non-inertial

frames of reference because the inertial forces are always external forces for any of the bodies in the system. Therefore, the laws of conservation of momentum (2.7.1), of angular momentum (4.4.1) and of energy (3.4.2) are not valid in non-inertial frames of reference.

7.3 Relative Motion in a Frame of Reference Fixed to the Earth. Gravity Force and Weight of a Body

7.3.1

A frame of reference fixed in the earth is non-inertial lor two reasons: firstly, due to the diurnal, or daily, rotation of the earth at constant angular velocity $\mathbf{\Omega}$ (where $\Omega = 2\pi$ rad per 24 hours $= 7.3 \times 10^{-5}$ rad/s) and, in the second place, due to the effects of the gravitational field of the sun, moon, planets and other astronomical bodies on the earth. This gravitational field is practically uniform within the limits of the earth and imparts the same acceleration of translational motion to the terrestrial frame of reference and all bodies moving with respect to this frame. This acceleration is $\mathbf{a}_0 = d\mathbf{v}_0/dt = \mathbf{g}_f$, where $\mathbf{g}_f$ is the intensity of the field (6.2.1).

It follows from the equations in (7.2.2) and (7.2.3) that the equation of relative motion of a particle of mass m in a frame of reference fixed to the earth is of the form

$$m\mathbf{a}_r = \mathbf{F} + \mathbf{F}_{gr} + \mathbf{I}_{cf} + \mathbf{I}_C$$

where $\mathbf{I}_{cf}$ and $\mathbf{I}_C$ are the centrifugal and Coriolis inertial forces. $\mathbf{F}_{gr}$ is the gravitation al attraction of the earth

for the particle (6.1.3), and **F** is the sum of all the other forces acting on the particle, with the exception of the gravitational forces.

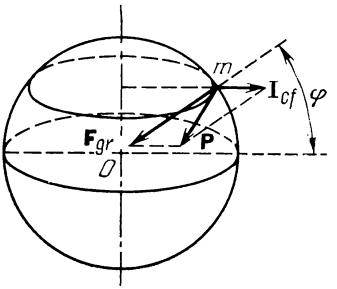


Figure 7.2: The Coriolis force in a non-inertial frame of reference.

7.3.2

The *gravity force of a body* is the force P, applied to the body and equal to the vector sum of the force $\mathbf{F}_{gr}$ of gravitational attraction of the earth for the body (6.1.3) and the centrifugal inertial force $\mathbf{I}_{cf}$ due to the diurnal rotation of the earth (Figure 7.2):

$$\mathbf{P} = \mathbf{F}_{gr} + \mathbf{I}_{cf}$$

i.e.

$$\mathbf{P} = -G\frac{mM_{\oplus}}{r^3}\mathbf{r} - m[\mathbf{\Omega}[\mathbf{\Omega}\mathbf{r}]]$$

Here m and $M_{\oplus}$ are the masses of the body and the earth, **r** is the radius vector drawn from the centre of the earth

to the place where the body is located, $\boldsymbol{\Omega}$ is the angular velocity of diurnal rotation of the earth, and G is the gravitational constant (6.1.1).

The gravity force causes an unsupported or unfastened body to fall to the earth. It equals the force with which the body presses against a horizontal support (or pulls on a vertical suspension) due to the gravitation attraction of the earth. It can be measured in the terrestrial frame of reference by means of a spring dynamometer. The point of application of the gravity force acting on a body, i.e. the point of application of the resultant of all the gravity forces acting on all the particles of the body is called the *centre of gravity of the body*. It coincides with its centre of mass (2.3.3).

7.3.3

The gravity force of a body is independent of its velocity of relative motion. It is proportional to the mass m of the body and can be presented in the form

$$\mathbf{P} = m\mathbf{g},$$

where $\mathbf{g}$ is the *acceleration due to gravity*, or *free fall acceleration* (see (7.3.5)). At a given place on earth, vector $\mathbf{g}$ is the same for all bodies and changes for other places.

The gravity force of a body coincides with the gravitational attraction of the earth for the body only at the earth's North and South poles because there the centrifugal inertial force $\mathbf{I}_{cf} = 0$. The greatest difference between the gravity force and the gravitational attraction is observed at the equator, where the force $\mathbf{I}_{cf}$ reaches its maximum value and is opposite in direction to $\mathbf{F}_{gr}$. Even at

the equator, however, the gravity force and the force of gravitational attraction differ by only 0.35 per cent. At all points of the earth's surface, except the equator and the poles, forces **P** and $\mathbf{F}_{gr}$ fail to coincide in direction as well (Figure 7.2), but the maximum angle between them does not exceed 6 minutes of arc. The gravity force decreases with the altitude. Near the earth's surface this decrease is approximately 0.034 per cent per kilometre above the surface.

7.3.4

The acceleration **g** near the earth's surface varies from 9.78 m/s^2 at the equator to 9.83 m/s^2 at the North and South poles. This is due, in the first place, to the dependence of the centrifugal inertial force on the geographical latitude of the location and, in the second, to the nonspherical shape of the earth, which is slightly flattened at the poles of its axis of rotation, having the shape of an ellipsoid of rotation, or spheroid (the polar and equatorial radii of the earth are $R_p = 6357$ km and $R_e = 6378$ km). The *standard value of free fall acceleration*, accepted in establishing a system of units and for barometric computation, is 9.806 65 m/s^2.

7.3.5

The *free fall of a body* is its motion due to the action of the gravitational held alone.

The free fall acceleration of a body falling to the earth, mea sured in a non-inertial frame of reference rotating together with the earth, can he found by the motion equa-

tion (7.3.1) by putting $\mathbf{F} = 0$, $\mathbf{F}_{gr} + \mathbf{I}_{cf} = m\mathbf{g}$ and $I_C = 2m[\mathbf{v}_r\boldsymbol{\Omega}]$. Then

$$\mathbf{a}_r = \mathbf{g} + 2[\mathbf{v}_r\boldsymbol{\Omega}]$$

If $\mathbf{v}_r = 0$, $\mathbf{a}_r = \mathbf{g}$.

Consequently, vector $\mathbf{g}$ is equal to the free fall acceleration of the body, measured with respect to the terrestrial frame of reference at the instant when the relative velocity of the body equals zero. For this reason, vector $\mathbf{g}$ is called the *free fall acceleration*.

If the relative velocity of a freely falling body $\mathbf{v}_r = 0$, its acceleration with respect to the earth is not $\mathbf{g}$: $\mathbf{g} = \mathbf{a}_r + 2[\mathbf{v}_r\boldsymbol{\Omega}]$. At velocities $v_r < 680\,\text{m/s}$, however, the values of $\mathbf{g}$ and $\mathbf{a}_r$ differ by less than one per cent. It can therefore be assumed in many cases that with respect to an observer on the earth free fall of a body is caused only by the gravity force acting on and imparting the acceleration g to the body. In a similar manner the action of the Coriolis inertial force on a freely falling body can be treated as a comparatively small perturbation. Thus, for instance, a freely falling body is deviated by the Coriolis force to the east of the direction assumed by a plumb line, i.e. from the direction of the vector $\mathbf{P} = m\mathbf{g}$. This deviation for a body falling freely without any initial velocity from the height h is equal, at the latitude φ, to

$$s = \frac{2}{3}\Omega h\sqrt{\frac{2h}{g}}\cos\varphi$$

If, for example, $h = 160\,\text{m}$ and $\varphi = 45°$, $s = 1.55\,\text{cm}$.

7.3.6

The *weight of a body* is the force $\mathbf{Q}$ with which it acts, due to the earth's gravity, on a support or suspension that prevents it from falling freely. It is assumed that the body and its support (or suspension) are fixed with respect to the frame of reference in which the weight is determined. The force $-\mathbf{Q}$ is exerted by the support or suspension on the body. It follows from the fundamental equation of dynamics of relative motion (7.2.2), where $\mathbf{a}_r = \mathbf{a}_C = 0$ and $\mathbf{F} = \mathbf{F}_{gr} - \mathbf{Q}$, that

$$\mathbf{Q} = \mathbf{F}_{gr} + \mathbf{I}_t$$

where $\mathbf{F}_{gr}$ is the force of gravitational attraction of the earth for the body, and $\mathbf{I}_t$ is the inertial force of transportation due to the non-inertial nature of the frame of reference.

Example 1. The weight of a body in a frame of reference fixed to the earth is equal to the gravity force of the body (7.3.2):

$$\mathbf{Q} = \mathbf{F}_{gr} + \mathbf{I}_{cf} = \mathbf{P}$$

Example 2. The weight of a body in a frame of reference attached to a lift (elevator) which travels with translational motion with respect to the earth with the acceleration $\mathbf{a}_0$:

$$\mathbf{Q} = \mathbf{P} - m\mathbf{a}_0$$

If the lift falls freely, $\mathbf{a}_0 = \mathbf{g}$, the free fall acceleration, and the weight of the body in the lift is $\mathbf{Q} = 0$, i.e. the body is in the state of weightlessness.

7.3.7

Weightlessness, or zero gravity, is the state of a mechanical system in which the external gravitational field acting on the system does not cause any pressure of the parts against one another or their deformation. Such a state occurs in a mechanical system complying with the following three conditions:

(a) no other external forces, except those of the gravitational field, act on the system;

(b) the size of the system is such that within its limits the external gravitational field can be regarded as being uniform; and

(c) the system is in translational motion.

The state of weightlessness is characteristic, for instance, of bodies in a spaceship because, during the main part of its path in a gravitational field, it is in free flight, i.e. coasting with the engines turned off.

7.4 Principle of Equivalence

7.4.1

The inertial forces acting on bodies in a non-inertial frame of reference are proportional to the masses of the bodies and, other conditions being equal, impart the same relative accelerations (7.1.3) to them. In other words, all bodies free of external action travel in an "inertial force field", i.e. with respect to a non-inertial frame of reference, in absolutely the same way, provided that the

initial conditions of their motion are also the same. A similar behaviour is observed for bodies travelling with respect to an inertial frame of reference and

subject to the action of the forces of a gravitational field. At each point in the field these forces, like the inertial forces, are proportional to the masses of the bodies and impart the same free fall acceleration to them. This acceleration is equal to the field intensity (6.2.1) at the point being considered. In a non-inertial frame of reference, for example, fixed to a lift (elevator) travelling with uniform acceleration vertically upward at the acceleration of transportation $\mathbf{a}_0 = \text{const}$, all free bodies fall in the absence of a gravitational field with the same relative acceleration $\mathbf{a}_r = -\mathbf{a}_0$. Free bodies behave in exactly the same way in the same lift when it travels at uniform velocity in a uniform gravitational field of the intensity $\mathbf{g}_f = -\mathbf{a}_0$. Thus, on the basis of experiments conducted on the free fall of bodies within a tightly closed lift, it is impossible to establish whether the lift is travelling at uniform velocity in a gravitational field of the intensity $\mathbf{g}_f = \mathbf{a}_r$ (in particular, the lift may also be at rest in this held) or whether it is travelling with constant acceleration of transportation $\mathbf{a}_t = -\mathbf{a}_r$ in the absence of a gravitational field.

7.4.2

The *local equivalence principle* states that the gravitational field in a limited region of space is equivalent to an "inertial force held" in a properly chosen non-inertial frame of reference. The region of space should be small enough so that the gravitational held in it can be assumed to be

uniform.

The equivalence principle is not to be understood as a statement of the identity of inertial forces and Newton's gravitational forces that act between bodies. As a matter of fact, the intensity of a true gravitational held, set up by bodies, decreases as the bodies recede farther and farther away, and approaches zero at infinity. Gravitational fields "equivalent" to inertial forces do not comply with this condition. For example, the intensity of a gravitational held, "equivalent" to the centrifugal inertial forces in a rotating frame of reference increases without limit at points farther and farther away from the axis of rotation. The intensity of a held "equivalent" to the inertial forces of transportation in a frame of reference having translational motion is the same at all points of the held.

7.4.3

A true gravitational held, in contrast to one "equivalent" to inertial forces, exists in both non-inertial and inertial frames of reference. No possible choice of a non-inertial frame of reference can completely eliminate a true gravitational held, i.e. compensate for it throughout space by means of a "field of inertial forces". This follows in any case, even if only from the different behaviour of "fields of inertial forces" and true gravitational fields at infinity. A true gravitational field can be eliminated only locally, i.e. for a small region of space within whose limits the field can he assumed uniform, and for a short time interval during which the field can be assumed constant. The non-inertial frame of reference corresponding to this operation should travel with an acceleration of transpor-

tation equal to the free fall acceleration of bodies in the region of the true gravitational field being considered. Thus, in a space ship in free flight in a gravitational field, the gravitational forces of attraction are compensated for by the inertial forces of transportation and do not lead to the relative motion of bodies in the ship.

Part II

Fundamentals of Molecular Physics and Thermodynamics

8 Ideal Gases

8.1 Subject Matter of Molecular Physics. Thermal Motion

8.1.1

The branch of physics called *molecular physics* deals with the dependence of the structure and physical properties of bodies on the nature of the motion of and the interaction be tween the particles making up these bodies.

Molecular physics is based on the *molecular-kinetic theory* of the structure of matter. According to this theory all bodies consist of submicroscopic particles – atoms, molecules or ions – in continuous chaotic motion, which is called *thermal motion*. Experimental confirmations of the molecular-kinetic theory are: the Brownian movement (**??**), the transport phenomenon in various aggregate states of matter and other phenomena.

8.1.2

The molecular-kinetic theory of the structure of matter is used to advantage in various branches of physics. A great variety of physical phenomena that proceed due to the interaction and motion of particles of matter are dealt with in this theory from a unified point of view. This theory, for example, enables the mechanism of elasticity in solids (**??**) to be understood, reveals the cause of internal friction in gases (**??**) and liquids, explains the difference between real gases (**??**) and ideal gases (8.4.1), and elucidates the mechanism of electrical conduction in conductors of electric current of various kinds, and the electrical and magnetic properties of matter.

8.1.3

The thermal motion of particles of matter in various states of aggregation is not the same. It depends upon the forces of attraction or repulsion between the atoms, molecules and ions.

There are practically no attractive forces between the atoms and molecules in sufficiently dilute gases. This is due to the fact that the particles of such gases are located at distances from one another exceeding the range of molecular action (**??**). The particles of such gases travel at uniform velocity and in a straight line until they collide with one another or with the walls of the container. These collisions are of a random nature. Each molecule in the volume of the gas may collide with any of the nearby particles, changing the direction of its motion in any arbitrary manner. The collision of a molecule (or atom) of the gas with a wall may occur at any angle θ

where $-\pi/2 \leqslant \theta \leqslant \pi/2$. As a result, the thermal motion of molecules of gas is a purely random one and, on the average, the same number of molecules are travelling at any instant of time in any arbitrary direction within the gas.

8.1.4

Solid crystalline bodies are characterized by considerable forces of interaction between their particles (atoms, molecules and ions). The joint action of the forces of attraction and repulsion between the particles (**??**) leads to the vibration of the particles of solids about their equilibrium positions, which are called the *sites*, or *points, of the crystal lattice* (**??**). Intermolecular interaction and violations of periodicity in crystals (40.1.1) make these vibrations anharmonic (**??**).

8.1.5

The thermal motion of molecules of a liquid is of a kind intermediate between the two preceding types of motion (8.1.3) and (8.1.4). For a definite length of time the molecules of a liquid vibrate about a.certain equilibrium position and are in a temporarily settled state. At the end of this time the equilibrium position of the molecules is displaced and a new settled arrange ment is formed. Simultaneously, a slow displacement of the molecules and their vibrations occur within small volumes of the liquid (see also (**??**)).

8.2 Statistical and Thermodynamic Methods of Investigation

8.2.1

There is a huge number of atoms (or molecules) in any body. For instance, in $1\,\mathrm{cm}^3$ of a gas whose properties are close to those of an ideal gas there are 2.7×10^{19} molecules at standard conditions. In the condensed states, liquid and solid, the amount is of the order of 10^{22} particles per cm^3. If we assume that each atom (or molecule) obeys Newton's second law (2.4.1) it is obviously quite impossible, not only to solve the differential equations of motion of the separate particles, but even to write these equations. Therefore, the behaviour of a separate molecule (or atom) of a body, for example, its path and the sequence of changes in its state, cannot be investigated by the methods used in classical mechanics.

8.2.2

The macroscopic properties of systems consisting of a very great number of particles are studied by the *statistical method*. It is based on the application of probability theory to definite models of the structure of the systems being investigated. The branch of theoretical physics in which the physical properties of systems are studied by the statistical method is called *statistical physics (statistical mechanics)*. Special laws, known as statistical laws are manifested in the joint behaviour of large numbers of particles. Certain *average values of the physical quantities* can be found for a system consisting of a great many par-

ticles. These values characterize the set of particles as a whole. In gases, for instance, there are average values of the velocity of thermal motion of the molecules (10.3.6) and of their energy (10.2.4). In a solid there is an average energy per degree of freedom of the vibrational motion of its particles (**??**), etc. All properties of a system of particles are due, not only to the individual properties of the particles themselves, but to the features of their joint motion and average values of their dynamic characteristics (average velocities, average energies, etc.).

In addition to statistical laws, there are also dynamical laws, describing the motion of separate particles. The relationship between dynamical and statistical laws is manifested in the fact that the laws of motion of individual particles have an effect of the description of a system of particles, which is being investigated by the statistical method.

8.2.3

Another procedure for studying physical phenomena is the *thermodynamic method*, which is not concerned with either the internal structure of the substances making up the bodies (or systems) being investigated or with the nature of the motion of the separate particles. The thermodynamic method is based on the study of the various conversions of energy that occur in a system. The conditions of these conversions and the relations between the various kinds of energy enable the physical properties of systems to be studied in a great variety of processes in which the systems participate. The branch of physics in which research in the physical properties of systems

is conducted by means of the thermodynamic method is called *thermodynamics (phenomenological thermodynamics)*. Thermodynamics is based on two experimentally established principles, or laws, of thermodynamics (9.3.1) and (**??**), as well as the Nernst. heat theorem, or third law of thermodynamics (**??**).

8.3 Thermodynamic Variables. Equations of State. Thermodynamic Processes

8.3.1

Thermodynamics deals with *thermodynamic systems*. There are macroscopic items (bodies and fields) that can interchange energy with one another as well as with the external medium, i.e. bodies and fields that are external with respect to the given system.

8.3.2

The physical quantities introduced to describe the state of a thermodynamic system are called *thermodynamic variables*, or *parameters (state variables)*, of the system. Usually the pressure, specific volume and temperature are taken as the thermo dynamic variables.

The *pressure p* is a physical quantity equal to the force acting on unit area of a body's surface in the direction normal to the surface:

$$p = \frac{\mathrm{d}F_n}{\mathrm{d}A}$$

where dF_n is the numerical value of the normal force acting on an infinitely small part of the surface of the body, the area of this part being dA.

The *specific volume* v is the reciprocal of the density ρ of a body: $v = 1/\rho$. The specific volume of a homogeneous body is equal to the volume of unit mass of the body.

8.3.3

The concept of temperature is of significance for equilibrium states of a thermodynamic system (8.3.4). An *equilibrium state (state of thermodynamic equilibrium)* is a state of a system which does not change with time (*steady state*), the steady nature of the state not being due to processes occurring in the external medium. An equilibrium state is set up in a system subject to constant external conditions and is maintained in the system for an arbitrary length of time. The temperature is the same in all parts of a thermodynamic system that is in a state of thermodynamic equilibrium. If two bodies with different temperatures come into contact, internal energy (9.1.2) is transferred from the body at the higher temperature to that at the lower temperature. The process ceases when the temperatures of the two bodies become equal.

8.3.4

The temperature of an equilibrium system is a measure of the intensity of thermal motion of its molecules (or atoms or ions). In an equilibrium system of particles that

obeys the laws of classical statistical mechanics (8.2.2), the average kinetic energy of the thermal motion of the particles is proportional to the absolute temperature (10.2.5).

Temperature can only be measured indirectly, making use of the fact that many physical properties of bodies, lending themselves to direct measurement, depend on the temperature. A change in the temperature of a body leads to changes in its length and volume, density, electric resistance, elastic properties, etc. A change in any of these properties can be used to measure temperature. For this purpose, it is necessary for a certain body, said to be *temperature-indicating*, that the functional dependence of the given property on the temperature be known. Temperature scales based on temperature-indicating bodies are said to be *empirical*.

In the *international centigrade (Celsius)* scale, temperature is measured in °C and is denoted by t (degree centigrade or degree Celsius). It is assumed that at a standard pressure of $1.013\,25 \times 10^5$ Pa (**??**) the melting point of ice and the boiling point of water are equal to 0 °C and 100 °C.

In the thermodynamic (absolute) temperature scale, the temperature is measured in kelvins (K) and is denoted by T.

The relation between the absolute temperature and the temperature measured by the Celsius scale is:

$$T = 273.15 + t$$

The temperature $T = 0\,\mathrm{K}$ (or $t = -273.15\,°\mathrm{C}$) is called absolute zero temperature (the impossibility of reaching absolute zero is discussed in (**??**)).

8.3.5

The state variables of a system are classified as being either external or internal. *External state variables* are physical quantities depending upon the position in space and various properties of bodies that are external with respect to the given system. For a gas, for instance, the volume V of the vessel in which the gas is contained is an external variable because the volume depends upon the position of external bodies: the walls of the vessel. For a dielectric in an electric field, the intensity of the field, set up by certain external sources, is an external variable. If a liquid is held in an open vessel, the atmospheric pressure, for example, is an external variable.

The *internal variables of a system* are physical quantities that depend both on the position of bodies external with respect to the system, and on the coordinates and velocities of the particles that make up the system. Internal variables of a gas, for instance, are its pressure and energy because they depend upon the coordinates and velocities of the moving molecules and on the density of the gas.

8.3.6

The state variables of a system that is in equilibrium (8.3.3) are not independent. The internal variables of such a system depend only on its external variables and on the temperature. The equilibrium state of a simple system[1] of given chemical composition and mass M is determined

[1] An example of a simple system is a gas in the absence of external fields. Mixtures of chemically homogeneous gases of constant composition, chemically pure liquids, etc. arc also simple systems.

by specifying two variables: the volume V and the temperature T. The equation of state (thermal equation of state) of a simple system is the functional dependence of the equilibrium pressure p in the system on the volume and temperature

$$p = f(V, T)$$

The equation of state is obtained experimentally in thermodynamics. In statistical physics (8.2.2), it is derived theoretically. This constitutes the interrelation between the statistical (8.2.2) and thermodynamical (8.2.3) methods of investigation.

8.3.7

If any of the external variables of the system is changed, the state of a thermodynamic system is changed. This is called a *thermodynamic process*. A thermodynamic process is said lo be equilibrium (an *equilibrium, quasi-static process*) if the system passes at an infinitely slow rate through a continuous succession of infinitely close thermodynamic equilibrium states (8.3.3). All processes that do not comply with these conditions are said to be *non-equilibrium* ones. Real processes are non-equilibrium ones because they proceed at a finite rate. But the slower they proceed, the closer they are to equilibrium processes

Isoprocesses are thermodynamic processes that proceed in a system of constant mass with a constant value of one of the state variables.

An *isothermal process* proceeds at constant temperature ($T = \text{const}$).

An *isochoric (isovolumic) process* proceeds at constant volume ($V = \text{const}$).

An *isobaric process* proceeds at constant pressure ($p = \text{const}$).

An *adiabatic process* is a thermodynamic process that occurs in a system without heat exchange (9.2.4) with external bodies (see also (9.5.10)).

8.3.8

State functions are physical quantities characterizing the state of a system. Variation in the state functions during thermodynamic processes is independent of the kind of process that occurs. State functions are uniquely determined by the values of the variables of the initial and final states of the system. The simplest state functions of a system are its internal energy U (9.1.2) and its entropy S (**??**).

8.4 Equation of State of an Ideal Gas

8.4.1

An *ideal*, or *perfect*, *gas* is one whose molecules do not interact with one another at a distance and are of vanishingly small size. In real gases (**??**) the molecules are subject to the forces of intermolecular interaction (**??**).

In collisions with one another and in striking the walls of the vessel the molecules of an ideal gas behave as perfectly elastic spheres (1.1.4) of diameter d (*elective diameter of the molecule*), which depends upon the chemical nature of the gas. The existence of an effective diameter d (where $d \approx 10^{11}\,\text{m}$) signifies that forces of mutual repulsion (**??**) act between the molecules. Intermolecular

forces of attraction (**??**) decrease drastically with an increase in the distance r between molecules and are practically negligible at $R_m > 10^9$ m. In real gases at low densities the average distance $\langle r \rangle$) between molecules exceeds R_m and, to a good approximation, they can be regarded as ideal. Hydrogen, helium, oxygen and nitrogen are assumed to be ideal gases at densities corresponding to standard conditions.

8.4.2

For a given mass of an ideal gas, the ratio of the product of the numerical values of pressure and volume to the absolute temperature is a constant value (*Clapeyron equation*):

$$\frac{pV}{T} = C = \text{const}$$

The numerical value of the *gas constant* C depends upon the choice of the units of measurement of p, V and T.

If v is the specific volume of the gas (8.3.2) and M is its mass, then $V = Mv$ and the Clapeyron equation assumes the form

$$pv = \frac{C}{M}T = BT$$

where $B = C/M$ is the *specific gas constant* referred to unit mass.

8.4.3

From the definition of the mole (**??**), it is evident that moles of different gases contain the same number of molecules. This is called *Avogadro's number* N_A (**??**).

The *molar mass* μ of a gas or any body is the physical quantity equal to the ratio of the mass of the gas (or any body) to the number of moles N it contains: $\mu = M/N$.

The molar mass is proportional to the relative mass of the molecules of a gas: $\mu = 10^{-3} m/m_0$, where m is the mass of the molecules of the given gas and m_0 is the atomic mass unit based on the carbon scale (**??**). The coefficient 10^{-3} is required because molar mass is measured in kg/mol in the SI (International System of Units).

The *molar volume* V_μ is the physical quantity equal to the ratio of the volume V of a gas to the number of moles N it contains: $V_\mu = V/N$. The mass of a mole is numerically equal to μ and therefore $V_\mu = \mu v$, where v is the specific volume (8.3.2).

8.4.4

The equation of state for a mole of ideal gas is

$$\frac{pV_\mu}{T} = \mu B = R \quad \text{or} \quad pV_\mu = RT$$

Here R is the *universal gas constant*, being the gas constant per mole of gas. The universal nature of R follows from *Avagadro's law*, which states that moles of all ideal gases occupy the same volume at the same pressure and the same temperature. Under standard conditions ($T = 273.15\,\mathrm{K} and \mathrm{p} = 1.0132 \times 10^5\,\mathrm{Pa} = 1\,\mathrm{atm} = 760\,\mathrm{mm\,Hg}$) a mole of any gas has a volume of $V_\mu = 22.415 \times 10^{-3}\,\mathrm{m}^3$. From this we can calculate the numerical values of R for various systems of units (**??**).

If a volume V of gas has the mass M kg, i.e. if it contains M/μ moles, then $V = M/\mu \times V_\mu$ and the equation of

state of the gas assumes the form called the *Mendeleev-Clapeyron equation*:

$$pV = \frac{M}{\mu}RT$$

8.4.5

Boltzmann's constant k is a physical quantity equal to the ratio of the universal gas constant R to Avogadro's number N_A, i.e. $k = R/N_a$. The values of k in various systems of units are listed in **??**.

The equation of state of an ideal gas, expressed by means of Boltzmann's constant, is of the form

$$p = \frac{kN_A}{V_\mu} = kn_0T$$

where $n_0 = N_A/V_\mu$ is the number of gas molecules in unit volume (molecule concentration).

At constant temperature, the pressure of a gas is proportional to the concentration of its molecules (or to the density of the gas).

9 First law of Thermodynamics

9.1 Total and Internal Energy of a System

9.1.1

An arbitrary thermodynamic system (8.3.1), in any thermodynamic state (8.3.3), has a *total energy* E , which is the sum of:

(a) the kinetic energy E_k^{Mech} of mechanical motion of the system as a whole (or of its macroscopic parts);

(b) the potential energy E_p^{Ext} (3.3.1) of the system in all external force fields (for instance, electromagnetic or gravitational fields); and

(c) the internal energy U:

$$E = E_k^{\text{Mech}} + E_p^{\text{Ext}} + U$$

9.1.2

The *internal energy* of a body or of a thermodynamic system (8.3.1) is the energy depending only upon the thermodynamic state of the body (or system). For a fixed (stationary) system, not located in any external force fields, the internal energy coincides with the total energy. The internal energy also coincides with the rest energy of a body (or system) (5.7.3) and includes the energy of all the types of internal motion of a body (or system) and the energy of interaction of all the particles (atoms, molecules, ions, etc.) making up the body (or system).

For example, the internal energy of a gas consisting of poly atomic molecules (ammonia, carbon dioxide, etc.) consists of:

(a) kinetic energy of thermal translational and rotary motion of the molecules;

(b) kinetic and potential energy of vibration of the atoms and molecules;

(c) potential energy due to intermolecular interactions;

(d) energy of the electron shells of the atoms and ions;

(e) kinetic energy and potential energy of interaction of the nucleons (**??**) in the nuclei of the atoms.

Terms (d) and (e) do not ordinarily change in processes that proceed at not very high temperatures so that ionization and excitation play no essential role. Under these conditions, terms (d) and (e) are not taken into account in the internal energy balance. Term (c) is also neglected for an ideal gas (8.4.1).

9.1.3

The internal energy is a single-valued function of the thermodynamic state of a system. The value of the internal energy in any state is independent of the process by means of which the system reached this state. The change in internal energy in the transition of a system from state 1 to state 2 equals $\Delta U = U_2|U_1$ and is independent of the kind of transition $1 \to 2$ that occurs. If the system accomplishes a cycle (**??**) the total change in its internal energy is equal to zero:

$$\oint \mathrm{d}U = 0$$

As is well known, such a relation mathematically signifies that an element of change in the internal energy $\mathrm{d}U$ is a complete (exact) differential. This property is also possessed by the entropy (**??**) and other state functions (8.3.8) (cf. (9.2.5)).

9.1.4

In a system that is in a state of thermodynamic equilibrium the internal energy depends only upon the temperature and the external variables (8.3.5). In particular, for a simple system (8.3.6) of constant mass M, the internal energy is a function of the temperature T and volume V of the system (*calorific equation of state* of a simple system):

$$U = \varphi(V, T).$$

Example 1. The internal energy of an ideal gas (8.4.1) depends only on its absolute temperature and is propor-

tional to the mass M of the gas:

$$U = \oint_0^T C_V \mathrm{d}T + U_0 = M\left(\oint_0^T c_V \mathrm{d}T + u_0\right)$$

where C_V and $c_V = C_V/M$ are the heat capacity (9.5.1) and specific heat (9.5.2) of the gas at constant volume (in an isochoric process) (8.3.7), $u_0 = U_0/M$ is the internal energy of unit mass of the gas at $T = 0\,\mathrm{K}$. For monatomic gases at ordinary temperatures C_V is independent of T, and $U = C_V T + U_0$.

Example 2. The internal energy of a van der Waals gas (**??**) is:

$$U = \oint_0^T C_V \mathrm{d}T - \frac{M^2}{\mu^2}\frac{a}{V} + U_0$$

where M is the mass of the gas, μ is its molar mass (8.4.3), and a is the van der Waals' coefficient (**??**).

9.1.5

In thermodynamics, the internal energy is determined to an accuracy of the constant term U_0, whose value depends upon the choice of the reference point for the quantity U, i.e. from a state with zero internal energy. The quantity U_0 plays practically no role in thermodynamical calculations in which the changes ΔU of internal energy, independent of U_0, are determined (see also (8.1.2)).

9.2 Heat and Work

9.2.1

An energy exchange between a thermodynamic system and external bodies takes place in two ways: either by doing work or by means of heat exchange.[1] The amount of energy transferred to the system by external bodies in force inter action between the system and the bodies is called *work* done on the system. The amount of energy transferred to the system by external bodies by heat transfer is called the *quantity of heat* imparted to the system.[2]

9.2.2

If the thermodynamic system is fixed (stationary), a necessary condition to do work is displacement of the external bodies interacting with it, i.e. it is necessary to change the external state variables of the system (8.3.5). In the absence of external force fields, energy exchange between the lixed system and the external medium may take place by doing work when the volume and shape of the system are changed. According to the law of conservation of energy, the work W' done on the system by external forces is numerically equal and opposite in sign to the work W that the system does on the external medium, i.e. against the external forces:

$$W' = -W$$

[1] A discussion of the third way of energy exchange – mass transfer – is beyond the scope of the present handbook.

[2] Sometimes the term "heat" is employed in place of the term "quantity of heat" if this leads to no confusion. It should be indicated, however, that heat, generally speaking, is to be under stood as a form of energy transfer (9.2.4)).

9.2.3

Expansion work is the work done by the system in over coming the external pressure. An element of expansion work $\delta W = p_{\text{ext}} \mathrm{d}V$, where p_{ext} is the uniformly distributed external pressure, and dV is an element of change of the system's volume. If the expansion process is an equilibrium (quasistatic) one (8.3.7), $p_{\text{ext}} = p$, where p is the pressure in the system. Then $\delta W = p \mathrm{d}V$.

The significance of the different ways used in denoting the elements of change δW and $\mathrm{d}U$ is explained in (9.4.3) and (9.1.3). The work done in equilibrium expansion of system from volume V_1 to volume V_2 is

$$W = \int_{V_1}^{V_2} p \mathrm{d}V$$

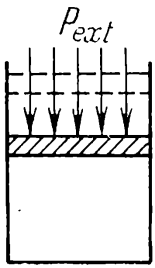


Figure 9.1: The work done by a gas in a vessel with a weightless movable piston.

Example. The work done by a gas in a vessel with a weightless movable piston (Figure 9.1). The pressure of the gas $p > 0$ and, therefore, in its expansion ($\mathrm{d}V > 0$)

the gas performs positive work ($\delta W > 0$). In compressing the gas ($dV < 0$ and $\delta W < 0$) the positive work is done on the p_{ext} gas by the forces of external pressure. Here the gas does negative work. The graphical representation of work is discussed in (9.4.2).

9.2.4

Heat exchange takes place between bodies (or parts of bodies) heated to different temperatures. There are three types of heat exchange: convection, conduction and radiation (radiative heat exchange).

Convective heat exchange is the transfer of energy in the form of heat between the unequally heated parts of gases or liquids or between gases, liquids and solids. Convective heat exchange in liquids takes place by motion of parts of the liquid with respect to one another or with respect to solids. For instance, in a radiator of a hot-water heating system, energy of the hot water flowing through the radiator is transferred by convection to the less heated walls of the radiator.

The phenomenon of *heat conduction* consists in the transfer of energy in the form of heat from one part of a non-uniformly heated body to another part. Thus, for example, energy is transferred through the walls of a heating radiator from the more heated inner surfaces to the less heated outer surfaces.

Radiative heat exchange takes place without direct contact of the bodies exchanging energy and consists in the emission and absorption of the energy of an electromagnetic field by the bodies. An immense amount of energy is delivered from the sun to the earth's surface by radia-

tive heat exchange.

9.2.5

Work and heat are the energy characteristics of processes for changing the state of thermodynamic systems and are meaningful only when such processes occur. Depending upon the kind of process the system undergoes in its transition from state 1 to state 2, it is necessary to perform different amounts of work and to deliver different quantities of heat to the system.

A comparison with (8.1.3) (on the change in the internal energy in processes) indicates that work and heat are not forms of energy and it is incorrect to speak of a "stockpile of work" or a "reserve of heat" in a body. For the same reasons, an element of the quantity of heat δQ and an element of work δW are not complete differentials (cf. (9.1.3)).

9.2.6

The performance of work on a system can change any type of its energy. For example, when a gas is rapidly com pressed in a vessel by means of a moving piston (9.2.3), the work done by external forces on the gas increases the internal energy of the gas. In the inelastic collision of two bodies (3.5.3), a part of the work is expended to change the kinetic energy of the bodies (3.2.2) and a part to change their internal energy.

If energy is delivered to a system in the form of heat, it is used only to increase the internal energy of the system. This is due to the fact that in any kind of heat

exchange (9.2.4), energy is transferred directly between chaotically moving particles of the bodies. This changes their internal energy. In the process of heat conduction, for instance, in a nonuniformly heated rigid body the particles of the body located in its more heated parts transmit some of their energy to particles located in its less heated parts. The result is an equalization of the internal energies of the various parts of the body, an equalization of their temperatures and the discontinuation of the heat conduction process.

It follows from the aforesaid that there is a qualitative difference and lack of equivalence between work and heat as a form of energy transmission. Frequently, these two forms of transmitting energy exist simultaneously. For example, in heating a gas in a vessel with a moving piston, the volume of the gas is increased and work is done against the external pressure simultaneously.

9.2.7

A thermodynamical system is said to be *closed* (or *isolated*) if there is no exchange of energy in any form between it and the external medium. Such systems comply with the *law of conservation of energy*: the total energy of the system remains unchanged regardless of the processes that occur within it.

9.2.8

A thermodynamical system is said to be *adiabatically isolated* (i.e. *thermally isolated*) if no heat exchange takes

place between it and the external medium. Such a system can per form work on external bodies. Moreover, external forces can do work on the system. An example of such a system is a cylinder with a moving piston, filled with gas and surrounded all over by a dense layer of heatproof felt. The absence of heat exchange with the external medium does not exclude the possibility of the gas doing expansion work (9.2.3), or the external pressure doing compression work on the gas.The properties of a system approximate those of an adiabatically isolated one if the change in its state is so rapid that there is no time, during the process, for heat exchange with the external medium. (For example, rapid expansion of a compressed gas released from a gas cylinder when the outlet valve is opened for a short time.)

9.3 First Law of Thermodynamics

9.3.1

The *first law of thermodynamics* states that the change in the internal energy ΔU_{1-2} of a system that occurs in the process 1 2 of transition of the system from state 1 to state 2 is equal to the sum of the work W'_{1-2} done by external forces on the system and the quantity of heat Q_{1-2} transmitted to the system:

$$\Delta U_{1-2} = W'_{1-2} + Q_{1-2}$$

where $W'_{1-2} = -W_{1-2}$ and W_{1-2} is the work done by the system to overcome external forces in the process $1 \rightarrow 2$. Hence

$$Q_{1-2} = \Delta U_{1-2} + W_{1-2}$$

The quantity of heat transferred to a system is expended in changing the internal energy of the system and in the work done by the system to overcome the external forces.

For an element of the quantity of heat δQ an element of work δW and an infinitely small change δU of the internal energy, the first law of thermodynamics assumes the form

$$\delta Q = \mathrm{d}U + \delta W.$$

9.3.2

If $\delta Q > 0$, then heat is being transferred to the system; if $Q < 0$, then heat is being removed from the system. In the finite process $1 \to 2$, elements of the quantity of heat can have both signs, and the total quantity of heat Q_{1-2} in the process $1 \to 2$ is equal to the algebraic sum of the quantities of heat transferred during all the parts of this process:

$$\delta Q_{1-2} = \int_1^2 \delta Q$$

It is customary to consider $\delta W > 0$ when the system performs work on external bodies, and $\delta W < 0$ when work is performed by external bodies on the system. The work W_{1-2} done by the system in the finite process $1 \to 2$ is equal to the algebraic sum of the amounts of work δW performed by the system during all the parts of this process:

$$W_{1-2} = \int_1^2 \delta W$$

9.3.3

If the system for instance, the working medium (**??**) in a periodic-action engine completes a cycle **1** → **1** (**??**), then $\Delta U_{1-1} = 0$ and $W_{1-1} = Q_{1-1}$. It is impossible to build an operating engine that could perform more work than the energy transferred to the system from outside. Such an engine is called a *perpetual motion machine of the first kind*. The impossibility of constructing a perpetual motion engine of the first bind is also a statement of the first law of thermodynamics.

9.4 Graphical Representation of Thermodynamic Processes and Work

9.4.1

The equation of state (8.3.6) of a thermodynamic system enables the value of any state variable, for example p, to be determined from any values of two other variables, such as V and T. Therefore, thermodynamic processes can be graphically represented in various two-dimensional coordinate systems. In addition to the most extensively used pV diagrams, pT and VT diagrams are also used.

In Figure 9.2, the thermodynamic process in the pV diagram is represented by the curve C_1C_2, and the points $C_1(p_1, V_1)$ and $C_2(p_2, V_2)$indicate the initial and final states of the thermodynamic system.

Only equilibrium processes (8.3.7) can be represented graphically. State variables for the whole body (or system) cannot be specified for non-equilibrium processes

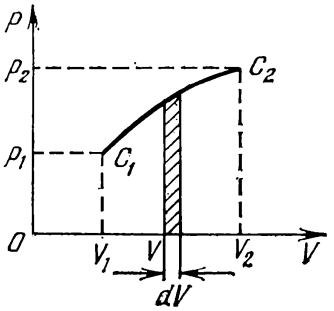


Figure 9.2: Graphical representation of an equilibrium process.

(8.3.7) because they differ at different parts of the body (or system). Hence such graphical representation of non-equilibrium processes is impossible.

9.4.2

An element of work δW done by a system (9.2.3) in an equilibrium process (8.3.7) can be measured by the area of the trapezoid having one curvilinear side, hatched in Figure 9.2. The work W_{1-2} done by the system in the process C_1C_2 is equal to

$$W_{1-2} = \int_{V_1}^{V_2} p\,\mathrm{d}V$$

and is measured by the area that is bounded by the curve representing process C_1C_2,the axis of abscissas and the ordinates p_1 and p_2 of points C_1 and C_2. The amount of work W_{1-2} done depends on how the system passes from state C_1 to state C_2, i.e. upon the kind of process C_1C_2

the system undergoes in changing its state. In the pV diagram of Figure 9.3 the amounts of work done by the system in the processes $C_1L_1C_2$, $C_1L_2C_2$ and $C_1L_3C_2$, and equal to WL_1, WL_2 and WL_3, respectively, are measured by three areas that differ in size: $WL_1 > WL_2 > WL_3$.

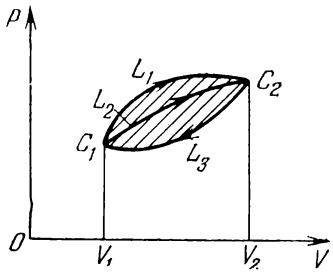


Figure 9.3: Finding work done via graphical representation.

After the system has completed the cycle (**??**) $C_1L_1C_2L_3C_1$ the total amount of work $W_{C_1-C_1}$ is not equal to zero. The positive expansion work in process $C_1L_1C_2$ exceeds the negative work done in the compression process $C_2L_3C_1$. The resultant positive work is measured by the hatched area in Figure 9.3.

9.4.3

The work W and the quantity of heat Q are not state functions (8.3.8). In various processes **1** → **2**, in which the state of a system is changed, various quantities of heat are transferred to the system, and various amounts of work are done. Elements of the changes in the physical quantities δQ and δW are not complete differentials

(9.1.3).

9.5 Heat Capacity of Matter. Applying the First Law of Thermodynamics to Isoprocesses in an Ideal Gas

9.5.1

The *heat capacity* C of a body is a physical quantity, numerically equal to the ratio of the quantity of heat δQ transferred to the body to the change $\mathrm{d}T$ in the temperature of the body in the thermodynamic process being considered:

$$C = \frac{\delta Q}{\mathrm{d}T}$$

The value of C depends upon the mass of the body, its chemical composition, thermodynamic state and the process in which the heat δQ is transferred to it.

9.5.2

The *specific heat* c is the heat capacity per unit mass of a substance. For a homogeneous body $c = C/M$, where M is the mass of the body.

The *molar heat capacity* C_μ is the heat capacity of a single mole (**??**) of the substance: $C_\mu = \mu c$, where μ is the molar mass (8.4.3) of the substance.

9.5.3

The element of the quantity of heat δQ that must be transferred to a body to change its temperature from T to $T + \Delta T$ is

$$\delta Q = C\mathrm{d}T$$

For a homogeneous body $\delta Q - Mc\,\mathrm{d}T = (M/\mu)C_\mu \mathrm{d}T$, where M is the mass of the body, μ is its molar mass and M/μ is the number of moles in the body.

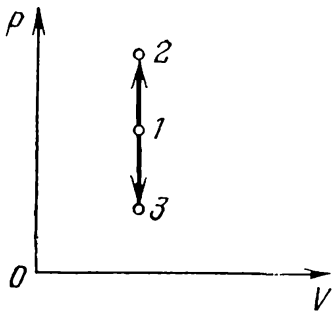


Figure 9.4: Finding work done via graphical representation.

9.5.4

For equilibrium (8.3.7) iso processes in gases (8.3.7), the first law of thermodynamics (9.3.1) is of the form

$$\frac{M}{\mu}C_\mu \mathrm{d}T = \mathrm{d}U + p\mathrm{d}V$$

In the isochoric process (8.3.7) of heating or cooling a gas (straight lines 1-2 and 1-3 in Figure 9.4), no element of

work $\delta W = p\mathrm{d}V$ is done (because $\mathrm{d}V = 0$). The whole quantity of heat transferred to the gas is expended in changing its internal energy: $\delta Q = \mathrm{d}U$. If C_{V_μ} is the molar heat capacity at constant volume,

$$\frac{M}{\mu} C_{V_\mu} \mathrm{d}T = \mathrm{d}U + p\mathrm{d}V \qquad (\star)$$

Within a definite temperature range it can be assumed that $C_{V_\mu} \approx \text{const}$ (for greater detail see (**??**)) and that the change in the internal energy of the gas ΔU_{1-2}, as its temperature is changed from T_1 to T_2, is equal to $\Delta U_{1-2} = U_2 - U_1 = (M/\mu) C_{V_\mu} (T_2 - T_1)$ and is due to the quantity of heat Q_{1-2} transferred to the gas in the isochoric process:

$$Q_{1-2} = \frac{M}{\mu} C_{V_\mu} (T_2 - T_1) \qquad (\star\star)$$

At $T_2 > T_1$, $Q_{1-2} > 0$ and a definite quantity of heat is transferred to gas; at $T_2 < T_1$, $Q_{1-2} < 0$ and a definite quantity of heat is conducted away from the gas.

9.5.5

For an ideal gas, equations $(\star)$ and $(\star\star)$ express the change in its internal energy in *any process* for changing the state of the gas in the temperature range $(T_2 - T_1)$. The internal energy of an ideal gas depends only on its chemical composition, mass and temperature.

In real gases (**??**) the internal energy includes the potential energy of interaction between the molecules (9.1.2), which depends upon the distance between them. This part of the internal energy of a real gas is changed with

changes in its volume. Therefore, equations (⋆) and (⋆⋆) express the change in the internal energy of a real gas only when it is heated or cooled in an isochoric (constant-volume) process.

9.5.6

In an arbitrary equilibrium process (8.3.7) that takes place in an ideal gas, the first law of thermodynamics assumes the form

$$\frac{M}{\mu} C_\mu \mathrm{d}T = \frac{M}{\mu} C_{V_\mu} \mathrm{d}T + p\mathrm{d}V$$

where C_μ is the molar heat capacity of an ideal gas in the given process. The isobaric process (8.3.7) of heating (straight line 1-2 in Figure 9.5) or cooling (straight line 1-3 in Figure 9.5) a gas can be carried out, for instance, in a vessel (cylinder) closed by a movable piston subject to a constant external pressure. An clement of work 6IF done by an ideal gas in an isobaric process is

$$\delta W = p\mathrm{d}V = \frac{M}{\mu} R\mathrm{d}T$$

in which use is made of the expression $\mathrm{d}V = (M/\mu)(R/p)\mathrm{d}T$ from the Mendeleev-Clapeyron equation (8.4.4) at $p =$ const. The universal gas constant R (8.4.4) is numerically equal to the work done by one mole of an ideal gas when it is heated isobarically by one degree:

$$R = \frac{\delta W}{(M/\mu)\mathrm{d}T}$$

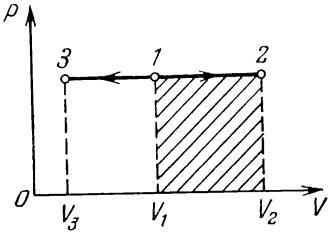


Figure 9.5: Finding work done via graphical representation.

The work W_{1-2} done by the gas in isobaric expansion 1-2 is

$$W_{1-2} = \int_{V_1}^{V_2} p\mathrm{d}V = p(V_2 - V_1)$$

It is represented by the hatched area in Figure 9.5. For an ideal gas the work W_{1-2} is also equal to

$$W_{1-2} = \frac{M}{\mu} R(T_2 - T_1).$$

9.5.7

The element of the quantity of heat δQ transferred to the gas in the isobaric process is

$$\delta Q = \frac{M}{\mu} C_{p\mu} \mathrm{d}T$$

where $C_{p\mu}$ is the molar heat capacity at constant pressure. If the quantity $C_{p\mu}$ can be assumed constant in the temperature range $(T_2 - T_1)$ then the quantity of heat

Q_{1-2} transferred to (or conducted away from) the gas is

$$Q_{1-2} = \frac{M}{\mu} C_{p\mu}(T_2 - T_1).$$

9.5.8

The molar heat capacities $C_{p\mu}$ and $C_{V\mu}$ of a gas are related by the *Mayer equation*

$$C_{p\mu} - C_{V\mu} = R$$

For specific heats c_p and c_V, the Mayer relation is of the form

$$c_p - c_V = \frac{R}{\mu}$$

where μ is the molar mass (8.4.3) of the gas.

For the heat capacities C_p and C_V

$$C_p - C_V = \frac{M}{\mu} R$$

where M is the mass of the gas and M/μ is the number of moles it contains.

The Mayer relation indicates that in heating a gas isobarically by one degree a greater quantity of heat must be transferred to the gas than in isochoric heating by the same amount. This difference between the quantities of heat should be equal to the work done by the gas in its isobaric expansion.

9.5.9

The isothermal process (8.3.7) of expanding (or compressing) a gas takes place in heat exchange (9.2.4) between

the gas and the external medium at a constant temperature difference. For this to be feasible, the heat capacity (9.5.1) of the external medium should be sufficiently great and the process of expansion (or compression) should proceed extremely slowly. First-order phase transitions (**??**), such as boiling and condensation, are isothermal processes if they proceed at constant external pressure.

In an isothermal process in an ideal gas the internal energy of the gas remains unchanged (9.5.5) and the whole quantity of heat Q_{1-2} transferred to the gas is expended in the work W_{1-2} done by the gas to overcome the external forces:

$$Q_{1-2} = W_{1-2} = \int_{V_1}^{V_2} p\mathrm{d}V = \frac{M}{\mu} RT \int_{V_1}^{V_2} \frac{\mathrm{d}V}{V} = \frac{M}{\mu} RT \ln \frac{V_2}{V_1}$$

Here $\frac{M}{\mu}$ is the number of moles contained in mass M of the gas, T is the constant temperature of the gas, and V_1 and V_2 are the initial and final volumes of the gas. If the gas isothermally expands ($V_2 > V_1$), heat is being transferred to it $Q_{1-2} > 0$)and it is doing positive work ($W_{1-2} > 0$) represented by the hatched area in Figure 9.6. In isothermal compression of the gas (process 1-3 in Figure 9.6), the work done by the gas is negative ($W_{1-3} < 0$) . Positive work ($W'_{1-3} = -W_{1-3} > 0$) is being done by the external forces to compress the gas. In compression, a certain quantity of heat is conducted away from the gas ($Q_{1-3} < 0$). The heat capacity of a substance in an isothermal process equals infinity ($\mathrm{d}T = 0$ and $\delta Q \neq 0$).

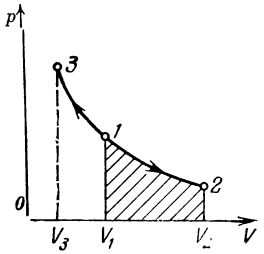


Figure 9.6: Finding work done via graphical representation.

9.5.10

An adiabatic process (8.3.7) proceeds under the condition that $\delta Q = 0$. It is essential that the condition $Q = 0$ does not define this process, because it does not specify the absence of heat exchange with the external medium. It only indicates that the algebraic sum of the quantities of heat transferred to and conducted away from the gas in various parts of the process equals zero. In an adiabatic process, work is done by an ideal gas at the expense of its internal energy:

$$\delta W = -\mathrm{d}U = -\frac{M}{\mu} C_{V\mu} \mathrm{d}T$$

where $C_{V\mu}$ is the molar heat capacity of the gas at constant volume, M/μ is the number of moles contained in mass M of the gas, and $\mathrm{d}T$ is an element of change in the temperature of the gas. If the gas adiabatically expands ($\delta W = p\mathrm{d}V > 0$), it is cooled ($\mathrm{d}T < 0$). In adiabatic compression the gas is heated: $\delta W = p\mathrm{d}V < 0$ and $\mathrm{d}T > 0$.

9.5.11

The *adiabatic*, or *Poisson*, *equation*: $pV^{\varkappa} = \text{const}$, is valid for an equilibrium adiabatic process (8.3.7).

Making use of the Mendeleev-Clapeyron equation (8.4.4), the relation between p and T, and also between V and T, in an adiabatic process can be found from the adiabatic equation:

$$pT^{\frac{\varkappa}{\varkappa-1}} = \text{const} \quad \text{and} \quad VT^{\frac{1}{\varkappa-1}} = \text{const}$$

In these equations the dimensionless quantity $\varkappa = C_{p\mu}/C_{V\mu} = c_p/c_V > 1$ is the *adiabatic exponent*. The full curve in Figure 9.7, called an *adiabat* or *adiabatic curve*, represents an adiabatic process in a pV diagram. The dashed curve on this diagram, called an *isotherm*, represents an isothermic process at the temperature corresponding to the initial state 1 of the gas. In an adiabatic process the pressure varies with the volume more rapidly than in an isothermic process (the curve is steep er). In adiabatic expansion the temperature of the gas and its pressure decrease faster than in the corresponding isothermal expansion. In adiabatic compression of the gas its pressure increases at a higher rate than in the corresponding isothermal compression. The reason is that the increased pressure is due both to the decrease in the volume of the gas and to the increase in its temperature.

9.5.12

The work W_{1-2} done by the gas in the adiabatic process $1 \to 2$ is measured by the hatched area in Figure 9.7. The amount of work W_{1-2} done in an adiabatic process is expressed by

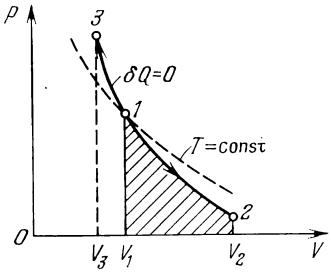


Figure 9.7: Finding work done via graphical representation.

In these equations M/μ is the number of moles contained in mass M of the gas, $C_{V\mu}$ is the molar heat capacity at constant volume, $\varkappa$ is the adiabatic exponent, and p, V and T are the state variables of the gas in the states 1 and 2 as indicated.

9.5.13

Table 9.1 lists summarized information on the isoprocesses in gases.

Thermodynamic Process	Isochoric	Isobaric	Isothermal	Adiabatic
Basic Condition	$V = \text{const}$	$p = \text{const}$	$T = \text{const}$	$\delta Q = 0$
Relationship between state variables	$\frac{p}{T} = \text{const}$	$\frac{V}{T} = \text{const}$	$pV = \text{const}$	$pV^{\varkappa} = \text{const}$ $pT^{\varkappa/(\varkappa-1)} = \text{const}$ $VT^{1/(\varkappa-1)} = \text{const}$
Work done in the process	$\delta W = 0$ $W = 0$	$\delta W = p\mathrm{d}V$ $W = p(V_2 - V_1)$	$\delta W = p\mathrm{d}V$ $W = \frac{M}{\mu} RT \ln \frac{V_2}{V_1}$	$\delta W = p\mathrm{d}V = -\mathrm{d}U$ $W = -\Delta U = C_V(T_2 - T_1)$
Quantity of heat transferred in the process	$\delta Q = C_V \mathrm{d}T$ $Q = C_V(T_2 - T_1)$	$\delta Q = C_p \mathrm{d}T$ $Q = C_p(T_2 - T_1)$	$\delta Q = \delta W$ $Q = W$	$\delta Q = 0$ $Q = 0$
Change in internal energy	$\mathrm{d}U = \delta Q$ $U = Q$	$\mathrm{d}U = C_V \mathrm{d}T$ $U = C_V(T_2 - T_1)$	$\mathrm{d}U = 0$ $\Delta U = 0$	$\mathrm{d}U = -\delta W = C_V \mathrm{d}T$ $\Delta U = -W = C_V(T_2 - T_1)$
Heat capacity	$C_V = \frac{M}{\mu} \frac{R}{\varkappa - 1}$	$C_p = \frac{M}{\mu} \frac{\varkappa R}{\varkappa - 1}$	$C_T = \infty$	$C_{ad} = 0$

Table 9.1: A table of isoprocesses.

10 Kinetic Theory of Gases

10.1 Certain Information on Classical Statistical Physics

10.1.1

The *kinetic theory of gases* deals with the structure and physical properties of gases and is based on the statistical method of investigation (8.2.2). Besides those mentioned in (8.2.1), the following propositions form the foundation for *classical statistical physics*.

10.1.2

A system of particles complies with the laws of conservation of energy (9.2.7), momentum (2.7.1) and of angular momentum (4.4.1). A system of charged particles complies with the law of conservation of electric charge (**??**).

10.1.3

All physical processes in a system of particles proceed *continuously* in space and time. A space-time description of any physical phenomena in classical mechanics and classical statistical physics assumes the possibility of continuous variation of all physical quantities that characterize the state of the system. For instance, the velocity and energy of any particle can vary continuously due to the action of various forces.

10.1.4

Each particle in a system is regarded as being "tagged]]. It can be distinguished from all other such particles (*distinguishability of identical particles in classical statistical physics*).

10.1.5

Any particle of a system can have arbitrary values for its coordinates and momenta (or velocities) independently of the values of these quantities for other particles of the system. If an infinitely small volume $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ is chosen in the coordinate system XYZ, any particle can be located within this volume regardless of the presence in this volume of any arbitrary number of other particles. Likewise, any particle may be located within an arbitrary element of "volume" $\mathrm{d}p_x\,\mathrm{d}p_y\,\mathrm{d}p_z$ of momentum space (or $\mathrm{d}u_x\,\mathrm{d}u_y\,\mathrm{d}u_z$ of velocity space) regardless of the presence in these "volumes" of any arbitrary number of other particles. This implies that any particle can have momentum components along the coordinate axes within the

limits from p_x to $p_x + \mathrm{d}p_x$, from p_y to $p_y + \mathrm{d}p_y$ and from p_z to $p_z + \mathrm{d}p_z$ (and, respectively, velocity components of the particle along the coordinate axes within the limits from u_x to $u_x + \mathrm{d}u_x$, u_y to $u_y + \mathrm{d}u_y$, u_z to $u_z + \mathrm{d}u_z$). The minimum values of the volumes $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ and $\mathrm{d}p_x\,\mathrm{d}p_y\,\mathrm{d}p_z$ are not restricted in any manner.

(10.1.2) and (10.1.3) apply not only to classical statistical physics, they are characteristic of all of classical physics. Sections (10.1.4) and (10.1.5) apply only to classical statistical physics; they do not hold for quantum statistics (**??**).

10.2 Basic Equation of the Kinetic Theory of Gases

10.2.1

The pressure of a gas (8.3.2) in a vessel is the result of the collisions of the gas molecules with the walls of the vessel. The pressure of a gas is the macroscopic effect of the thermal motion of molecules (8.1.1). As the molecules of the gas strike the walls of the vessel (8.4.1) they impart their momentum (2.3.4) to the walls. This change in momentum of the molecules exerts the pressure on the walls. For an ideal gas the collisions between the molecules within the vessel has no effect on the pressure exerted by the gas on the walls. Owing to the chaotic nature of thermal motion of the molecules (8.1.3) the pressure of the gas is the same on all the walls and, by definition (8.3.2), is the average force exerted normal to unit area of the wall surface.

10.2.2

The *basic equation of the kinetic theory of gases* is

$$pV = \frac{2}{3}E_k$$

where p is the pressure of the gas, V is its volume and $E_k = \sum_{i=1}^{N} \frac{m_i u_i^2}{2}$ is total kinetic energy of translational motion of all the N molecules of gas in the vessel.

10.2.3

In a homogeneous gas $m_i = m$, i.e. the mass of each molecule is the same, hut their velocities u_i differ (10.3.1), and

$$E_k = \frac{m}{2}\sum_{i=1}^{N} u_i^2$$

It proves expedient to introduce the root-mean-square velocity v_{rms} of translational motion of gases:

$$v_{\text{rms}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} u_i^2}$$

where N is the total number of molecules in volume V of the gas. The root-mean-square velocity pertains to the whole population of molecules, but is meaningless in application to a single molecule or to a small number of molecules. Making use of v_{rms} the expression for E_k assumes the form

$$E_k = \frac{1}{2}Nmv_{\text{rms}}^2 \quad \text{and} \quad pV = \frac{1}{3}Nmv_{\text{rms}}^2 = \frac{1}{3}Mv_{\text{rms}}^2$$

where $M = Nm$ is the mass of the gas.

The basic equation for the pressure of a gas is

$$p = \frac{2}{3}E_{k_0} = \frac{1}{3}n_0 m v_{\text{rms}}^2 = \frac{1}{3}\rho v_{\text{rms}}^2$$

where $E_{k_0} = E_k/V$, $n_0 = N/V$ is the number of molecules in unit volume, and $\rho = n_0 m$ is the density of the gas.

10.2.4

From a comparison of the Mendeleev-Clapeyron equation (8.4.4) with the basic equation (10.2.3) $RT = \mu v_{\text{rms}}^2/3$ it follows that

$$v_{\text{rms}} = \sqrt{\frac{3RT}{\mu}} = \sqrt{\frac{3RT}{mN_A}} = \sqrt{\frac{3kT}{m}} = 1.73\sqrt{pv}$$

where k is Boltzmann's constant (8.4.5), m is the mass of a molecule, N_A is Avogadro's number (**??**), p is the pressure of the gas, and v is its specific volume.

10.2.5

The *average kinetic energy of translational motion of the molecules in an ideal gas* is

$$v_{\text{rms}} = \sqrt{\frac{3RT}{\mu}} = \sqrt{\frac{3RT}{mN_A}} = \sqrt{\frac{3kT}{m}} = 1.73\sqrt{pv}$$

The average kinetic energy $\langle E_k^m \rangle$ is proportional to the absolute temperature and is independent of any other variables (Figure 10.1). The absolute, or thermodynamic,

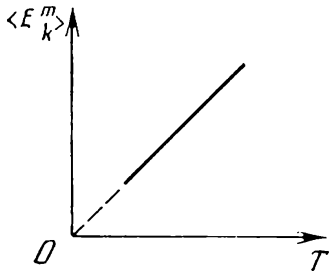


Figure 10.1: The relation of average kinetic energy and absolute temperature.

temperature is a measure of the average kinetic energy of translational motion of the molecules in an ideal gas. This consequence of the kinetic theory of gases is not valid in the region of ultra-low temperatures, close to absolute zero (8.3.4).

10.3 Maxwell's Molecular Velocity and Energy Distribution Law (Maxwell's Distribution Law)

10.3.1

The *law of the distribution of molecular velocities for an ideal gas*, theoretically established by Maxwell, determines the number $\mathrm{d}n$ of molecules of a homogeneous monatomic ideal gas, out of the total number n_0 of its molecules in unit volume, that has a velocity, at a given temperature, lying within the range from u to $u + \mathrm{d}u$. This law is

valid for gases in a state of thermodynamic equilibrium (8.3.3). The velocity distribution of the molecules of such a gas is a steady-state value. The Maxwellian distribution is based on pairwise collisions of the randomly moving gas molecules. This assumes uniform distribution of the molecules throughout the volume of the vessel, i.e. constant density of the gas.

10.3.2

In its most customary form, the law of the distribution of molecular speeds (magnitudes of the molecular velocities) is presented as

$$\mathrm{d}n = n_0 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{mu^2}{2kT}\right) 4\pi u^2 \mathrm{d}u$$

Here u is the magnitude of the molecular velocity (speed of a molecule), m is the mass of a molecule, k is Boltzmann's constant, and T is the absolute (thermodynamic) temperature. Molecular velocity distribution curves for various temperatures $T_1 < T_2 < T_3$ are shown in Figure 10.2. As is evident from these curves, the most probable velocity u_p (10.3.4) increases with temperature and the fraction of molecules travelling at this velocity decreases.

10.3.3

Another form of the Maxwellian distribution is

$$\mathrm{d}n = n_0 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{mu^2}{2kT}\right) \mathrm{d}u_x \mathrm{d}u_y \mathrm{d}u_z$$

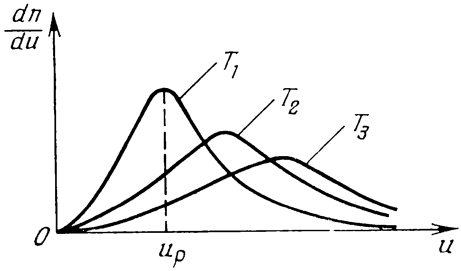


Figure 10.2: Molecular velocity distribution for various temperatures.

where u_x, u_y and u_z are projections of the molecular velocity on the coordinate axes. Owing to the random nature of thermal motion of the molecules (8.1.1), distributions of the molecules according to the projections u_i of their velocities on the coordinate axes ($i = x, y, z$) are mutually independent. Hence

$$\mathrm{d}n = n_0 f(u_x) f(u_y) f(u_z) \mathrm{d}u_x \mathrm{d}u_y \mathrm{d}u_z$$

where

$$f(u_i) = \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{mu_i^2}{2kT}\right) \quad (i = x, y, z)$$

is a distribution function of the projections of the molecular velocities.

The Maxwellian distribution is isotropic. This is evident from the fact that the function $f(u_x, u_y, u_z)$ depends only upon the speed (magnitude of the velocity), and because the function $f(u_i)$ is the same along all the axes.

10.3.4

The law of the distribution of molecular velocities (10.3.2) can be used to determine the *most probable velocity* u_p of the molecules, which corresponds to the maximum of the function

$$f(u) = \frac{dn}{du} = 4\pi n_0 \left(\frac{m}{4\pi kT}\right)^{1/2} \exp\left(-\frac{mu^2}{2kT}\right)$$

The condition

$$\left[\frac{d}{du}\left(\exp\left(-\frac{mu^2}{2kT}\right)u^2\right)\right]_{u=u_p} = 0$$

yields

$$u_p = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2RT}{\mu}} = v_{rms}\sqrt{\frac{2}{3}}$$

10.3.5

A third form of the law of the distribution of molecular velocities is

$$dn = \frac{4n_0}{\sqrt{\pi}} \exp\left(-\frac{u^2}{u_{p^2}}\right)\left(\frac{u}{u_p}\right)^2 \frac{du}{du_p}$$

The fraction dn/n_0 of the molecules of a gas having velocities lying within the range from u to $u+du$ is numerically equal to the hatched area dS under the curve in Figure 10.3, which represents the dependence of $(u_p/n_0)(dn/du)$ on u/u_p:

$$dS = \left(\frac{u_p}{n_0}\frac{dn}{du}\right)\frac{du}{u_p} = \frac{dn}{n_0}$$

The area under the curve in Figure **??**, i.e. bounded by the curve and the axis of abscissas, is equal to unity. This area represents the fraction of the molecules having all possible velocity values from 0 to ∞.

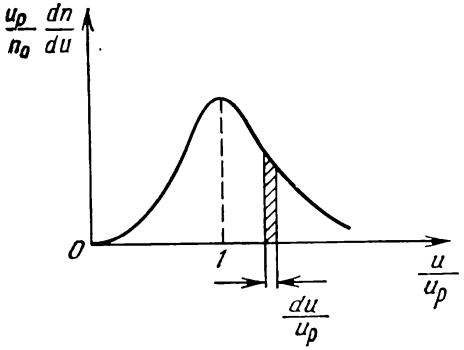


Figure 10.3: A velocity distribution curve for molecules.

10.3.6

The arithmetic mean velocity $\langle u \rangle$ of translational motion of the molecules in an ideal gas, calculated by means of the distribution law (10.3.2), is

$$\langle u \rangle = v_{\text{rms}} \sqrt{\frac{8}{3\pi}} = \sqrt{\frac{8RT}{\pi\mu}} = \sqrt{\frac{8kT}{\pi m}} = 1.60\sqrt{pv}$$

10.3.7

The energy distribution of the molecules in an ideal gas determines the fraction $\mathrm{d}n_{E^m}/n_0$ of molecules, of their

total number n_0, having kinetic energies $E_k^{\mathrm{m}} = mu^2/2$ lying within the range from E_k^{m} to $E_k^{\mathrm{m}} + \mathrm{d}E_k^{\mathrm{m}}$:

$$\mathrm{d}n_{E^{\mathrm{m}}} = \frac{2n_0}{\sqrt{\pi}} (kT)^{-3/2} \exp\left(-\frac{E_k^{\mathrm{m}}}{kT}\right) \sqrt{E_k^{\mathrm{m}}}\,\mathrm{d}E_k^{\mathrm{m}}$$

Here $\mathrm{d}n_{E^{\mathrm{m}}}/n_0 = f(E_k^{\mathrm{m}})\,\mathrm{d}E_k^{\mathrm{m}}$, where $f(E_k^{\mathrm{m}})$ is the distribution function for the molecular energies in an ideal gas.

Example. The average kinetic energy $\langle E_k^{\mathrm{m}} \rangle$ of a molecule in an ideal gas is

$$\begin{aligned}\langle E_k^{\mathrm{m}} \rangle &= \int_0^\infty E_k^{\mathrm{m}} f(E_k^{\mathrm{m}})\,\mathrm{d}E_k^{\mathrm{m}} \\ &= \frac{2}{\sqrt{\pi}(kT)^{3/2}} \int_0^\infty E_k^{\mathrm{m}} \exp\left(-\frac{E_k^{\mathrm{m}}}{kT}\right) \sqrt{E_k^{\mathrm{m}}}\,\mathrm{d}E_k^{\mathrm{m}} \\ &= \frac{3}{2}kT\end{aligned}$$

(cf. (10.2.4)).

10.3.8

The relative motion (7.1.2) of two particles with the masses m_1 and m_2 is equivalent to the motion of a single particle with the reduced mass $m_{\mathrm{red}} = m_1 m_2/(m_1 + m_2)$. For a homogeneous gas $m_1 = m_2 = m$ and $m_{\mathrm{red}} = m/2$. The distribution of molecules according to their relative velocities establishes the fraction $\mathrm{d}n_{u_{\mathrm{rel}}}/n_0$ of molecules, of their total number n_0, having relative velocities u_{rel} lying within the range from u_{rel} to $u_{\mathrm{rel}} + \mathrm{d}u_{\mathrm{rel}}$:

$$\mathrm{d}n_{u_{\mathrm{rel}}} = n_0 \left(\frac{m}{4\pi kT}\right)^{3/2} \exp\left(-\frac{mu_{\mathrm{rel}}^2}{4kT}\right) 4\pi u_{\mathrm{rel}}^2 \mathrm{d}u_{\mathrm{rel}}$$

Here $\mathrm{d}n_{u_{\mathrm{rel}}}/n_0 = f(u_{\mathrm{rel}})\mathrm{d}u_{\mathrm{rel}}$, where

$$f(u_{\mathrm{rel}}) = 4\pi\left(\frac{m}{4\pi kT}\right)^{3/2}\exp\left(-\frac{mu_{\mathrm{rel}}^2}{4kT}\right)u_{\mathrm{rel}}^2$$

is the relative velocity distribution of the molecules in an ideal gas.

Example. The *average relative velocity* $\langle u_{\mathrm{rel}}\rangle$ of the molecules is

$$\langle u_{\mathrm{rel}}\rangle = \int_0^\infty u_{\mathrm{rel}} f(u_{\mathrm{rel}})\,\mathrm{d}u_{\mathrm{rel}} = \sqrt{2}\sqrt{\frac{8kT}{\pi m}} = \sqrt{2}\langle u\rangle$$

where $\langle u\rangle$ is the arithmetic mean velocity of the molecules (10.3.6).

10.4 Particle Distribution in a Potential Force Field (Boltzmann Distribution)

10.4.1

Gases are ordinarily in the potential (conservative) gravitational fields of the earth (6.2.1). If there were no such gravitational field, the atmospheric air would be dispersed in the universe. If, on the other hand, there were no thermal motion, the molecules of atmospheric air would drop to the earth. Gravity and thermal motion lead to a steady state of gases, in which their concentration and pressure decrease with an increase of their altitude above the earth's surface.

10.4.2

If a gas (or other system of particles) is located in an external potential (conservative) force field (3.1.6), the distribution of the particles throughout the volume is described by the *Boltzmann distribution law*. This law establishes the number of particles $\mathrm{d}n(x, y, z)$ that have coordinates within the ranges from x to $x + \mathrm{d}x$, from y to $y + \mathrm{d}y$, and from z to $z + \mathrm{d}z$. In other words, $\mathrm{d}n(x, y, z)$ is the number of molecules contained in the element of volume $\mathrm{d}V = \mathrm{d}x\mathrm{d}y\mathrm{d}z$. The Boltzmann distribution law is of the form

$$\mathrm{d}n\,(x, y, z) = \text{const}\,\exp\left(-\frac{E_{\mathrm{p}}^{\mathrm{m}}}{kT}\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$

where $E_{\mathrm{p}}^{\mathrm{m}}(x, y, z)$ is the potential energy of a particle in the external force field, k is Boltzmann's constant, and T is the absolute temperature. The value of the constant in the preceding equation is determined from the normalization condition:

$$\int \mathrm{d}n = n_0$$

where n_0 is the total number of particles in unit volume.

Example. For particles of mass m, located in the earth's gravitational field, $E_{\mathrm{p}}^{\mathrm{m}} = mgh$ (3.3.3), where g is the acceleration due to gravity (7.3.3), and h is the altitude. At each altitude there is a Maxwellian molecular velocity distribution (10.3.1). The number of molecules contained in the volume $\mathrm{d}V$ is

$$\mathrm{d}n\,(x,y,z) = \text{const}\exp\left(-\frac{mgh}{kT}\right)\mathrm{d}V.$$

The gas density $\rho = m\mathrm{d}n/\mathrm{d}V$ decreases with an increase in the altitude according to the exponential law

$$\rho = \text{const}\exp\left(-\frac{mgh}{kT}\right)$$

The value of the constant can be determined from the condition $\rho = \rho_0 = \text{const}$ at $h = 0$. The density of the gas or its pressure varies according to the *barometric height formula*

$$\rho = \rho_0 \exp\left(-\frac{mgh}{kT}\right) \quad \text{and} \quad p = p_0 \exp\left(-\frac{mgh}{kT}\right)$$

10.4.3

The *Maxwell-Boltzmann law*, or *distribution*, describes the distribution of the coordinates and velocities of gas molecules subject to the action of an arbitrary potential (conservative) force field:

$$\mathrm{d}n = \text{const}\exp\left(-\frac{p_x^2 + p_y^2 + p_z^2}{2mkT}\right)\mathrm{d}p_x\mathrm{d}p_y\mathrm{d}p_z$$
$$\times \exp\left(-\frac{E_\mathrm{p}^\mathrm{m}(x,y,z)}{kT}\right)\mathrm{d}x\mathrm{d}y\mathrm{d}z$$

where $\mathrm{d}n$ is the number of molecules contained in six-dimensional space within the element of volume $\mathrm{d}\Gamma = \mathrm{d}x\mathrm{d}y\mathrm{d}z\,\mathrm{d}p_x\mathrm{d}p_y\mathrm{d}p_z$, and $E_\mathrm{p}^\mathrm{m}(x,y,z)$ is the potential energy of a molecule in the external force field at a point with the coordinates x, y and z, and the projections p_x, p_y and p_z of the momenta on the axes. The Maxwell-Boltzmann

law is the product of two distribution functions. One of them describes the distribution with respect to coordinates, and the other with respect to momenta (or velocities).

10.5 Mean Free Path of Molecules

10.5.1

Gas molecules have finite dimensions (8.4.1) and are continuously colliding with one another in thermal motion. Between two successive collisions, molecules travel at uniform velocity in a straight line over certain distances called their free paths λ. The mean free path $\langle\lambda\rangle$ is the average distance that a molecule travels without a collision. The mean free path is a characteristic of the whole totality of gas molecules at given values of p and T.

10.5.2

In unit time each molecule undergoes a mean number of collisions $\langle z\rangle$ (*average collision frequency*) equal to

$$\left\langle z^2 \right\rangle = \pi d^2 n_0 \langle u_{\text{rel}}\rangle = \sqrt{2}\pi d^2 n_0 \langle u\rangle$$

Here d is the effective diameter of the molecule (8.4.1), n_0 is the number of molecules in unit volume of the gas, $\langle u_{\text{rel}}\rangle$ is their average relative velocity (10.3.8), and $\langle u\rangle$ is their arithmetic mean velocity (10.3.6).

10.5.3

The average distance travelled by a molecule in unit time is numerically equal to $\langle u \rangle$. Therefore $\langle u \rangle = \langle \lambda \rangle \langle z \rangle$. The mean free path $\langle \lambda \rangle$ is

$$\langle \lambda \rangle = \frac{\langle u \rangle}{\langle z \rangle} = \frac{1}{\sqrt{2} d^2 n_0}$$

At constant temperature, n_0 is proportional to the pressure p of the gas (8.4.5). Hence, for a given gas the mean free path is inversely proportional to the pressure:

$$p_1 \langle \lambda_1 \rangle = p_2 \langle \lambda_2 \rangle = \text{const}$$

in which the subscripts 1 and 2 refer to two states of the gas.

10.5.4

If molecules are ejected from some source of particles and a beam is formed by means of a diaphragm, the free path distribution law is valid for the molecules of the beam:

$$N = N_0 \exp\left(-\frac{x}{\langle \lambda \rangle}\right)$$

where N is the number of molecules in the beam that have travelled a distance x without collisions, and N_0 is the number of molecules in the beam at $x = 0$, i.e. at the output of the diaphragm.

10.6 Principle of the Equipartition of Energy. Internal Energy of an Ideal Gas

10.6.1

The *number of degrees of freedom* of a body is the minimum number of coordinates (number of independent coordinates) that it is necessary to specify in order to completely determine the position of the body in space.

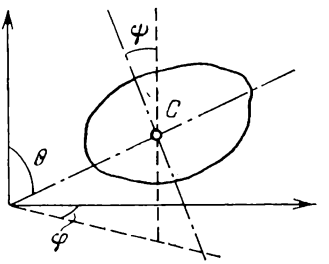


Figure 10.4: Degrees of freedom for a particle.

A particle, for instance, travelling along one of the coordinate axes has a single degree of freedom. The same particle travelling in a plane has two degrees of freedom. The position of a particle travelling freely in space is determined by three degrees of freedom: the coordinates x, y and z. A rigid body (1.1.4) has six degrees of freedom: its position in space is determined by the three coordinates of its centre of mass (2.3.3), two angles, θ and φ, determining the direction of a certain axis, fixed

in the body and passing through its centre of mass (Figure 10.4), and, in addition, angle ψ, determining the direction of a second axis, fixed in the body and perpendicular to the first axis. Changes in the three coordinates of the centre of mass at given constant angles θ, φ and ψ correspond to translational motion of the rigid body. The coordinates of the centre of mass are the three degrees of freedom of translational motion. Changes in the angle θ, φ or ψ at a constant position of the centre of mass lead to rotation of the rigid body. Hence, the correspond ing degrees of freedom are said to be rotary. Additional degrees of freedom of oscillatory, or vibratory, motion (10.6.5) are introduced to determine the position in space of an imperfectly rigid body, whose various parts can be displaced with respect to one another.

10.6.2

In many problems a *molecule of a monatomic gas* can be regarded as a particle (cf. (8.4.1)). This is based on the fact that practically all the mass of such an atom is concentrated in its nucleus (**??**), which is of extremely small size. The molecule of monatomic gas has three degrees of freedom of translational motion. Its average kinetic energy $\left\langle E_{\mathrm{k}}^{\mathrm{m}} \right\rangle$ is equal to the kinetic energy of molecules travelling at a velocity equal to the root-mean-square velocity v_{rms} (10.2.4):

$$\left\langle E_{\mathrm{k}}^{\mathrm{m}} \right\rangle = \frac{3}{2} kT$$

Owing to the random nature of thermal motion of the molecules, this energy is uniformly distributed between

the three degrees of freedom. Hence, on an average each degree of freedom of translational motion of a monatomic molecule accounts for the same amount of kinetic energy $\left\langle E^{\mathrm{m}}_{\mathrm{k}_0} \right\rangle = \left\langle E^{\mathrm{m}}_{\mathrm{k}} \right\rangle / 3 = kT/2$, where k is Boltzmann's constant and T is the absolute temperature.

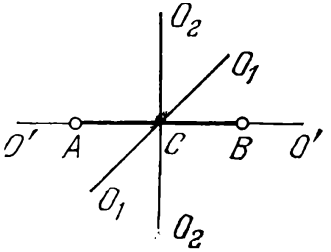


Figure 10.5: Degrees of freedom of A diatomic molecule.

10.6.3

As a first approximation, a molecule of a diatomic gas consists of two atoms, A and B, rigidly held together by an incompressible, undeformable bond (Figure 10.5). In addition to the three degrees of freedom of translational motion, the centre of mass C (2.3.3) of such a molecule has two more degrees of freedom of rotary motion about axes $O_1 - O_1$ and $O_2 - O_2$. Rotation about the third axis $O' - O'$ makes no contribution to the molecule's energy because the moment of inertia (4.2.1) of the atoms with respect to this axis is negligibly small. Thus, a diatomic molecule has five degrees of freedom. Molecules consisting of three or more atoms have six degrees of free-

dom: three translational and three rotary degrees (Figure 10.6).

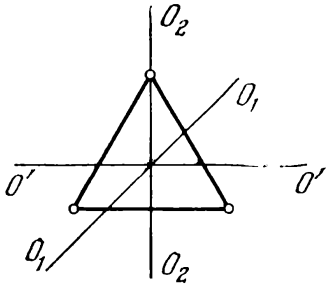


Figure 10.6: Degrees of freedom of molecules with three or more atoms.

10.6.4

The principle of equipartition of energy states that on an average each degree of freedom of a molecule accounts for the same amount of kinetic energy, equal to $kT/2$. If a molecule has i degrees of freedom its average kinetic energy $\left\langle E_{\mathrm{k}}^{\mathrm{m}} \right\rangle = ikT/2$.

The principle of equipartition of energy leads to the conclusion that all degrees of freedom of the molecule are equally justified: each makes the same contribution to its average energy. This conclusion actually has only a limited field of application, and has been revised for quantum statistics (**??**).

10.6.5

In real molecules, which are not absolutely rigid, vibratory degrees of freedom make a contribution to the energy of the molecule in accordance with the principle of equipartition of energy (10.6.4). Each vibratory degree of freedom accounts, not only for kinetic energy $\left\langle E_{\mathrm{k}}^{\mathrm{m}} \right\rangle$ but for potential energy $\left\langle E_{\mathrm{p}}^{\mathrm{m}} \right\rangle$ as well. For harmonic vibrations (**??**), $\left\langle E_{\mathrm{k}_0}^{\mathrm{m}} \right\rangle = \left\langle E_{\mathrm{p}_0}^{\mathrm{m}} \right\rangle$. According to (10.6.4)

$$\left\langle E_{\mathrm{k}_0}^{\mathrm{m}} \right\rangle = \left\langle E_{\mathrm{p}_0}^{\mathrm{m}} \right\rangle = \frac{1}{2} kT$$

where $\left\langle E_{\mathrm{k}_0}^{\mathrm{m}} \right\rangle$ and $\left\langle E_{\mathrm{p}_0}^{\mathrm{m}} \right\rangle$ are the average kinetic and potential energies of a molecule per vibratory degree of freedom.

The average value $\left\langle E_0^{\mathrm{m}} \right\rangle$ of the total energy of a molecule per vibratory degree of freedom is

$$\left\langle E_0^{\mathrm{m}} \right\rangle = \left\langle E_{\mathrm{k}_0}^{\mathrm{m}} \right\rangle + \left\langle E_{\mathrm{p}_0}^{\mathrm{m}} \right\rangle = 2 \left\langle E_{\mathrm{k}_0}^{\mathrm{m}} \right\rangle = kT$$

This equation is valid for harmonic vibrations of particles (atoms, molecules or ions) in the crystal lattice sites of solids (8.1.4).

10.6.6

The internal energy (9.1.2) of a polyatomic gas constitutes the kinetic energy of all kinds of motion of its particles. For one mole of such a gas

$$U = \left\langle E_{\mathrm{k}}^{\mathrm{m}} \right\rangle N_A = \frac{i}{2} k N_A T = i \frac{RT}{2}$$

where N_A is Avogadro's number (**??**), i is the number of degrees of freedom of the molecule (10.6.4), and R is the universal gas constant (8.4.4).